2 CHEMISTRY MOCK TEST: DAYS 1-20

Time: 75 min | **Max. marks:** 40 | **Questions:** 22, all compulsory
Marks per question: A (MCQ) 1, B (very short) 2, C (short) 3, D (long) 5, E (case-based) 4

TIME PLAN FOR 75 MINUTES

Section A 15 min, B 12 min, C 15 min, D 10 min, E 8 min; keep the last 15 min for checking.



Timer lagao, phone door rakho, aur real board exam ki tarah ek baar mein poora paper likho!

SECTION A: MULTIPLE CHOICE — 12 × 1 = 12 marks

1. In the reaction $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$, the substance oxidised is: ★★★

(a) Fe_2O_3 (b) Al (c) Al_2O_3 (d) Fe

2. The brown fumes given off when lead nitrate is heated are of: ★★★

(a) O_2 (b) NO (c) NO_2 (d) N_2O

3. Which of the following is an endothermic process? ★★☆

(a) Respiration (b) Burning natural gas
(c) Heating $CaCO_3$ (d) Quicklime + water

4. A solution turns red litmus blue. Its pH is most likely: ★★☆

(a) 1 (b) 4 (c) 5 (d) 10

5. The chemical formula of Plaster of Paris is: ★★★

(a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(c) $CaSO_4$ (d) $CaCO_3$

6. Tooth decay starts when the pH of the mouth falls below: ★★☆

(a) 7.4 (b) 6.5 (c) 5.5 (d) 8.5

7. Which of these metals is the best conductor of heat? ★☆☆

(a) Lead (b) Mercury (c) Silver (d) Iron

8. Which element forms an amphoteric oxide? ★★☆

(a) Sodium (b) Magnesium (c) Aluminium (d) Sulphur

9. The next higher homologue of propanol, C_3H_7OH, is: ★★☆

(a) C_2H_5OH (b) C_4H_9OH (c) C_3H_7CHO (d) C_4H_{10}

10. Which of these burns with a yellow, sooty flame? ★★☆

(a) Methane (b) Ethane
(c) Naphthalene (d) Ethanol

ASSERTION–REASON OPTIONS: Q11, Q12

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

11. **Assertion (A):** Dry HCl gas does not change the colour of dry blue litmus paper.
Reason (R): HCl gas produces H^+ ions only in the presence of water. ★★★

12. **Assertion (A):** Copper displaces zinc from zinc sulphate solution.
Reason (R): Zinc is more reactive than copper. ★★★

SECTION B: VERY SHORT ANSWER — 5 × 2 = 10 marks

13. Balance: (a) $Al + CuCl_2 \rightarrow AlCl_3 + Cu$ (b) $BaCl_2 + Al_2(SO_4)_3 \rightarrow BaSO_4 + AlCl_3$ [2] ★★★

14. Write the chemical name, formula and one use of: (a) bleaching powder (b) baking soda. [2] ★★★

15. (a) Why is sodium kept immersed in kerosene? (b) Why do ionic compounds conduct electricity in the molten state but not in the solid state? [2] ★★☆

16. What is a homologous series? Write the general formula of alkenes and name the second member. [2] ★★☆

17. Chips packets are flushed with nitrogen gas. Why? Name the process that this prevents. [2] ★★☆

SECTION C: SHORT ANSWER — 3 × 3 = 9 marks

18. Write balanced equations for: (a) the chlor-alkali process (b) the preparation of bleaching powder (c) the preparation of Plaster of Paris from gypsum. [3] ★★★

19. Zinc is extracted from zinc blende (ZnS) and from zinc carbonate. Name the process used for each ore to get zinc oxide and write all equations, including the reduction of zinc oxide with carbon. [3] ★★★

20. Explain the cleansing action of soap with the help of micelles. Why does soap not form lather with hard water? [3] ★★★

SECTION D: LONG ANSWER — 1 × 5 = 5 marks

21. Write a balanced equation with state symbols and name the type of reaction for: (a) water added to quicklime (b) heating ferrous sulphate crystals (c) an iron nail dipped in copper sulphate solution (d) sodium sulphate solution mixed with barium chloride solution (e) hydrogen passed over heated copper(II) oxide; also name the substance oxidised and the substance reduced in (e). [5] ★★★

SECTION E: CASE-BASED — 1 × 4 = 4 marks

22. Riya added equal-sized pieces of four metals P, Q, R and S separately to dilute hydrochloric acid and recorded her observations. [4] ★★★

Metal	Observation with dilute HCl
P	Brisk effervescence; test tube becomes warm
Q	A few bubbles form slowly
R	No change at all
S	Moderate, steady effervescence

- (i) Arrange P, Q, R and S in decreasing order of reactivity. [1]
- (ii) Which metal could be copper? Give a reason. [1]
- (iii) P is magnesium. Write the balanced equation for its reaction with dilute HCl and give the test for the gas evolved. [2]

END OF QUESTION PAPER

Stop the timer. Check every answer with the answer key and marking points on pages 3 and 4, then fill in the score tracker.

3 ANSWER KEY AND MARKING POINTS

Boss Level: MCQ answers

Q	Ans	Why
1	(b)	2 C–C bonds + 8 C–H bonds = 10
2	(a)	Both are alkenes, C_nH_{2n}; they differ by CH_2
3	(b)	Aldehyde group –CHO; suffix "-al"
4	(c)	N≡N triple bond = 3 shared pairs
5	(c)	C_3H_4 is propyne, C_nH_{2n-2} (triple bond)
6	(b)	$2Na + 2C_2H_5OH \rightarrow 2C_2H_5ONa + H_2$
7	(c)	Addition of hydrogen with Ni or Pd catalyst
8	(b)	Tails dissolve in oil; ionic heads face the water
9	(a)	Free electrons in graphite carry the current
10	(c)	R is false: detergents are sulphonate or ammonium salts; the sodium salts of carboxylic acids are soaps

Boss Level: model answers, Q11–12

11 Why carbon shares electrons (1 + 1 + 1)

- C^{4-}: a nucleus with 6 protons cannot hold 10 electrons
- C^{4+}: removing 4 electrons needs a very large amount of energy; so carbon **shares** electrons and forms covalent bonds
- **(b) Catenation** (long chains, branches, rings) and **tetravalency** (bonds with O, H, N, S, Cl and others)

12 IUPAC names (1 each)

Formula	IUPAC name	Functional group
CH_3CH_2Br	Bromoethane	Halo (–Br)
CH_3COCH_3	Propanone	Ketone (>C=O)
CH_3CH_2CHO	Propanal	Aldehyde (–CHO)

Boss Level: model answer, Q13

13 Test: 1, observation: 1, equation: 1

- Add sodium hydrogencarbonate to both: **ethanoic acid** gives brisk effervescence; the gas turns lime water milky (CO_2)
- **Ethanol:** no effervescence. (Ethanoic acid also turns blue litmus red.)

$CH_3COOH_{(aq)} + NaHCO_{3(s)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$

Boss Level: model answer, Q14 (2 + 2 + 1)

- **A** = ethanol, CH_3CH_2OH; **B** = ethene, $CH_2=CH_2$
- **C** = ethanoic acid, CH_3COOH; **D** = ethyl ethanoate, $CH_3COOC_2H_5$

$CH_3CH_2OH_{(l)} \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2=CH_{2(g)} + H_2O_{(l)}$

Dehydration; B decolourises bromine water

$CH_3CH_2OH_{(aq)} + 2[O] \xrightarrow[\text{Heat}]{\text{Alkaline } KMnO_4} CH_3COOH_{(aq)} + H_2O_{(l)}$

Oxidation

$CH_3COOH_{(l)} + CH_3CH_2OH_{(l)} \xrightarrow{\text{Acid}} CH_3COOC_2H_{5(l)} + H_2O_{(l)}$

Esterification: D is the sweet-smelling ester

$CH_3COOC_2H_{5(l)} + NaOH_{(aq)} \rightarrow C_2H_5OH_{(aq)} + CH_3COONa_{(aq)}$

Saponification: D + NaOH gives ethanol and sodium ethanoate

Mock Test: Section A answers

Q	Ans	Why
1	(b)	Al gains oxygen, so Al is oxidised; Fe_2O_3 is reduced
2	(c)	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
3	(c)	$CaCO_3 \rightarrow CaO + CO_2$ needs heat
4	(d)	Bases turn red litmus blue; pH above 7
5	(b)	$CaSO_4 \cdot ½H_2O$
6	(c)	Below pH 5.5 the tooth enamel corrodes
7	(c)	Silver and copper are the best; lead and mercury are poor
8	(c)	Al_2O_3 reacts with both acids and bases
9	(b)	Add CH_2: C_4H_9OH, butanol
10	(c)	Unsaturated and aromatic compounds give a sooty flame
11	(a)	Without water, HCl gives no H^+ ions
12	(d)	A is false: Cu is less reactive than Zn; R is true

Mock Test: Section B, Q13–14

13 Balanced equations (1 + 1)

$2Al_{(s)} + 3CuCl_{2(aq)} \rightarrow 2AlCl_{3(aq)} + 3Cu_{(s)}$

Al 2 = 2, Cu 3 = 3, Cl 6 = 6 ✔

$3BaCl_{2(aq)} + Al_2(SO_4)_{3(aq)} \rightarrow 3BaSO_{4(s)}\downarrow + 2AlCl_{3(aq)}$

Ba 3 = 3, Cl 6 = 6, Al 2 = 2, S 3 = 3, O 12 = 12 ✔

14 Bleaching powder and baking soda (½ each)

- **(a)** Calcium oxychloride, $CaOCl_2$: disinfecting drinking water, bleaching cotton and linen
- **(b)** Sodium hydrogencarbonate, $NaHCO_3$: antacid, baking powder, soda-acid fire extinguisher

Mock Test: Section B, Q15–17

15 Sodium and ionic compounds (1 + 1)

- **(a)** Sodium reacts so vigorously with oxygen and moisture that it catches fire in air; kerosene keeps air away
- **(b)** In the solid, ions are held in a rigid lattice and cannot move; in the melt, ions move freely and carry charge

16 Homologous series (1 + 1)

- Compounds with the **same functional group** and similar chemical properties; successive members differ by CH_2 (14 u)
- Alkenes: C_nH_{2n}; second member: **propene**, $CH_3CH=CH_2$

17 Nitrogen flushing (1 + 1)

- Nitrogen is unreactive and replaces oxygen, so the fats and oils in chips are not oxidised
- It prevents **rancidity** (bad smell and taste)

Mock Test: Section C, Q18

18 Chemicals from common salt (1 each)

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$

(a) Chlor-alkali: Cl_2 at the anode, H_2 at the cathode

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$

(b) Bleaching powder

$CaSO_4 \cdot 2H_2O_{(s)} \xrightarrow{373\ K} CaSO_4 \cdot ½H_2O_{(s)} + 1½H_2O_{(g)}$

(c) Plaster of Paris from gypsum

Mock Test: Section C, Q19

19 Extraction of zinc (1 each)

$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{Heat} 2ZnO_{(s)} + 2SO_{2(g)}$

Roasting: sulphide ore heated strongly in excess air

$ZnCO_{3(s)} \xrightarrow{Heat} ZnO_{(s)} + CO_{2(g)}$

Calcination: carbonate ore heated strongly in limited air

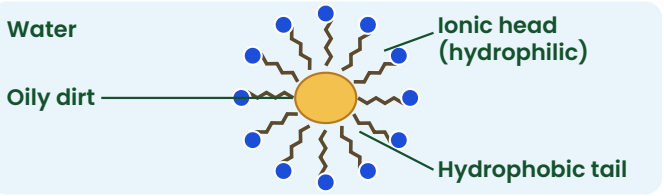
$ZnO_{(s)} + C_{(s)} \xrightarrow{Heat} Zn_{(s)} + CO_{(g)}$

Reduction of zinc oxide with carbon

Mock Test: Section C, Q20

20 Cleansing action of soap (2 + 1)

- A soap molecule has an **ionic (hydrophilic) head** that dissolves in water and a long **hydrocarbon (hydrophobic) tail** that dissolves in oil
- In water the molecules form **micelles**: tails point inwards and trap the oily dirt, heads face the water outside
- The micelles stay suspended and the dirt is rinsed away with the water
- **Hard water:** Ca^{2+} and Mg^{2+} ions react with soap to form an insoluble **scum**, so soap is wasted and little lather forms



Mock Test: Section D marking points (1 each)

$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$

(a) Combination; exothermic

$2FeSO_{4(s)} \xrightarrow{Heat} Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$

(b) Thermal decomposition; green crystals turn reddish-brown

$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$

(c) Displacement; blue solution turns light green

$Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$

(d) Double displacement; white precipitate

$CuO_{(s)} + H_{2(g)} \xrightarrow{Heat} Cu_{(s)} + H_2O_{(g)}$

(e) Redox: H_2 is oxidised and CuO is reduced

Mock Test: Section E marking points

- **(i)** P > S > Q > R (1)
- **(ii)** R; copper is below hydrogen in the reactivity series, so it cannot displace hydrogen from dilute acid (1)
- **(iii)** Bring a burning splinter near the mouth of the test tube: the gas burns with a pop sound, so it is hydrogen (1 + 1)

$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}\uparrow$



★★★ *Balancing, redox, pH, reactivity series, esterification: ye sab pakka aayenge, Nitesh Bhaiya ki guarantee hai!*

Find your weak chapter: score tracker

Chapter	Questions	Max.	Mine	Revise
1. Chemical reactions and equations	1, 2, 3, 13, 17, 21	12	___	Days 1–4
2. Acids, bases and salts	4, 5, 6, 11, 14, 18	9	___	Days 5–9
3. Metals and non-metals	7, 8, 12, 15, 19, 22	12	___	Days 10–14
4. Carbon and its compounds	9, 10, 16, 20	7	___	Days 15–19
Mock test total	22 questions	**40**	___	
Chapter 4 Boss Level	14 questions	**24**	___	Days 15–19

How to use the marking points

- Give yourself marks **only** for points that match the key; partial marks are fine
- A balanced equation without state symbols loses ½ mark in many papers: always add them
- Write every wrong question in a "mistake notebook" and re-solve it after 3 days

Day 20 checklist: tick before you sleep

- ☐ I finished the Chapter 4 Boss Level (24 marks) in 40 minutes
- ☐ I wrote the full 40-mark mock test in one sitting of 75 minutes
- ☐ I checked every answer against the key and marking points
- ☐ I found my weak chapter from the marks table above
- ☐ I re-read the Days of that chapter and corrected my mistakes

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Weak chapter ke Days phir se padho.



LEVEL 20 COMPLETE

+200 XP earned. Chemistry cleared. Next: Level 21



BUJHO *Samajh gaya! Chemistry ke chaaron chapter clear, Boss bhi haar gaya!*

PAHELI *Shabaash! Ab Physics ki duniya mein milte hain.*



NITESH SIR SAYS

Mock ke marks kam aaye toh tension mat lo. ***Galtiyon ki list hi asli revision list hai!*** *Chemistry complete, ab Physics ki taiyari.*

CHEMISTRY | CHAPTER 4 COMPLETE | CHEMISTRY COMPLETE

CHAPTER 4 SWAHAA!

Carbon and its Compounds | Days 15–20



BUJHO *Covalent bond, isomers, soap ka micelle, mock test bhi! Chapter 4 Swahaa, poori Chemistry Swahaa!*

PAHELI *Trophy tumhari hai, Bujho! 20 levels clear. Ab Physics ki taraf chalte hain.*

CHAPTER XP EARNED
+750 XP

LEVELS CLEARED
6 levels (Days 15–20)

BOOK PROGRESS
Level 20 / 60 2500 XP total

WHAT YOU HAVE MASTERED

- ✔ Covalent bonding, tetravalency, catenation
- ✔ Functional groups and IUPAC names
- ✔ Ethanol, ethanoic acid, esterification
- ✔ Hydrocarbons, isomers, homologous series
- ✔ Combustion, oxidation, addition, substitution
- ✔ Soaps, detergents and micelles

NEXT CHAPTER Day 21: Physics begins

Chemistry cleared!

DAY 21 · LEVEL 21

PHYSICS | CHAPTER 5 | PART 1 · ASTRA SERIES

LIGHT & REFLECTION

Goal: master reflection and the terms of spherical mirrors.



by Nitesh Prakash

01 Light | 02 Reflection Laws | 03 Plane Mirror | 04 Spherical Mirrors | 05 Mirror Terms | +125 XP today | Level 21 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT IS LIGHT?

DEFINITION

- **Light** is a form of **energy** that produces the **sensation of vision** in our eyes.
- It is an **electromagnetic wave**, so it needs **no medium**: it travels even through vacuum.
- Speed of light in vacuum (or air): $c = 3 \times 10^8$ **m/s**, the fastest speed known.

How light behaves around us

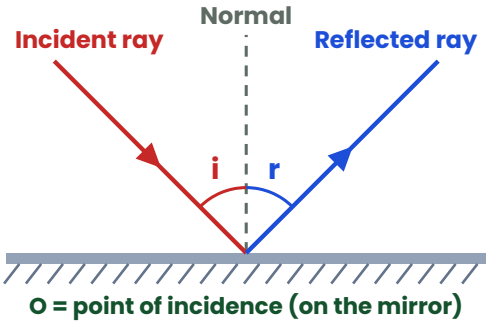
- ✔ **Travels in straight lines** (rectilinear propagation). Proof: an opaque object casts a **sharp shadow**.
- ✔ We see an object only when light from it (its own or reflected) **enters our eyes**.
- ✔ Shiny, polished surfaces such as a mirror **reflect most of the light** falling on them.
- ✔ Sunlight takes about **8 minutes** to reach the Earth.

2 REFLECTION OF LIGHT ★★★ MOST ASKED

DEFINITION

- **Reflection:** the bouncing back of light into the **same medium** when it falls on a polished surface.

The terms in one figure



- **Incident ray:** the ray that falls on the surface; **reflected ray:** the ray that bounces back
- **Normal:** the line at 90° to the surface at the point of incidence O
- **Angle of incidence (∠i):** between incident ray and normal; **angle of reflection (∠r):** between reflected ray and normal

Laws of reflection ★★★

1. The **angle of incidence is equal to the angle of reflection**: ∠i = ∠r.
2. The incident ray, the normal at the point of incidence and the reflected ray **all lie in the same plane**.

These laws hold for **every reflecting surface**, plane or spherical.



★★★ *Dono laws of reflection har saal 1 mark mein aate hai: Nitesh Bhaiya ki guarantee, ∠i = ∠r ratna nahi, samajhna hai!*



BUJHO *Ray seedhi mirror pe 90° pe giri, toh ∠i = 90° hoga na?*

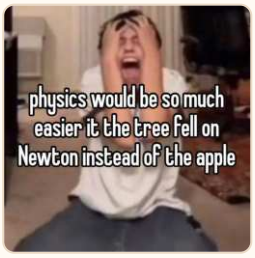
PAHELI *Nahi! Angles normal se naapte hai. Normal ke along giri ray ka ∠i = 0°, aur wo usi raste wapas aati hai.*



Solved Example 1: Angle work ★★☆

Q. A ray strikes a plane mirror making an angle of 30° with the **mirror surface**. Find ∠i, ∠r and the angle between the incident and reflected rays.

- Angle with the mirror = 30°, so angle with the normal: ∠i = 90° − 30° = **60°**
- By the first law, ∠r = ∠i = **60°**
- Angle between incident and reflected rays = ∠i + ∠r = **120°**



MEME BREAK

Newton ke sir pe seb gira toh gravity mili. Hamare sir pe sirf syllabus girta hai! Chalo, Light chapter pyaar se shuru karte hai.

3 IMAGE: REAL OR VIRTUAL?

DEFINITION

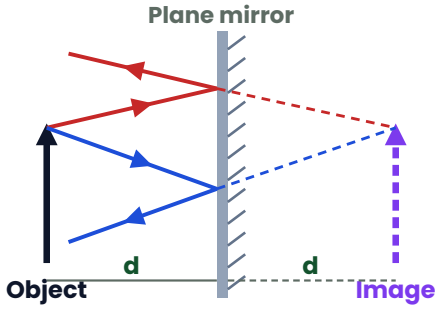
- **Image:** the point where rays from a point of the object **actually meet**, or **appear to meet**, after reflection or refraction.

Real image vs virtual image ★★★

Real image	Virtual image
Rays **actually meet**	Rays only **appear to meet** (when produced backwards)
Can be **obtained on a screen**	**Cannot** be obtained on a screen
Always **inverted**	Always **erect**
Example: picture on a cinema screen	Example: your face in a plane mirror

4 IMAGE IN A PLANE MIRROR ★★★ MOST ASKED

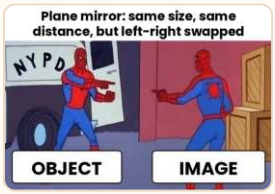
How the image forms



Reflected rays diverge; produced backwards (dashed), they meet behind the mirror at the image.

Characteristics of the image (plane mirror) ★★★

1. **Virtual and erect**; it cannot be caught on a screen.
2. **Same size** as the object (magnification m = +1).
3. As far **behind** the mirror as the object is **in front** of it.
4. **Laterally inverted**: left looks right and right looks left.



MEME BREAK

*Object aur image ek doosre ko point kar rahe hai: same size, same distance. Bas left-right ulta! Yahi hai **lateral inversion**.*

Lateral inversion in daily life

- Raise your right hand in front of a mirror: the image raises its **left** hand.
- Ambulances carry **AMBULANCE** written in mirror letters, so drivers ahead read it correctly in their rear-view mirror.

Solved Example 2: Plane mirror distances ★★☆

Q. Riya stands 2 m in front of a plane mirror. (a) How far is her image from her? (b) She walks 0.5 m towards the mirror. Now how far is the image from her?

- (a) Image is 2 m behind the mirror, so distance = 2 + 2 = **4 m**
- (b) New object distance = 1.5 m, so image distance from her = 1.5 + 1.5 = **3 m**

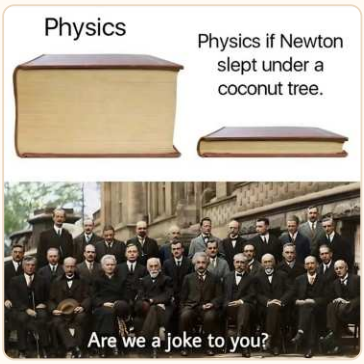
5 SPHERICAL MIRRORS ★★★ MOST ASKED

DEFINITION

- **Spherical mirror:** a mirror whose reflecting surface is a **part of a hollow sphere** of glass.

Concave vs convex

	Concave mirror	Convex mirror
Reflecting surface	curved **inwards**, facing the centre of the sphere	curved **outwards**
Polished (painted) side	outer surface	inner surface
Action on parallel rays	**converging** mirror	**diverging** mirror
Spoon test	inner side of a shiny spoon	back (outer) side of the spoon

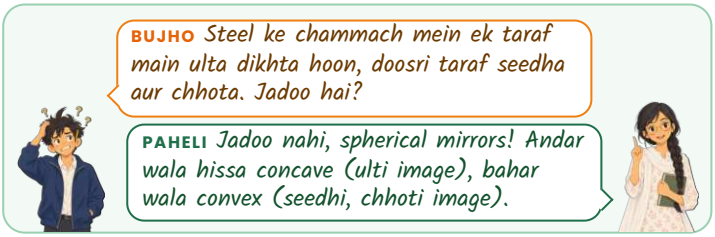


MEME BREAK

Physics ki kitaab ki motai dekh ke darr lag raha hai? Tension nahi, roz ek chhota topic: aaj sirf mirrors!

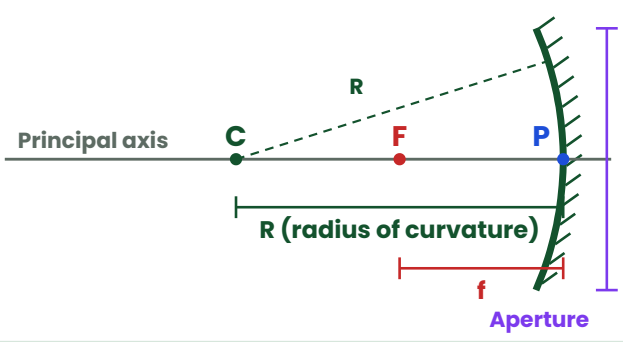
REMEMBER: CAVE GOES IN

Con**cave** is hollowed in like a **cave**; con**vex** bulges out. Concave **converges** light, convex **diverges** it.



6 MIRROR TERMS ★★★ MOST ASKED

Concave mirror with all its terms



Know each term

- **Pole (P):** the centre of the reflecting surface of the mirror
- **Centre of curvature (C):** the centre of the sphere of which the mirror is a part; it lies **in front** of a concave mirror and **behind** a convex mirror
- **Radius of curvature (R):** the radius of that sphere, i.e. the distance PC
- **Principal axis:** the straight line through P and C; it is normal to the mirror at P
- **Aperture:** the diameter of the reflecting surface

Principal focus and focal length ★★★

- **Concave:** rays parallel to the principal axis, after reflection, **actually meet** at a point F on the axis: the **principal focus** (real, in front)
- **Convex:** such rays, after reflection, **appear to diverge** from a point F behind the mirror (virtual focus)
- **Focal length (f):** the distance PF between the pole and the principal focus

R = 2f: the link ★★★

For spherical mirrors of **small aperture**, the principal focus lies midway between P and C:

$$R = 2f \text{ or } f = R / 2$$

Solved Example 3: Using R = 2f ★★☆

Q. The radius of curvature of a spherical mirror is 20 cm. What is its focal length? NCERT

- f = R / 2 = 20 cm / 2 = **10 cm**
- For a concave mirror F is 10 cm in front; for a convex mirror 10 cm behind

Q. A concave mirror has f = 15 cm. Where is C? **Ans.** R = 2 × 15 = **30 cm** in front of the pole.

ACTIVITY: FIND THE FOCUS WITH SUNLIGHT

- Hold a concave mirror facing the Sun and move a sheet of paper in front of it until a **sharp, bright spot** appears.
- Hold it there for a minute: the paper starts to smoke and may burn. The spot is the **image of the Sun**, formed at the **principal focus**.
- Distance from mirror to paper ≈ focal length. Caution: never look at the Sun or its reflection directly.



★★★ *Newton ka fruit salad ho ya mirror ka R = 2f, basics strong toh numerical easy! Nitesh Bhaiya ki guarantee: R = 2f wala 1 mark pakka.*



BUJHO *Plane mirror bhi toh mirror hai, uska focal length kitna hoga?*

PAHELI *Plane mirror ko infinite radius wala sphere samjho: R = ∞, toh f = R/2 = ∞. Isliye uski image na badi hoti hai, na chhoti!*



COMMON MISTAKES

- Measuring ∠i from the **mirror surface**: always measure it from the **normal**.
- Writing that a plane mirror image is "real": it is **virtual**, so no screen can catch it.
- Mixing up sides: a concave mirror reflects from its **inner, curved-in** side.

Exam tips

- ✔ Draw the **normal as a dashed line** in every reflection figure and label ∠i and ∠r.
- ✔ For "characteristics of a plane mirror image", write all four points.
- ✔ In the definition of focus, always say **parallel to the principal axis**.

QUICK RECALL: ONE-MARK QUESTIONS

1. Speed of light in vacuum? **3×10^8 m/s**
2. ∠i for a ray along the normal? **0°**
3. Magnification of a plane mirror? **+1**
4. Which mirror is diverging? **Convex**
5. f of a mirror with R = 24 cm? **12 cm**

7 PRACTICE MCQS

Buff doge banna hai toh 10 MCQs khud solve karo, answer key baad mein!



1. The angle of incidence for a ray falling along the normal on a plane mirror is: ★★☆

(a) 90° (b) 45°
(c) 0° (d) 180°

2. A plane mirror forms an image that is: ★★★

(a) real and inverted (b) virtual, erect, same size
(c) virtual and diminished (d) real and magnified

3. Which of these mirrors is called a converging mirror? ★★☆

(a) plane (b) convex
(c) concave (d) both convex and concave

4. The radius of curvature of a spherical mirror is 32 cm. Its focal length is: ★★★

(a) 64 cm (b) 32 cm
(c) 16 cm (d) 8 cm

5. The image formed by a convex mirror can never be obtained on a screen because it is: ★★☆

(a) real (b) virtual
(c) magnified (d) inverted

6. A ray makes an angle of 25° with the surface of a plane mirror. The angle of reflection is: ★★☆

(a) 25° (b) 65°
(c) 90° (d) 115°

7. The centre of the reflecting surface of a spherical mirror is called its: ★★☆

(a) centre of curvature (b) principal focus
(c) pole (d) aperture

8. AMBULANCE is written in mirror letters on ambulances because a plane mirror image is: ★☆☆

(a) magnified (b) laterally inverted
(c) real (d) diminished

9. **Assertion (A):** The focal length of a spherical mirror of radius of curvature 30 cm is 15 cm.
Reason (R): For a spherical mirror of small aperture, the principal focus lies midway between the pole and the centre of curvature. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The image formed by a plane mirror can be taken on a screen.
Reason (R): A plane mirror forms a virtual image. ★☆☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. List four characteristics of the images formed by plane mirrors. CBSE 2017 **(2 marks)**
2. Define: (i) pole (ii) centre of curvature (iii) principal focus of a concave mirror. Write the relation between R and f. **(3 marks)**
3. Distinguish between a real image and a virtual image (any three points). Which kind does a plane mirror form? **(3 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Along the normal, the angle with the normal is 0°.
2	(b)	Virtual, erect, same size, laterally inverted.
3	(c)	Parallel rays meet at F after reflection.
4	(c)	f = R/2 = 32/2 = 16 cm.
5	(b)	Rays only appear to meet; no screen catches it.
6	(b)	∠i = 90° − 25° = 65°, so ∠r = 65°.
7	(c)	The pole is the centre of the reflecting surface.
8	(b)	Left and right swap in a mirror.
9	(a)	f = R/2 = 15 cm; R explains A.
10	(d)	A virtual image cannot be taken on a screen.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Dual nature of light (from class notes): light behaves as a **wave** and also as a stream of tiny energy packets called **photons** (particle nature). Class 10 uses only the ray model.

Model answers: short and long answer

1 Plane mirror image
- virtual and erect; same size as the object
- as far behind the mirror as the object is in front
- laterally inverted

2 Definitions
- **Pole:** centre of the reflecting surface
- **C:** centre of the sphere of which the mirror is a part
- **F (concave):** point on the principal axis where rays parallel to it meet after reflection; **R = 2f**

3 Real vs virtual
Real: rays actually meet, can be taken on a screen, inverted. Virtual: rays only appear to meet, cannot be taken on a screen, erect. A plane mirror forms a **virtual** image.

Day 21 checklist: tick before you sleep

- ☐ I can define light, reflection, ∠i and ∠r
- ☐ I can state both laws of reflection
- ☐ I can list four features of a plane mirror image
- ☐ I can tell a concave mirror from a convex one
- ☐ I can define P, C, R, F, f and use R = 2f

My score: ____ / 5
5/5 = Jalwa hai! 4 se kam? Page 1–3 phir se padho.

LEVEL 21 COMPLETE +125 XP earned. Next: Level 22



BUJHO *Samajh gaya! Angle normal se, plane mirror ki image virtual, aur R = 2f. Mirror ka chakkar clear!*

PAHELI *Shabaash! Kal Level 22: concave mirror ke chhe cases.*

Day 21 hit-list: Eduspark PYQ analysis

Topic	Rating
Laws of reflection	★★★
Characteristics of plane mirror image	★★★
R = 2f and focal length	★★★
Real vs virtual image	★★☆
Pole, C and principal focus definitions	★★☆
Concave vs convex mirror	★★☆
Lateral inversion (ambulance)	★☆☆



NITESH SIR SAYS
Terms ka diagram bana ke yaad karo, ratke nahi. ***Kal milte hai Day 22 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 22 · LEVEL 22

PHYSICS | CHAPTER 5 | PART 2 · ASTRA SERIES

RAY DIAGRAMS & MIRROR IMAGES

Goal: draw any mirror image with just two rays.



by Nitesh Prakash

01 Ray Rules | 02 Concave Cases | 03 Ray Diagrams | 04 Convex Cases | 05 Uses | +125 XP today | Level 22 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

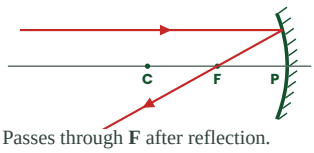
1 RULES FOR DRAWING RAY DIAGRAMS

★★★ MOST ASKED

To locate an image, take **any two** of these rays from the top of the object. Where the reflected rays **meet** (real) or **appear to meet** (virtual) is the image of that point.

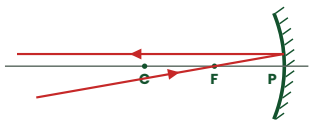
The four rays (concave mirror shown)

Rule 1: Parallel ray



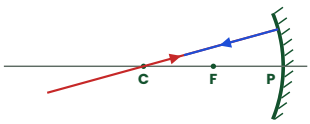
Passes through **F** after reflection.
Convex: appears to come from F behind.

Rule 2: Ray through F



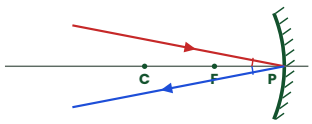
Comes out **parallel** to the axis.
Convex: ray aimed at F comes out parallel.

Rule 3: Ray through C



Hits along the normal and **retraces** its path.
Convex: ray aimed at C retraces its path.

Rule 4: Ray at the pole



Reflects **symmetrically**: ∠i = ∠r about the axis.
Convex: same rule.



BUJHO *Ray diagram mein teen-teen rays banani padegi kya?*

PAHELI *Nahi, koi bhi do rays kaafi hai. Jahan dono reflected rays milti hai, wahi image ka top hai.*



DRAWING CHECKLIST

- Arrow on every ray in the direction light travels
- Virtual rays and virtual images **dashed**
- Label P, F, C and the principal axis; object AB, image A′B′



★★★ *Ray diagram ka 2 ya 3 mark ka sawal har saal aata hai: Nitesh Bhaiya ki guarantee! Object F par toh image infinity par.*

2 IMAGES BY A CONCAVE MIRROR

★★★ MOST ASKED

All six positions at a glance ★★★

Object at	Image at	Size	Nature
Infinity	F	highly diminished, point-sized	real, inverted
Beyond C	between F and C	diminished	real, inverted
C	C	same size	real, inverted
Between C and F	beyond C	enlarged	real, inverted
F	infinity	highly enlarged	real, inverted
Between P and F	behind the mirror	enlarged	**virtual, erect**

PATTERN TO REMEMBER

Bring the object from infinity towards F: the real image moves **from F out to infinity** and grows **bigger**. Cross F (between P and F) and the image jumps **behind** the mirror: virtual, erect, enlarged.

Concave, case 1. Object at infinity

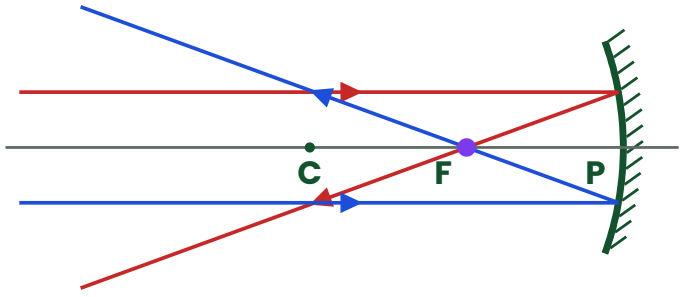


Image at F: point-sized, real, inverted



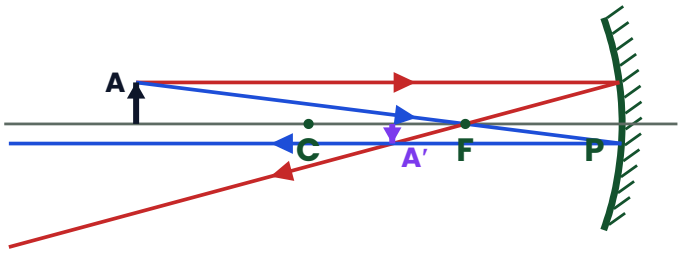
Steel ke chammach mein ulti shakal

Concave mirror: real, inverted image beyond C

MEME BREAK

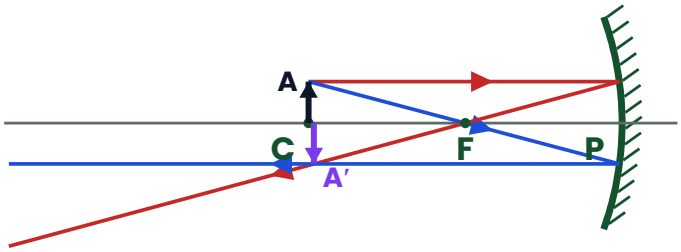
Chammach mein ulta chehra? Wo concave mirror ka case 2 hai: object beyond C, image real, inverted aur chhoti!

Concave, case 2. Object beyond C ★★★



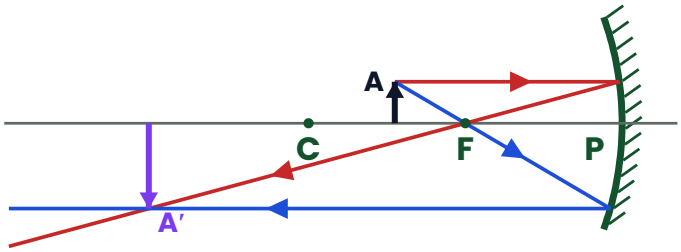
Between F and C: diminished, real, inverted

Concave, case 3. Object at C



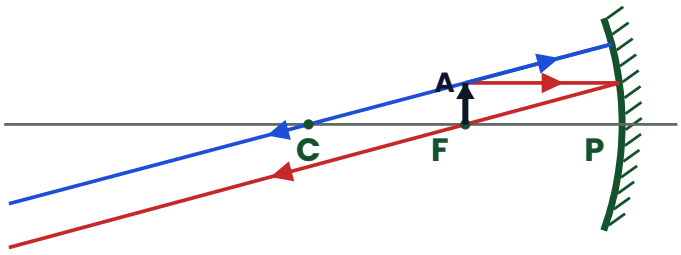
At C: same size, real, inverted

Concave, case 4. Between C and F ★★★



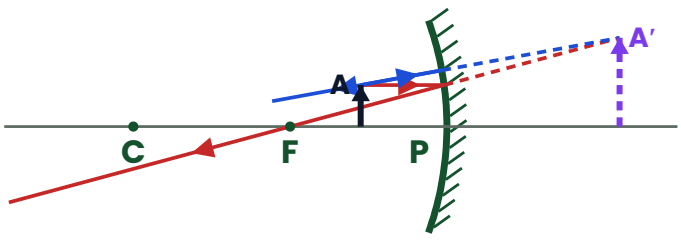
Beyond C: enlarged, real, inverted

Concave, case 5. Object at F

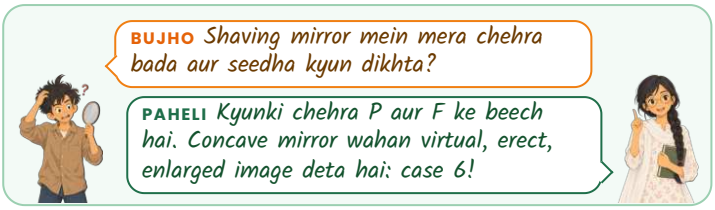


At infinity: highly enlarged, real, inverted

Concave, case 6. Between P and F ★★★



Behind mirror: enlarged, virtual, erect. This is how a shaving mirror works.



3 IMAGES BY A CONVEX MIRROR ★★☆

Only two cases, always virtual

Object at	Image at	Size	Nature
Infinity	F, behind the mirror	highly diminished, point-sized	virtual, erect
Between infinity and P	between P and F, behind	diminished	virtual, erect

Convex, case 1. Object at infinity

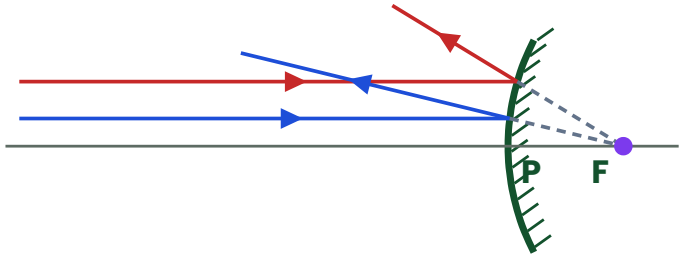


Image at F behind the mirror: point-sized, virtual, erect

Convex, case 2. Object at a finite distance ★★★

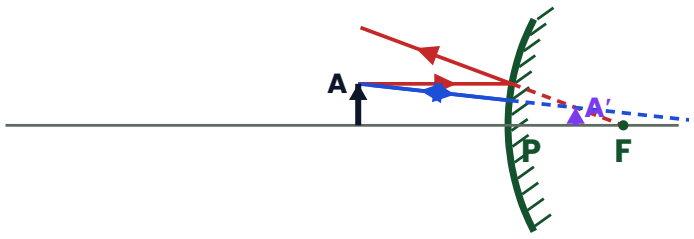


Image between P and F behind the mirror: diminished, virtual, erect



MEME BREAK

Bike mirror mein truck chhota dikha, panic! Phir yaad aaya convex mirror hamesha chhoti image deta hai. Par truck sach mein paas hai, phir panic!

4 USES OF SPHERICAL MIRRORS ★★★ MOST ASKED

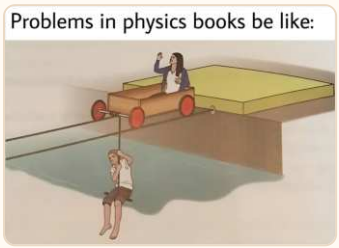
Concave mirror uses ★★★

- **Torches, vehicle headlights, searchlights:** bulb at F gives a **powerful parallel beam**
- **Shaving mirrors:** face between P and F gives an **enlarged, erect** image
- **Dentists:** see a **large image** of the teeth
- **Solar furnaces:** sunlight is **concentrated** at F to produce heat

Convex mirror uses ★★★

- **Rear-view (wing) mirrors** in vehicles
- **Big convex mirrors** in shops and at sharp road bends, for security and safety

Why convex for rear view? (1) It always gives an **erect**, diminished image. (2) It is curved outwards, so it gives a **much wider field of view**: the driver sees a large area behind.



MEME BREAK

Physics ke problems mein ladka ice pe phisalta hai, hum ray diagram mein. Practice se dono sambhal jaate hai!

Solved Example 1: Which mirror? ★★★

Q. Name the mirror that gives: (a) an erect, enlarged image; (b) an erect image of the same size; (c) an erect, diminished image of any object. **NCERT**

- (a) **Concave** mirror, object between P and F
- (b) **Plane** mirror
- (c) **Convex** mirror, for every position

ACTIVITY: FIND THE IMAGE POSITIONS

- Fix a concave mirror on a stand and place a lit candle far away; move a screen until a sharp image appears (near F).
- Bring the candle closer step by step: beyond C, at C, between C and F. Each time, find the image on the screen and note its size.
- With the candle between P and F, no screen catches the image: look into the mirror to see an enlarged, erect, virtual image.

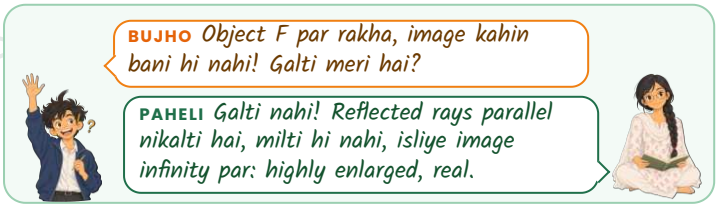
Solved Example 2: Range for a magnified image ★★★

Q. Rohit wants an erect, magnified image of an object using a converging mirror of focal length 40 cm. (a) In what range should the object be placed? (b) State one use of this kind of image formation. **Mod. CBSE SQP 2022**

- A converging mirror is concave; an erect, magnified (virtual) image forms only when the object is between P and F
- (a) Place the object **between 0 and 40 cm** from the mirror
- (b) Shaving or make-up mirror, dentist's mirror

COMMON MISTAKES

- Drawing the virtual image with a **solid** line: use dashes.
- Writing "image at F" for an object at F: the image is at **infinity**.
- Saying a convex mirror can form a real image: it **never** can.



Exam tips

- ✔ Ray diagram questions usually ask for **two rays**: choose the parallel ray and the ray through F or C.
- ✔ Write position, size and nature as three separate words: "beyond C, enlarged, real and inverted".
- ✔ For "why convex mirror in vehicles", write both reasons: erect image and wider field of view.

5 PRACTICE MCQS

Aisa haal na ho! 10 MCQs khud solve karo, phir answer key se milao.



1. For which ray is the angle of incidence on a concave mirror zero? **CBSE SQP 2024** ★★★

(a) a ray parallel to the principal axis
(b) a ray passing through the centre of curvature
(c) a ray passing through the principal focus
(d) a ray incident obliquely at the pole

2. An object is placed at C of a concave mirror. The image is: ★★★

(a) at F, diminished
(b) at C, same size, inverted
(c) beyond C, enlarged
(d) behind the mirror, erect

3. The magnification produced by a concave mirror when the object is between the pole and the focus is: **CBSE SQP 2024** ★★☆

(a) m = +1
(b) m = −1
(c) m > +1
(d) m < −1

4. A concave mirror forms a real image of the same size as the object when the object is at: ★★☆

(a) F
(b) C
(c) infinity
(d) between P and F

5. The mirror used in vehicle headlights is: ★★☆

(a) plane
(b) convex
(c) concave
(d) any spherical mirror

6. The image formed by a convex mirror is always: ★★★

(a) real and inverted
(b) virtual, erect, diminished
(c) virtual, erect, enlarged
(d) real and diminished

7. To get a highly enlarged real image from a concave mirror, the object is placed: ★★☆

(a) at C
(b) just beyond C
(c) between C and F, close to F
(d) between P and F

8. Sunlight is focused by a concave mirror. The bright spot forms at the: ★★☆

(a) pole
(b) principal focus
(c) centre of curvature
(d) infinity

9. **Assertion (A):** A convex mirror is used as a rear-view mirror in vehicles.
Reason (R): A convex mirror gives an erect, diminished image and a wider field of view. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** A concave mirror always forms a real image.
Reason (R): A concave mirror is a converging mirror. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Case-based question: the reflecting telescope

Large telescopes collect starlight with a big **concave mirror**. A small **plane mirror** turns the reflected beam sideways to an eyepiece. CBSE SQP 2024

1. Where does the concave mirror form the image of a distant star, and what is its nature?
2. Why is a concave and not a convex mirror used to collect the light?
3. What does the plane mirror do to the beam?

Short and long answer questions

1. Draw a labelled ray diagram to show the image formed by a concave mirror when the object is placed between its pole and principal focus. CBSE Compartment 2024 **(2 marks)**
2. Draw ray diagrams for a concave mirror with the object (i) beyond C (ii) at F. State the position and nature of the image in each case. **(3 marks)**
3. Why do we prefer a convex mirror as a rear-view mirror? State two uses of concave mirrors. NCERT **(3 marks)**

TRY YOURSELF

Draw the ray diagram for a concave mirror with the object at C and state where the image forms.

Answer key: MCQs

Q	Ans	Why
1	(b)	It hits along the normal and retraces its path.
2	(b)	Object at C gives image at C, same size.
3	(c)	Virtual, erect, enlarged image: m > +1.
4	(b)	Same size, real, inverted at C.
5	(c)	Bulb at F gives a parallel beam.
6	(b)	Every position: virtual, erect, diminished.
7	(c)	Close to F, the image goes far beyond C, very big.
8	(b)	Parallel rays from the Sun meet at F.
9	(a)	R explains why it is preferred.
10	(d)	Object between P and F gives a virtual image.

Case-based answers

- (i) At the principal focus F; real, inverted and point-sized (object at infinity)
- (ii) A concave mirror converges light and forms a real image; a convex mirror only spreads light out
- (iii) It reflects the beam sideways towards the eyepiece without changing its size; the image is laterally inverted

Model answers

1 Object between P and F

Ray parallel to the axis reflects through F; ray along the line from C hits normally and retraces. The reflected rays diverge; produced backwards (dashed) they meet **behind the mirror**: image virtual, erect, enlarged (case 6 figure).

2 Two cases

(i) Beyond C: image between F and C, real, inverted, diminished. (ii) At F: reflected rays are parallel, image at infinity, real, inverted, highly enlarged (cases 2 and 5).

3 Rear view and uses

Convex mirror: always erect, diminished image and a wider field of view. Concave uses: shaving mirror, headlights or torches (any two of those listed).

Day 22 checklist: tick before you sleep

- ☐ I can state the four rules for ray diagrams
- ☐ I can fill the six-case concave mirror table from memory
- ☐ I can draw any concave mirror ray diagram with two rays
- ☐ I know the two convex mirror cases
- ☐ I can give uses of concave and convex mirrors with reasons

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Table phir se likho.

LEVEL 22 COMPLETE +125 XP earned. Next: Level 23



BUJHO *Samajh gaya! Do rays, ek milan point, wahi image. Six cases bhi yaad!*

PAHELI *Shabaash! Kal Level 23: sign convention aur mirror formula.*

Day 22 hit-list: Eduspark PYQ analysis

Topic	Rating
Concave mirror image table (six cases)	★★★
Ray diagram: object between P and F	★★★
Convex mirror as rear-view mirror	★★★
Uses of concave mirrors	★★★
Rules for drawing ray diagrams	★★☆
Ray diagram: object beyond C or at C	★★☆
Convex mirror image cases	★★☆



NITESH SIR SAYS

*Diagram mein arrow aur dashed line hi marks dilate hai. Roz ek diagram banao. **Kal milte hai Day 23 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 23 · LEVEL 23

PHYSICS | CHAPTER 5 | PART 3 · ASTRA SERIES

MIRROR FORMULA & NUMERICALS

Goal: solve any mirror numerical with the right signs.



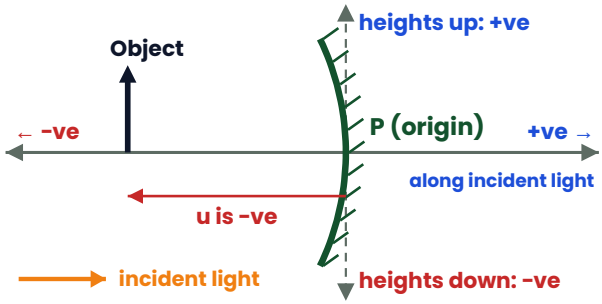
by Nitesh Prakash

01 Sign Convention | 02 Mirror Formula | 03 Magnification | 04 Solved Numericals | 05 Practice | +125 XP today | Level 23 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 NEW CARTESIAN SIGN CONVENTION ★★★ MOST ASKED

Pole = origin, principal axis = x-axis



DEFINITION

- The object is always placed to the **left** of the mirror, so light falls from left to right.
- All distances are measured from the **pole (P)**.
- Distances in the direction of **incident light** (to the right) are **+ve**; distances against it (to the left) are **−ve**.
- Heights **above** the principal axis are **+ve**; heights **below** it are **−ve**.

CLASS TRICK: GAANTH BAANDH LO

"Numericals solve karne ke liye isse gaanth baandh lo"

Mirror	u	v	f	h_o	h_i
Concave	−ve	depends	−ve	+ve	depends
Convex	−ve	+ve	+ve	+ve	+ve

"Depends": real image (in front) → v and h_i −ve; virtual image (behind) → v and h_i +ve. A convex mirror always forms a virtual, erect image, so its v and h_i are always +ve.



MEME BREAK

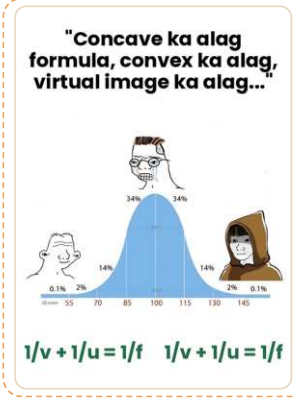
u ka sign? Seagull ki tarah yaad karo: real object hamesha mirror ke left, isliye u hamesha −ve.

2 MIRROR FORMULA ★★★ MOST ASKED

The relation between u, v and f ★★★

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

- **u** = object distance, **v** = image distance, **f** = focal length; all measured from P
- Valid for **all** spherical mirrors and all object positions, if signs are put correctly
- Also remember f = R/2. **Derivation is not required** for the board exam



★★★ *Mirror formula ka 3 mark ka numerical har paper mein: Nitesh Bhaiya ki guarantee! Formula ek, bas sign sahi daalo.*

Five steps for every numerical

1. Draw a quick sketch: mirror, P, object on the left.
2. Write the given values **with signs** (gaanth wali table).
3. Write the formula: 1/v + 1/u = 1/f.
4. Substitute with brackets, e.g. 1/(−27); solve for the unknown.
5. Read the sign of the answer: v −ve = real, in front; v +ve = virtual, behind.

ACTIVITY: FOCAL LENGTH OF A CONCAVE MIRROR

- Focus the image of a far-away window or tree on a screen using a concave mirror.
- Measure the distance from the mirror to the screen: that is the **focal length** (object at infinity, image at F).
- Example: mirror stand at the 10 cm mark, sharp image on the screen at the 40 cm mark: f = 40 − 10 = **30 cm**, so R = 2f = **60 cm**.

3 MAGNIFICATION ★★★ MOST ASKED

DEFINITION

- **Magnification (m):** the ratio of the height of the image to the height of the object.
- $m = h_i / h_o = -v / u$; it has **no unit**.

What the value of m tells you ★★★

m	Meaning
negative	image real and inverted
positive	image virtual and erect
$\|m\| > 1$	image enlarged
$\|m\| < 1$	image diminished
$m = +1$	same size, erect: plane mirror
$m = -1$	same size, real, inverted: object at C of a concave mirror



BUJHO *Question mein "real image, 3 times bada" likha hai. m = +3 lun ya −3?*

PAHELI *Real image hamesha ulti hoti hai, isliye m = −3. Virtual image hoti toh m = +3 leti.*



Signs at a glance ★★☆

Situation	v	h_i	m
Concave, real image (object beyond F)	−	−	−
Concave, object at C (image at C)	−	−	−1
Concave, virtual image (object between P and F)	+	+	+
Convex, any position	+	+	+ (< 1)
Plane mirror	+	+	+1

FORMULA CARD: REVISE IN 30 SECONDS

$f = R / 2$ **$1/v + 1/u = 1/f$**

$m = h_i / h_o = -v / u$ **$h_i = m \times h_o$**

Plane mirror: m = +1, image as far behind as object in front

4 SOLVED NUMERICALS ★★★ MOST ASKED

Solved Numerical 1: Screen position (from class notes) ★★★

Q. An object 7.0 cm in size is placed at 27 cm in front of a concave mirror of focal length 18 cm. Where should a screen be placed to get a sharp image? Find the size and nature of the image. NCERT

Given (with signs): u = −27 cm, f = −18 cm, h_o = +7.0 cm

Formula: 1/v + 1/u = 1/f

- 1/v = 1/f − 1/u = 1/(−18) − 1/(−27) = −1/18 + 1/27
- = (−3 + 2)/54 = −1/54, so **v = −54 cm**
- m = −v/u = −(−54)/(−27) = **−2**; $h_i = m \times h_o = -2 \times 7.0 =$ **−14 cm**

Answer: Screen at **54 cm in front** of the mirror; image 14 cm tall, **real, inverted, enlarged**.

Solved Numerical 2: Convex mirror (from class notes) ★★★

Q. An object is placed at 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image. NCERT

Given (with signs): u = −10 cm, f = +15 cm

Formula: 1/v + 1/u = 1/f

- 1/v = 1/f − 1/u = 1/15 − 1/(−10) = 1/15 + 1/10
- = (2 + 3)/30 = 5/30 = 1/6, so **v = +6 cm**
- m = −v/u = −(6)/(−10) = **+0.6**

Answer: Image **6 cm behind** the mirror; **virtual, erect, diminished**.

Solved Numerical 3: Concave mirror, image size ★★★

Q. An object is placed 25 cm in front of a concave mirror of focal length 15 cm. Find the position of the image. If the object is 4.0 cm tall, also find the nature and size of the image. CBSE Compartment 2025

Given (with signs): u = −25 cm, f = −15 cm, h_o = +4.0 cm

Formula: 1/v + 1/u = 1/f

- 1/v = 1/(−15) − 1/(−25) = −1/15 + 1/25 = (−5 + 3)/75 = −2/75
- **v = −37.5 cm**
- m = −v/u = −(−37.5)/(−25) = **−1.5**; h_i = −1.5 × 4.0 = **−6.0 cm**

Answer: Image **37.5 cm in front**; real, inverted, enlarged, 6.0 cm tall.

Solved Numerical 4: Rear-view mirror ★★★

Q. A convex rear-view mirror has a radius of curvature of 3.00 m. A bus is 5.00 m from it. Find the position, nature and size of the image. NCERT

Given (with signs): R = +3.00 m, so f = R/2 = +1.50 m; u = −5.00 m

Formula: 1/v + 1/u = 1/f

- 1/v = 1/1.50 + 1/5.00 = 0.667 + 0.200 = 0.867 m^{-1}
- **v = +1.15 m**
- m = −v/u = −(1.15)/(−5.00) = **+0.23**

Answer: Image **1.15 m behind** the mirror; virtual, erect, 0.23 times the size of the bus.

Solved Numerical 5: Find f from magnification ★★★

Q. A concave mirror forms a real image three times the size of an object placed 10 cm in front of it. Where is the image, and what is the focal length? NCERT

Given (with signs): u = −10 cm, real image so m = −3

Formula: m = −v/u, then 1/v + 1/u = 1/f

- −3 = −v/(−10), so **v = −30 cm**
- 1/f = 1/v + 1/u = −1/30 − 1/10 = −4/30
- **f = −7.5 cm**

Answer: Image **30 cm in front** of the mirror; focal length 7.5 cm (concave).



MEME BREAK

u = +30 likha? Batman ka thappad pakka! Real object ka u ***hamesha negative.***

Solved Numerical 6: Erect, magnified image ★★☆

Q. An object is placed 6 cm in front of a concave mirror of focal length 12 cm. Find the position, nature and magnification of the image.

Given (with signs): u = −6 cm, f = −12 cm
Formula: 1/v + 1/u = 1/f

- 1/v = 1/(−12) − 1/(−6) = −1/12 + 2/12 = 1/12
- **v = +12 cm** (positive: behind the mirror)
- m = −v/u = −(12)/(−6) = **+2**

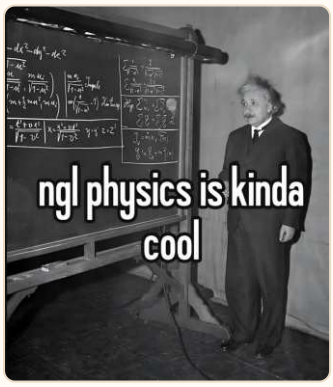
Answer: Image **12 cm behind** the mirror; virtual, erect, twice the size (object between P and F).

COMMON MISTAKES

- Writing f of a concave mirror as +ve: it is **−ve**.
- Forgetting brackets: 1/(−27), not −1/27 mixed with +1/27.
- Stopping at v: always state **nature** and **size** as well.

Exam tips

- ✔ Write units with every answer (cm or m) and the sign meaning in words.
- ✔ Use fractions with an LCM; avoid long decimals until the last step.
- ✔ If v comes out +ve for a concave mirror, the object is between P and F: the image is virtual.



MEME BREAK
Sign convention samajh aaya toh physics sach mein cool lagti hai! Ab numericals khud try karo.

TRY YOURSELF: NUMERICALS

1. Concave mirror, f = 20 cm, object at 30 cm. Find v and m.
2. Convex mirror, f = 20 cm, object at 20 cm. Find v and m.
3. Concave mirror, f = 10 cm, object at 5 cm. Find v and m; state the nature.

QUICK RECALL: ONE-MARK QUESTIONS

1. Sign of u for a real object? **Always −ve**
2. Sign of f for a convex mirror? **+ve**
3. Unit of magnification? **None**
4. m for an image at C of a concave mirror? **−1**
5. v is +ve for a concave mirror when? **Object between P and F**

Case-based question: the dentist's mirror

A dentist holds a small concave mirror of focal length 4 cm at a distance of 3 cm from a tooth.

1. Find the position of the image of the tooth.
2. Find the magnification. Describe the image.
3. Why does the dentist not use a convex mirror?

5 PRACTICE MCQS

Sign galat, answer galat, dil toota! 10 MCQs dhyan se karo.



1. In the New Cartesian sign convention, the focal length of a concave mirror is taken as: ★★★
(a) positive (b) negative
(c) zero (d) positive or negative

2. The magnification produced by a plane mirror is: ★★☆
(a) +1 (b) −1
(c) 0 (d) infinite

3. A negative value of magnification means the image is: ★★★
(a) virtual and erect (b) real and inverted
(c) always diminished (d) always enlarged

4. An object is 20 cm from a concave mirror of focal length 10 cm. The image distance is: ★★☆
(a) −10 cm (b) −20 cm
(c) +20 cm (d) −40 cm

5. The radius of curvature of a convex mirror is 32 cm. Its focal length is: ★★☆
(a) −16 cm (b) +16 cm
(c) +32 cm (d) −32 cm

6. For a convex mirror, the image distance v is always: ★★☆
(a) negative (b) positive
(c) zero (d) equal to u

7. A mirror gives m = +0.5 for every position of the object. The mirror is: ★★☆
(a) plane (b) concave
(c) convex (d) none of these

8. For a concave mirror, u = −15 cm and v = −30 cm. The magnification is: ★★☆
(a) +2 (b) −2
(c) +0.5 (d) −0.5

9. Assertion (A): The focal length of a convex mirror is taken as positive.
Reason (R): The principal focus of a convex mirror lies behind the mirror, in the direction of incident light. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** A concave mirror can produce a magnification of +2.
Reason (R): A concave mirror forms an enlarged, virtual, erect image when the object is between P and F. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1 An object 5.0 cm long is placed 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position, nature and size of the image. NCERT **(3 marks)**

2 State the New Cartesian sign convention for spherical mirrors. Write the mirror formula and the formula for magnification. **(3 marks)**

3 The magnification of a spherical mirror is −1. Name the mirror and state where the object is placed. **(2 marks)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Focus is in front, against incident light.
2	(a)	Same size and erect.
3	(b)	Minus sign: real, inverted.
4	(b)	1/v = −1/10 + 1/20 = −1/20; at C.
5	(b)	f = R/2 = +16 cm, behind.
6	(b)	Image always behind the mirror.
7	(c)	Always erect, diminished: convex.
8	(b)	m = −v/u = −(−30)/(−15) = −2.
9	(a)	F lies behind, along incident light.
10	(a)	m = +2 is virtual, erect, enlarged.

Model answers

1 **Convex mirror, R = 30 cm**
f = +15 cm, u = −20 cm. 1/v = 1/15 + 1/20 = 7/60, v = **+8.6 cm** (behind). m = −v/u = 8.57/20 = +0.43; $h_i = 0.43 \times 5.0 =$ **+2.1 cm**. Virtual, erect, diminished.

2 **Sign convention**
State the four rules on page 1; mirror formula 1/v + 1/u = 1/f; m = h_i/h_o = −v/u.

3 **m = -1**
Concave mirror; real, inverted image of the same size, so the object is at **C** (u = −2f).

4 **Try-yourself answers**
(1) v = −60 cm, m = −2. (2) v = +10 cm, m = +0.5. (3) v = +10 cm, m = +2: virtual, erect, enlarged.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Where the mirror formula comes from (derivation deleted from the syllabus): draw the ray diagram for an object beyond C. Two pairs of **similar triangles** give A′B′/AB = B′P/BP and A′B′/AB = B′F/FP. Equating them and putting in the signed distances (BP = −u, B′P = −v, FP = −f) leads to **1/v + 1/u = 1/f**.

Case-based answers

- (i) u = −3 cm, f = −4 cm: 1/v = −1/4 + 1/3 = 1/12, so **v = +12 cm** (behind the mirror)
- (ii) m = −v/u = −12/(−3) = **+4**: virtual, erect, 4 times enlarged
- (iii) A convex mirror always gives a diminished image; the dentist needs an enlarged one

Day 23 checklist: tick before you sleep

- ☐ I can state the four sign convention rules
- ☐ I know the gaanth wali sign table by heart
- ☐ I can write 1/v + 1/u = 1/f and m = −v/u
- ☐ I can solve a mirror numerical with signs and units
- ☐ I can read nature and size from the signs of v and m

My score: ____ / 5
5/5 = Jalwa hai! 4 se kam? Solved numericals phir se karo.

Coming up: Day 24

Refraction of light: why a pencil looks bent in water, the laws of refraction, refractive index and the glass slab.

LEVEL 23 COMPLETE +125 XP earned. Next: Level 24



BUJHO *Samajh gaya! Pehle sign, phir formula, phir answer ka matlab. Numericals ab aasan!*

PAHELI *Shabaash! Kal Level 24: refraction of light.*

Day 23 hit-list: Eduspark PYQ analysis

Topic	Rating
Mirror formula numericals	★★★
New Cartesian sign convention	★★★
Magnification m = −v/u and its sign	★★★
Image size from magnification	★★★
Finding f from magnification	★★☆
Convex mirror numericals (rear view)	★★☆
Signs of v, h_i and m by image type	★★☆



NITESH SIR SAYS
Har numerical mein pehle signs likho, phir formula. Sign sahi toh answer sahi. ***Kal milte hai Day 24 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 24
LEVEL 24

PHYSICS | CHAPTER 5 | PART 4 ASTRA SERIES

REFRACTION & REFRACTIVE INDEX

Goal: know why light bends, Snell's law and refractive index.



by Nitesh Prakash

01 Refraction | 02 Snell's Law | 03 Refractive Index | 04 Glass Slab | 05 Numericals | +125 XP today | Level 24 / 60

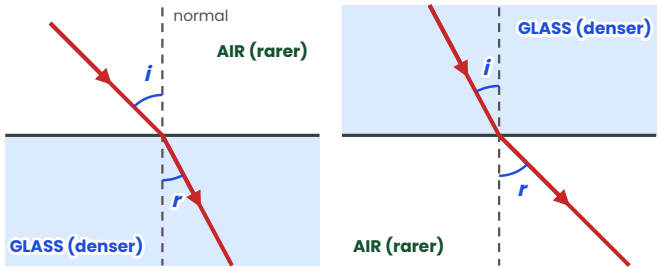
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 REFRACTION OF LIGHT

DEFINITION

- **Refraction:** the change in the **direction** of light when it passes **obliquely** from one transparent medium into another
- It happens because light has a **different speed** in different media: about 3×10^8 **m/s** in vacuum, slower in water and glass
- **Optically denser** medium: light is slower there (glass is denser than air). **Optically rarer**: light is faster

Figure 1: Which way does the ray bend? ★★★



(a) Bends TOWARDS normal (b) Bends AWAY from normal

Angles are always measured from the **normal** (the dashed line at 90° to the surface), never from the surface.

Rule: rarer and denser

- ✔ **Rarer → denser** (air → glass): speed falls, ray bends **towards** the normal, so $r < i$
- ✔ **Denser → rarer** (glass → air): speed rises, ray bends **away from** the normal, so $r > i$



★★★ *Towards ya away from normal: ye ek line har saal MCQ ya 1-mark mein aati hai, Nitesh Bhaiya ki guarantee!*

NCERT in-text questions: quick answers ★★☆

1 Air into water, obliquely: towards or away?

Towards the normal. Water is optically denser, so light slows down on entering it.

2 Kerosene 1.44, turpentine 1.47, water 1.33: where is light fastest?

In **water**, the medium with the lowest refractive index ($v = c/n$).

When does refraction NOT take place? ★★☆

1. **Normal incidence:** the ray falls along the normal ($i = 0°$). It goes straight through with $r = 0°$. The speed still changes, but the direction does not.
2. **Same optical density:** both media bend light equally (same refractive index), so the ray goes straight. Example: a glass rod dipped in glycerine almost disappears.



MEME BREAK

*Ray normal pe aayi? **i = 0°, r = 0°**. Na bend, na drama: seedha andar, seedha bahar. Newton bhi khush!*



BUJHO *Agar light normal pe aaye to speed bhi same rehti hai na?*

PAHELI *Nahi Bujho! Speed to glass mein phir bhi kam hoti hai. Sirf **direction** nahi badalti, kyunki koi tirchha angle hi nahi hai.*



2 REFRACTION AROUND US

Everyday examples

- A **pencil** partly dipped in water looks **bent** at the water surface
- A swimming pool or tank looks **shallower** than it really is (bottom appears raised)
- A **lemon** in a glass of water looks **bigger** from the side
- Letters under a thick glass slab look **raised**
- All happen because light from the object bends at the surface before it reaches our eye



MEME BREAK

*Pencil paani mein **toota hua** dikha? Scroll of truth: pencil theek hai, light ne surface pe **direction** badli hai!*

3 LAWS OF REFRACTION

The two laws ★★★

1. The incident ray, the refracted ray and the normal at the point of incidence **all lie in the same plane**.
2. **Snell's law:** for a given pair of media and a given colour of light, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a **constant**.

$$n_{21} = \sin i / \sin r = \text{constant}$$

The constant is the **refractive index of medium 2 with respect to medium 1**. It has **no unit** (it is a ratio).

CLASS TRICK: SINE VALUES YOU NEED

Refractive Index: darna nahi hai! Kam sawaal aate hain, aur aate bhi hain to max 2 marks.

Angle	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1

Solved Example 1: Using Snell's law ★★☆

Q. A ray in air falls on glass at $i = 60°$. The refractive index of the glass is √3. Find the angle of refraction.

- **Given:** $i = 60°$, $n = \sqrt{3}$
- **Formula:** $n = \sin i / \sin r \Rightarrow \sin r = \sin i / n$
- **Substitute:** $\sin r = (\sqrt{3}/2) \div \sqrt{3} = 1/2$
- **Answer:** $r = \mathbf{30°}$ (smaller than 60°: bent towards the normal, as expected)

4 REFRACTIVE INDEX

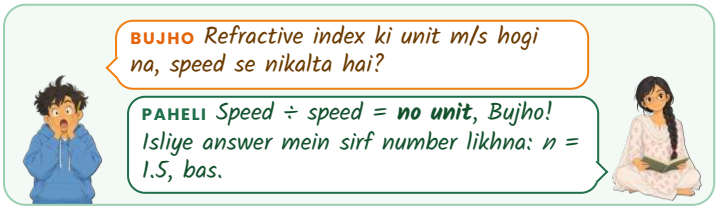
DEFINITION

- **Absolute refractive index** of a medium: speed of light in **vacuum (or air)** divided by the speed of light in that medium
- $\boldsymbol{n_m = c / v}$, where $c = 3 \times 10^8$ m/s. Medium 1 is **fixed**: always vacuum or air
- It is a pure number (no unit) and is always more than 1 for a material medium

Solved Example 5: Relative refractive index ★★☆

Q. $n_{glass} = 1.50$ and $n_{water} = 1.33$. Find (i) n of glass w.r.t. water (ii) n of water w.r.t. glass.

- **Formula:** $n_{21} = n_2 / n_1$
- (i) $n_{gw} = 1.50 / 1.33 \approx \mathbf{1.13}$
- (ii) $n_{wg} = 1.33 / 1.50 \approx \mathbf{0.89}$ (less than 1: water is rarer than glass)



Remember: n depends on the nature of the medium and the colour of light, not on the angle of incidence.

Relative refractive index ★★★

Refractive index of medium 2 **with respect to** medium 1 (here nothing is fixed; both media can be anything):

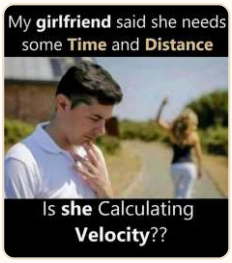
$$n_{21} = v_1 / v_2 = n_2 / n_1$$

Also $n_{12} = v_2 / v_1 = 1 / n_{21}$. Example: n of glass w.r.t. water = 1.50 / 1.33 ≈ 1.13.

CLASS TRICK: N.B. FROM SIR'S NOTES

- Jitna jyada denser medium, utni kam speed of light, isliye $\mathbf{n \propto 1/v}$.
- Jitna jyada denser medium, utna jyada refractive index; aur $\mathbf{n \propto}$ **bending**: n bada, bending zyada.
- Absolute mein fix thing: air ya vacuum (medium 1). Relative mein koi fix nahi, dono medium alag alag.

Note (NCERT): "denser" here means **optical density**, not mass density. Kerosene ($n = 1.44$) is optically denser than water (1.33) but lighter in mass density.



MEME BREAK

Speed = distance / time yaad hai? Yahan bhi wahi khel: $\boldsymbol{n = c / v}$. Medium mein light jitni slow, n utna bada!

Solved Examples 2 to 4: Speed and n ★★★

2 Speed becomes one-third

Light's speed in a medium is $c/3$. Then $n = c \div (c/3) = \mathbf{3}$.

3 Glass, n = 1.50 (NCERT)

$v = c/n = (3 \times 10^8) / 1.50 = \mathbf{2 \times 10^8}$ **m/s**.

4 Carbon disulphide, n = 1.63

Light travels **1.63 times slower** in CS_2 than in vacuum: $v = 3 \times 10^8 / 1.63 \approx \mathbf{1.84 \times 10^8}$ **m/s**.

NCERT table: absolute refractive index ★★☆

Medium	n	Medium	n
Air	1.0003	Canada balsam	1.53
Ice	1.31	Rock salt	1.54
Water	1.33	Carbon disulphide	1.63
Alcohol	1.36	Dense flint glass	1.65
Kerosene	1.44	Ruby	1.71
Fused quartz	1.46	Sapphire	1.77
Turpentine oil	1.47	Diamond	**2.42**
Benzene	1.50	Crown glass	1.52

Highest optical density: **diamond**. Lowest: **air**.

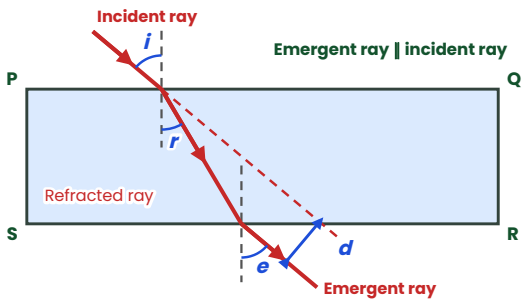
TRY YOURSELF: QUICK NUMERICALS

1. Speed of light in a medium is 2.25×10^8 m/s. Find n.
2. n of a medium is 2. Find the speed of light in it.
3. $n_A = 1.2$, $n_B = 1.5$. Find n_{BA}.

Answers: 1.33; 1.5×10^8 m/s; 1.25.

5 REFRACTION THROUGH A GLASS SLAB

Figure 2: Rectangular glass slab PQRS ★★★



What happens at each surface

1. At **PQ** (air → glass): the ray bends **towards** the normal ($r < i$).
2. At **SR** (glass → air): the ray bends **away from** the normal and comes out as the **emergent ray**.
3. The bending at the two faces is **equal and opposite**, so **angle of emergence *e* = angle of incidence *i***.
4. The emergent ray is **parallel** to the incident ray but **shifted sideways**.

DEFINITION

> **Lateral displacement (d):** the **perpendicular distance** between the incident ray (produced forward) and the emergent ray

Lateral displacement depends on ★★☆

1. **Angle of incidence** (larger *i*, larger shift)
2. **Thickness** of the slab (thicker slab, larger shift)
3. **Refractive index** of the slab (larger *n*, larger shift)



MEME BREAK
Andar ghusi, mudi, bahar nikli, phir mudi: e = i, emergent ray incident ke parallel. Perfectly balanced!

Activity (NCERT): tracing the path

- Fix a glass slab on paper and draw its outline PQRS
- Draw a slanting line on PQ as the incident ray; fix two pins on it
- Look from the other side and fix two more pins in line with the images of the first two
- Join the points: the emergent ray is parallel to the incident ray; measure *i*, *r* and *e*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

$n_{21} \times n_{12} = 1$, because $(v_1/v_2) \times (v_2/v_1) = 1$. A relative index can be less than 1; an absolute index of a material is always more than 1.



BUJHO *Slab se nikal kar ray parallel kyun aati hai? Andar to mud gayi thi!*

PAHELI *Dono faces **parallel** hain, Bujho. PQ pe jitna normal ki taraf mudi, SR pe utna hi door mudi. Isliye **e = i**, bas thoda side shift.*





★★★ *Glass slab ka labelled diagram with i, r, e aur lateral displacement: Nitesh Bhaiya ki guarantee!*

6 COMMON MISTAKES AND EXAM TIPS

× WRONG	✓ RIGHT
Angle measured from the surface	Measure *i*, *r*, *e* from the normal
Writing a unit for *n*	*n* is a ratio: no unit
Denser = heavier liquid	Denser = optically denser (higher *n*)
Lateral displacement = shift along the slab	Perpendicular distance between the two rays
Emergent ray drawn in the same line as the incident ray	Parallel to it, but shifted sideways

Exam tips

- ✔ In diagrams show arrows on every ray and dashed normals at both faces
- ✔ Light bends towards the normal when it **slows down**; away when it **speeds up**
- ✔ Higher *n* means lower speed: $v = c / n$
- ✔ For relative index write the order carefully: $n_{21} = n_2 / n_1$
- ✔ Write *n* as a plain number, with no unit

7 PRACTICE MCQS

1. A ray of light passes from air into glass obliquely. It: ★★★
(a) bends away from the normal (b) bends towards the normal
(c) goes straight (d) is reflected back

2. If $n = 1.5$ for a medium, the speed of light in it is: ★★★
(a) 1.5×10^8 m/s (b) 2×10^8 m/s
(c) 3×10^8 m/s (d) 4.5×10^8 m/s

3. A ray falls normally on a glass slab. Angle of refraction: ★★☆
(a) 0° (b) 45°
(c) 90° (d) equal to the critical angle

4. From the NCERT table, light travels fastest in: NCERT ★★☆
(a) kerosene (b) turpentine oil
(c) water (d) benzene

5. For a rectangular glass slab, the emergent ray is: ★★★
(a) perpendicular to the incident ray (b) parallel to the incident ray
(c) along the normal (d) in line with the refracted ray

6. The SI unit of refractive index is: ★☆☆
(a) m/s (b) metre
(c) dioptre (d) it has no unit

7. Light goes from medium 1 ($v = 2 \times 10^8$ m/s) to medium 2 ($v = 1.5 \times 10^8$ m/s). n_{21} is: ★★☆

(a) 0.75 (b) 1.33
(c) 1.5 (d) 2

8. Lateral displacement does NOT depend on: ★☆☆

(a) angle of incidence (b) thickness of slab
(c) refractive index of slab (d) colour of the slab's frame

9. **Assertion (A):** A ray of light bends towards the normal when it goes from water into air.
Reason (R): Light travels faster in air than in water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The refractive index of diamond (2.42) is the highest in the NCERT table.
Reason (R): Light travels slowest in diamond among these media. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. State the laws of refraction of light. If the speed of light in vacuum is 3×10^8 m/s, find the absolute refractive index of a medium in which light travels at 1.5×10^8 m/s. **(3 marks)**
2. Draw a labelled ray diagram for refraction through a rectangular glass slab. Mark i, r, e and the lateral displacement. Why is the emergent ray parallel to the incident ray? **(3 marks)**
3. The refractive index of diamond is 2.42. What is the meaning of this statement? **(2 marks)** NCERT

LEVEL 24 COMPLETE +125 XP earned. Next: Level 25



BUJHO *Samajh gaya! Slow hui to normal ki taraf, fast hui to door. Aur n = c/v, bas itna sa!*

PAHELI *Shabaash! Kal Level 25: lens aur unki images.*

Day 24 hit-list: Eduspark PYQ analysis

Topic	Rating
Towards / away from normal with reason	★★★
Glass slab diagram, $e = i$, lateral displacement	★★★
$n = c/v$ numericals	★★★
Laws of refraction, Snell's law	★★★
Reading the NCERT n table	★★☆
Relative refractive index	★★☆
Everyday examples of refraction	★☆☆

Answer key: MCQs

Q	Ans	Why
1	(b)	Air → glass is rarer → denser: towards the normal.
2	(b)	$v = 3 \times 10^8 / 1.5 = 2 \times 10^8$ m/s.
3	(a)	Normal incidence: no bending, $r = 0°$.
4	(c)	Water has the lowest n (1.33) of the four.
5	(b)	$e = i$ as the faces are parallel.
6	(d)	A ratio of two speeds: no unit.
7	(b)	$n_{21} = v_1/v_2 = 2/1.5 \approx 1.33$.
8	(d)	Only i, thickness and n matter.
9	(d)	Water → air bends **away**; R is true.
10	(a)	Highest n means lowest speed: R explains A.

Model answers: short and long answer

1 Laws and n

- Incident ray, refracted ray and normal lie in one plane
- $\sin i / \sin r$ = constant (Snell's law)
- $n = c/v = (3 \times 10^8)/(1.5 \times 10^8) = \mathbf{2}$

2 Glass slab

- Diagram as in Figure 2, arrows on all rays, normals dashed
- Bending at PQ and SR is equal and opposite as the faces are parallel, so $e = i$ and the emergent ray is parallel, shifted by d

3 Diamond, $n = 2.42$

- Speed of light in diamond is 1/2.42 of its speed in vacuum: $v = 3 \times 10^8 / 2.42 \approx \mathbf{1.24 \times 10^8}$ **m/s**

Day 24 checklist: tick before you sleep

- ☐ I can say why light bends and which way (towards or away from the normal)
- ☐ I can state both laws of refraction and use Snell's law
- ☐ I can use $n = c/v$ and $n_{21} = v_1/v_2$
- ☐ I can draw the glass slab diagram with i, r, e and d
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.





NITESH SIR SAYS
*Refraction ka ek hi funda: light ki speed badli to direction badli. Diagram mein normal aur arrows kabhi mat bhoolna. **Kal milte hai Day 25 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Light chapter board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 25 · LEVEL 25

PHYSICS | CHAPTER 5 | PART 5 · ASTRA SERIES

SPHERICAL LENSES & IMAGES

Goal: draw lens ray diagrams and know all image cases.



by Nitesh Prakash

01 Lens Basics | 02 Lens Terms | 03 Ray Rules | 04 Convex Cases | 05 Concave Cases | +125 XP today | Level 25 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 SPHERICAL LENSES

DEFINITION

- **Lens:** a piece of transparent material (glass or plastic) bound by two surfaces, of which **one or both are spherical**
- **Convex lens:** both surfaces bulge **outwards**; it is **thicker at the middle** than at the edges
- **Concave lens:** both surfaces curve **inwards**; it is **thicker at the edges** than at the middle

Convex vs concave lens ★★★

	Convex lens	Concave lens
Shape	thick middle, thin edges	thin middle, thick edges
Action on light	**converging**: bends parallel rays to meet	**diverging**: spreads parallel rays out
Focus	real (rays actually meet)	virtual (rays only seem to come from it)
Image (real object)	real or virtual	always virtual, erect, diminished
Focal length sign	positive (+)	negative (−)

2 TERMS USED FOR LENSES

Know these terms ★★★

- **Centre of curvature (C_1, C_2):** centres of the two spheres of which the lens surfaces are parts
- **Radius of curvature:** radius of the sphere of which a lens surface is a part
- **Principal axis:** the imaginary straight line through both centres of curvature
- **Optical centre (O):** the central point of the lens; a ray through it passes **without any deviation**
- **Aperture:** the effective diameter of the circular outline of the lens
- **Principal focus (F):** convex: the point where rays parallel to the axis **meet** after refraction; concave: the point from which they **appear to diverge**. A lens has two foci, F_1 and F_2
- **Focal length (f):** distance between the optical centre and the principal focus

Note: $2F_1$ and $2F_2$ mark twice the focal length from O on either side; they play the role that C plays for a mirror.



MEME BREAK

*"Converging lens" ka mask utaaro, andar se nikla **convex lens**! Teen naam, ek hi lens, focal length **positive**. Fancy naam se darna mat.*



BUJHO *Lens ka bhi mirror jaisa ek hi focus hota hai na?*

PAHELI *Nahi Bujho! Light lens ke dono taraf se aa sakti hai, isliye **do foci** hote hain, F1 aur F2, dono optical centre se barabar doori par.*



Quick test: which lens is it? ★☆☆

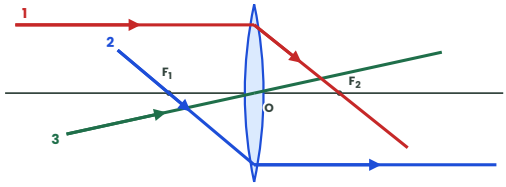
- ✔ Hold it close over printed letters: **bigger** letters mean convex, **smaller** letters mean concave
- ✔ Focus sunlight on paper: only a **convex** lens gives a bright spot (it can burn the paper, so never look at the Sun through it)
- ✔ Touch it: thick middle is convex, thick edges is concave



★★★ *Convex = converging, concave = diverging: ye pair har saal 1-mark mein aata hai, Nitesh Bhaiya ki guarantee!*

3 RULES FOR DRAWING RAY DIAGRAMS

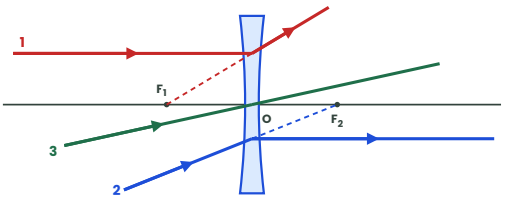
Figure 1: The three rules, convex lens



Rules in words (both lenses) ★★★

1. A ray **parallel to the principal axis**: after refraction it passes **through F_2** (convex) or **appears to come from F_1** (concave).
2. A ray **through F_1** (convex), or **directed towards F_2** (concave): after refraction it emerges **parallel** to the principal axis.
3. A ray through the **optical centre O** goes straight, **without deviation**.

Figure 2: The three rules, concave lens





MEME BREAK

Convex ho ya concave, optical centre se jaane wali ray pe dono agree: ***seedha jao, no bending!***

4 IMAGE FORMATION BY A CONVEX LENS

The six cases at a glance ★★★

Object at	Image at	Size	Nature
Infinity	at F_2	highly diminished, point-sized	real, inverted
Beyond $2F_1$	between F_2 and $2F_2$	diminished	real, inverted
At $2F_1$	at $2F_2$	same size	real, inverted
Between F_1 and $2F_1$	beyond $2F_2$	enlarged	real, inverted
At F_1	at infinity	infinitely large, highly enlarged	real, inverted
Between F_1 and O	same side as object	enlarged	**virtual, erect**

SHORTCUT: OBJECT COMES CLOSER, IMAGE RUNS AWAY

Convex lens mein object jaise jaise F_1 ki taraf aata hai, real image door jaati hai aur badi hoti jaati hai. F_1 ke andar aate hi image virtual ho jaati hai.

SHORTCUT: ANY TWO RAYS ARE ENOUGH

Two rays from the top of the object fix the top of the image. Show real rays as solid lines and back-extended (virtual) rays as dashed lines.

Case 1: Object at infinity

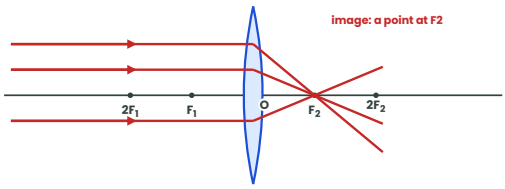


Image: real, inverted, point-sized, at F_2. Use: burning paper with sunlight.

Case 2: Object beyond $2F_1$

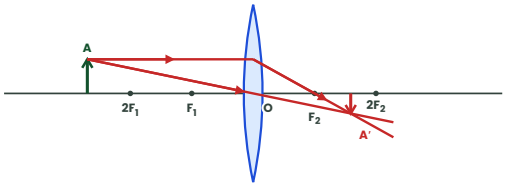


Image: real, inverted, diminished, between F_2 and $2F_2$. Use: camera.

Case 3: Object at $2F_1$

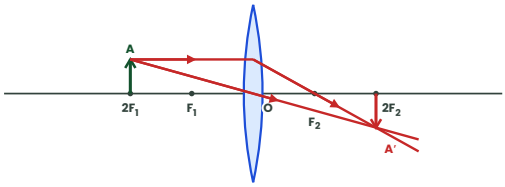


Image: real, inverted, same size, at $2F_2$.



MEME BREAK

Object F_1 pe rakha aur image balloon ki tarah ***infinity*** *ki taraf udd gayi! Rays parallel nikalti hain.*

Case 4: Object between F_1 and $2F_1$

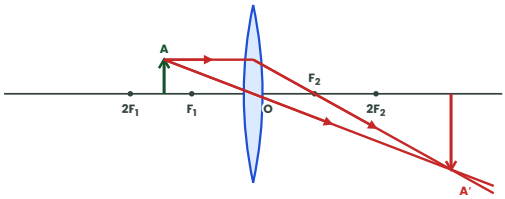


Image: real, inverted, enlarged, beyond $2F_2$. Use: projector.

Case 5: Object at F_1

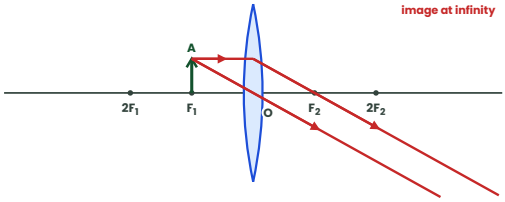


Image: real, inverted, highly enlarged, at infinity (rays emerge parallel).

Case 6: Object between F_1 and O

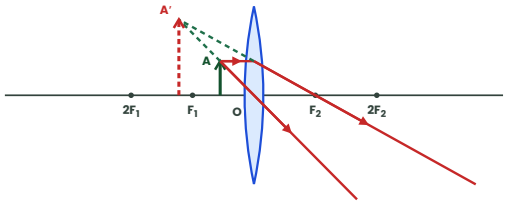


Image: virtual, erect, enlarged, on the object's side. Use: **magnifying glass**.



★★★ *Six-case table + ek ray diagram (mostly case 4 ya case 6): Nitesh Bhaiya ki guarantee!*

Solved Example: name the case ★★☆

Q. A convex lens of focal length 10 cm. Where is the image, and what is it like, for an object at (i) 25 cm (ii) 15 cm (iii) 6 cm?

- (i) 25 cm is beyond 2F (20 cm): image between F_2 and $2F_2$, real, inverted, diminished
- (ii) 15 cm is between F and 2F: image beyond $2F_2$, real, inverted, enlarged
- (iii) 6 cm is inside F: virtual, erect, enlarged, on the object's side

REMEMBER: THE MAGNIFYING GLASS

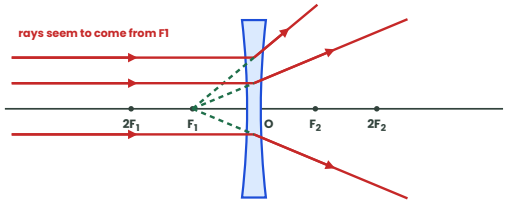
A reading lens is a **convex lens** held so that the object is **inside F_1** (case 6): the image is virtual, erect and enlarged.

5 IMAGE FORMATION BY A CONCAVE LENS

Concave lens: two cases ★★☆

Object at	Image at	Size	Nature
Infinity	at F_1	highly diminished, point-sized	virtual, erect
Between infinity and O	between F_1 and O	diminished	virtual, erect

Concave Case 1: Object at infinity



Concave Case 2: Object at a finite distance

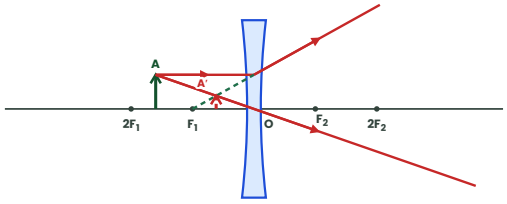


Image: virtual, erect, diminished, between F_1 and O, on the same side as the object.

BUJHO *Concave lens se screen pe real image le sakte hain?*

PAHELI *Kabhi nahi! Real object ke liye concave lens hamesha* ***virtual, erect aur chhoti*** *image banata hai, jo screen pe nahi aati.*



MEME BREAK

Concave lens ho to: ***virtual image, virtual image everywhere!*** *Real object ke liye ye* ***screen pe kabhi nahi aati.***

Uses of lenses (from class notes) ★★☆

Convex lens	Concave lens
Magnifying glass (reading small print, watch repair)	Spectacles for **myopia** (short-sightedness)
Camera, projector, microscope, telescope	Door peephole (wide view of outside)
Spectacles for **hypermetropia**, that is long-sightedness	Galilean telescope eyepiece
Focusing sunlight to burn paper	Combined with convex lenses in cameras to sharpen the image

× WRONG	✓ RIGHT
Concave lens forms a real image too	Concave lens (real object): always virtual and erect
Object at $2F_1$ gives image at F_2	Object at $2F_1$ gives image at $2F_2$, same size
Ray through O drawn bent	Ray through the optical centre goes straight
Virtual rays drawn as solid lines	Back-extended rays are always dashed

6 PRACTICE MCQS

1. A convex lens is also called a: ★★★

(a) diverging lens (b) converging lens
(c) plane lens (d) cylindrical lens

2. A ray passing through the optical centre of a lens: ★★★

(a) bends towards the axis (b) bends away from the axis
(c) goes undeviated (d) is reflected

3. An object is placed at $2F_1$ of a convex lens. The image is formed: ★★★

(a) at F_2 (b) at $2F_2$
(c) beyond $2F_2$ (d) at infinity

4. A magnifying glass uses a convex lens with the object placed: ★★★

(a) beyond $2F_1$ (b) at $2F_1$
(c) at F_1 (d) between F_1 and O

5. For a real object, a concave lens always forms an image that is: ★★★

(a) real and enlarged
(b) virtual, erect, diminished
(c) real, inverted, same size
(d) virtual and enlarged

6. For a real, inverted and enlarged image, the object must be placed: ★★☆

(a) between F_1 and $2F_1$
(b) beyond $2F_1$
(c) between F_1 and O
(d) at infinity

7. A convex lens forms the image at infinity when the object is: ★★☆

(a) at $2F_1$
(b) at F_1
(c) beyond $2F_1$
(d) at the optical centre

8. A ray parallel to the principal axis, after passing through a concave lens: ★★☆

(a) passes through F_2
(b) appears to come from F_1
(c) goes undeviated
(d) becomes parallel to the lens

9. **Assertion (A):** A convex lens can form both real and virtual images.
Reason (R): The nature of the image formed by a convex lens depends on the position of the object. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The image formed by a concave lens can be obtained on a screen.
Reason (R): A concave lens diverges the rays falling on it. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Draw a ray diagram to show the image formed by a convex lens when the object is placed between F_1 and $2F_1$. State the position, size and nature of the image. **(3 marks)**
2. Name the lens that can give (i) a real, inverted image of the same size (ii) a virtual, erect, diminished image. Where must the object be in each case? **(2 marks)**
3. Define principal focus and focal length of a concave lens. Why is its focus called virtual? **(3 marks)**



★★★ *Har ray pe arrow, virtual ray dashed: diagram ke poore marks yahi dilate hain, Nitesh Bhaiya ki guarantee!*

LEVEL 25 COMPLETE +125 XP earned. Next: Level 26



BUJHO *Samajh gaya! Do rays kaafi hain, aur concave lens hamesha chhoti virtual image deta hai.*

PAHELI *Shabaash! Kal Level 26: lens formula aur power.*

Answer key: MCQs

Q	Ans	Why
1	(b)	It bends parallel rays to meet at F_2.
2	(c)	Rule 3: no deviation at O.
3	(b)	Case 3: at $2F_2$, same size.
4	(d)	Case 6: virtual, erect, enlarged.
5	(b)	Concave lens, real object: always so.
6	(a)	Case 4: image beyond $2F_2$.
7	(b)	Case 5: rays emerge parallel.
8	(b)	Rule 1 for a concave lens.
9	(a)	R explains why both kinds occur.
10	(d)	Its image is virtual; R is true.

Model answers: short and long answer

1 Object between F_1 and $2F_1$

- Diagram as in Case 4: parallel ray through F_2, ray through O undeviated
- Image beyond $2F_2$, **enlarged, real and inverted**

2 Which lens?

- (i) Convex lens, object at $2F_1$
- (ii) Concave lens, object anywhere in front of it

3 Focus of a concave lens

- Rays parallel to the axis appear to diverge from a point F_1 on the axis after refraction: the principal focus
- Distance OF_1 is the focal length
- It is virtual because the rays do not actually pass through it; they only seem to come from it

Day 25 checklist: tick before you sleep

- [ ] I can tell a convex lens from a concave lens in three ways
- [ ] I can define optical centre, principal focus and focal length
- [ ] I can use the three ray rules for both lenses
- [ ] I can draw all six convex cases and both concave cases
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 ke diagrams phir se banao.





NITESH SIR SAYS
Lens ka diagram ratna nahi, banana hai. Har din ek case pencil se banao, ek hafte mein haath khud chalega. ***Kal milte hai Day 26 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Light chapter board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 26 LEVEL 26

PHYSICS | CHAPTER 5 | PART 6 ASTRA SERIES

LENS FORMULA & POWER

Goal: solve lens numericals with correct signs and find power.



by Nitesh Prakash

01 Sign Rules | 02 Lens Formula | 03 Magnification | 04 Power | 05 Numericals | +125 XP today | Level 26 / 60

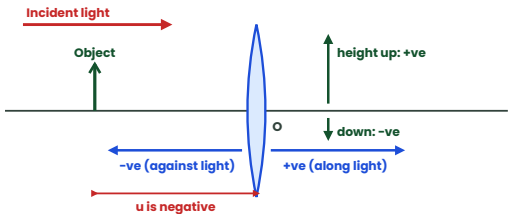
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 SIGN CONVENTION FOR LENSES

New Cartesian sign convention (5 points) ★★★

1. All distances are measured from the **optical centre O**.
2. Distances measured **in the direction of incident light** (left to right) are **positive**.
3. Distances measured **against** the incident light are **negative**, so u is always negative for a real object.
4. Heights measured **upwards** from the principal axis are **positive**.
5. Heights measured **downwards** are **negative**.

Figure 1: Signs at a glance



Sign table (from class notes) ★★★

Quantity	Convex lens	Concave lens
Object distance u	−	−
Image distance v	+ (real image)	−
Focal length f	+	−
Object height h	+	+
Image height h'	− (real image)	+

Note: the convex-lens row in the notes is for a **real** image. When a convex lens forms a **virtual** image (object inside F), v is negative and h' positive.

5 steps for every lens numerical ★★★

1. Write data with signs: u negative; f + for convex, − for concave.
2. Write the formula first: $1/v - 1/u = 1/f$.
3. Put values **in brackets**: $1/v = 1/(+10) + 1/(-15)$.
4. Take LCM, solve, keep the sign of the answer.
5. Read the sign: v + means real image, v − means virtual image.

2 LENS FORMULA AND MAGNIFICATION

DEFINITION

> **Lens formula:** relation between object distance u, image distance v and focal length f (derivation not required)

$$1/v - 1/u = 1/f$$

CLASS TRICK: FORMULA KO ULTA-SEEDHA KARO

1/v − 1/u = 1/f ko seedha 1/v = 1/f + 1/u likh lo; v nikaalna ho to yahi form sabse fast hai.

Magnification (m) ★★★

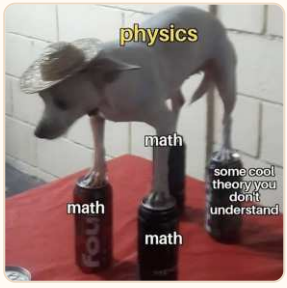
The ratio of the height of the image to the height of the object; for a lens it also equals v/u. It has **no unit**.

$$m = h'/h = v/u$$

If	It means
m negative	real and inverted image
m positive	virtual and erect image
$\|m\| > 1, = 1, < 1$	enlarged, same size, diminished

COMPARE: MIRROR VS LENS

Mirror: $1/v + 1/u = 1/f$ and $m = -v/u$. Lens: $1/v - 1/u = 1/f$ and $m = +v/u$. Do not mix them up.



MEME BREAK

Lens formula physics hai, par chalti maths pe hai: 1/v − 1/u = 1/f. Sign galat, poora answer galat!



BUJHO *Object ki doori 15 cm hai, to u = +15 likh doon?*

PAHELI *Nahi Bujho! Object light ke ulta side hai, isliye u = −15 cm. Real object ke liye u hamesha negative.*



3 SOLVED NUMERICALS: LENS FORMULA

Example 1: convex lens, real image NCERT ★★★

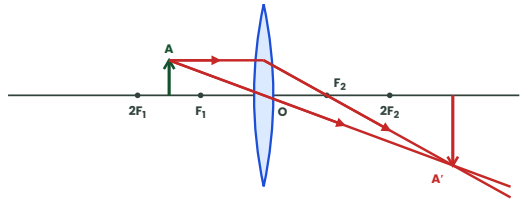
Q. A 2.0 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. Find the position, size and nature of the image.

Given: h = +2.0 cm, u = −15 cm, f = +10 cm

- $1/v = 1/f + 1/u = 1/10 + 1/(−15) = (3 − 2)/30 = 1/30$, so v = +30 cm
- $m = v/u = (+30)/(−15) = −2$
- $h' = m \times h = (−2)(2.0) = −4.0$ cm

Answer: Image 30 cm on the other side of the lens; real, inverted, twice the size (4.0 cm).

Figure 2: Example 1 as a ray diagram



Object between F_1 and $2F_1$ (u = −15 cm, f = 10 cm): image beyond $2F_2$ (v = +30 cm), real, inverted, enlarged.



MEME BREAK

*Signs pehle likho, formula baad mein. Tab numerical **first attempt** mein solve hoga!*

Example 2: concave lens NCERT ★★★

Q. A concave lens has focal length 15 cm. At what distance should the object be placed so that it forms an image 10 cm from the lens? Also find the magnification.

Given: f = −15 cm, v = −10 cm (virtual image, same side)

- $1/u = 1/v − 1/f = 1/(−10) − 1/(−15) = (−3 + 2)/30 = −1/30$, so u = −30 cm
- $m = v/u = (−10)/(−30) = +1/3 \approx +0.33$

Answer: Object 30 cm in front of the lens; image virtual, erect, one-third the size.



BUJHO *v ka answer negative aaya, matlab maine galti ki?*

PAHELI *Bilkul nahi! Negative v ka matlab image **virtual** hai, object wali side. Sign hi to answer bata raha hai.*



Example 3: find size of image NCERT ★★★

Q. A 5 cm tall object is placed 25 cm from a converging lens of focal length 10 cm. Find the position, size and nature of the image.

Given: h = +5 cm, u = −25 cm, f = +10 cm

- $1/v = 1/10 + 1/(−25) = (5 − 2)/50 = 3/50$, so v = +16.7 cm
- $m = 16.7/(−25) = −0.67$, so $h' = −0.67 \times 5 = −3.3$ cm

Answer: Image 16.7 cm behind the lens; real, inverted, diminished (3.3 cm).

Example 4: the magnifying glass case ★★☆

Q. An object is placed 6 cm from a convex lens of focal length 10 cm. Find the image distance and magnification.

Given: u = −6 cm, f = +10 cm

- $1/v = 1/10 + 1/(−6) = (3 − 5)/30 = −2/30$, so v = −15 cm
- $m = (−15)/(−6) = +2.5$

Answer: v negative: image virtual, erect, 2.5 times enlarged, on the object side (case 6).

Example 5: camera lens (PYQ) CBSE SQP 2025 ★★★

Q. A flower 80 mm tall is 150 mm from a camera lens. Its image on the sensor is 20 mm tall. Find the focal length of the lens.

Given: h = +80 mm, h' = −20 mm (real, inverted image), u = −150 mm

- $m = h'/h = −20/80 = −1/4$
- $v = m \times u = (−1/4)(−150) = +37.5$ mm
- $1/f = 1/v − 1/u = 1/37.5 + 1/150 = (4 + 1)/150 = 1/30$

Answer: f = +30 mm, so it is a convex (converging) lens.

Example 6: same-size image NCERT ★★★

Q. A convex lens forms a real, inverted image of a needle 50 cm from it. Where is the needle placed if the image is the same size as the object? Find the power of the lens.

Given: v = +50 cm, m = -1 (real, inverted, same size)

- $m = v/u = −1$, so u = −50 cm (needle at $2F_1$)
- $1/f = 1/50 − 1/(−50) = 2/50$, so f = +25 cm
- $P = 100/25 = +4$ D

Answer: Needle 50 cm in front of the lens; P = +4 D.

Formula sheet: mirror vs lens ★★★

Item	Spherical mirror	Spherical lens
Main formula	$1/v + 1/u = 1/f$	$1/v − 1/u = 1/f$
Magnification	$m = −v/u$	$m = v/u$
Measured from	pole P	optical centre O
f positive for	convex mirror	convex lens
Power used?	no	yes, $P = 1/f$

UNIT ALERT

Keep u, v, f in one unit; use metre only for power.

4 POWER OF A LENS

DEFINITION

- **Power (P):** the degree of convergence or divergence of light rays achieved by a lens. It is the reciprocal of focal length.
- **SI unit:** dioptre (D). **1 dioptre** is the power of a lens whose focal length is 1 metre (1 D = 1 m^{-1}).

$$P = 1/f \text{ (in metre)} = 100/f \text{ (in cm)}$$

CLASS TRICK: N.B.

Concave lens ka power hamesha −ve, convex lens ka power +ve. Sign dekho, lens pehchano!



BUJHO *Focal length 20 cm hai, to power = 1/20 = 0.05 D?*

PAHELI *Galti! f metre mein lo: 20 cm = 0.2 m, to* ***P = 1/0.2 = +5 D****. Shortcut: 100/20 = 5 D.*



Example 7: power and focal length NCERT ★★☆

Q. (i) Find the power of a convex lens of focal length 50 cm. (ii) A lens has power −2.0 D. Find its focal length and type.

Given: (i) $f = +50$ cm (ii) $P = -2.0$ D

- (i) $P = 100/f(\text{cm}) = 100/50 = +2$ D
- (ii) $f = 1/P = 1/(-2.0) = -0.5$ m $= -50$ cm

Answer: (i) +2 D (ii) −50 cm; negative, so it is a concave lens.

Example 8: dioptre (PYQ) CBSE 2020 ★★☆

Q. Define 1 dioptre power of a lens. Find the focal length of a lens of power +0.5 D.

Given: $P = +0.5$ D

- 1 D is the power of a lens of focal length 1 m
- $f = 1/P = 1/(+0.5) = +2$ m

Answer: $f = +2$ m (200 cm); positive, so a convex lens.

Power of a combination of lenses ★★★

When lenses are placed **in contact**, the net power is the **algebraic sum** of the individual powers:

$$P = P_1 + P_2 + P_3 + \dots$$

- ✔ Opticians use this: they put test lenses together to find the power of the spectacles you need.
- ✔ Cameras and microscopes use lens combinations to get a larger, sharper image.

Example 9: combination ★★☆

Q. A convex lens of power +2.0 D is kept in contact with a concave lens of power −0.5 D. Find the power and focal length of the combination.

Given: $P_1 = +2.0$ D, $P_2 = -0.5$ D

- $P = P_1 + P_2 = +2.0 + (-0.5) = +1.5$ D
- $f = 1/P = 1/1.5 = +0.67$ m $= +66.7$ cm

Answer: +1.5 D and 66.7 cm; the combination behaves like a convex lens.



MEME BREAK
Ramanujan wali maths nahi chahiye: bas ***P = 1/f****, aur f metre mein!*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

A spectacle prescription is written in dioptres. A reading of **−1.5 D** means a concave lens of $f = 1/(-1.5) = -0.67$ m; a reading of **+2.0 D** means a convex lens of $f = +0.5$ m. You will meet these lenses again in the chapter on the human eye.

Activity: find f of a convex lens in 2 minutes ★★☆

1. Hold a convex lens facing a far-off object such as a tree or a window across the room.
2. Move a paper screen behind the lens till a sharp, inverted image appears.
3. Measure the distance from lens to screen. Rays from a far object are nearly parallel, so this distance is about the focal length.
4. Work out the power: if $f = 20$ cm, then $P = 100/20 = +5$ D.

Caution: never look at the Sun through a lens.

× WRONG	✓ RIGHT
$u = +15$ cm for a real object	$u = -15$ cm (against the light)
Lens: $m = -v/u$	Lens: $m = +v/u$; the minus sign is for mirrors
$P = 1/f$ with f in cm	$P = 1/f$ in metre, or $100/f$ in cm
Powers in contact are multiplied	Powers add: $P = P_1 + P_2$
Concave lens power is positive	Concave lens power is negative

Try it yourself (answers in brackets) ★★☆

1. Convex lens, $f = 15$ cm, object at 30 cm. Find v. (Ans: +30 cm)
2. Concave lens, $f = 10$ cm, object at 10 cm. Find v. (Ans: −5 cm)
3. Find the power of a lens of $f = -40$ cm. (Ans: −2.5 D)
4. Lenses of +5 D and −3 D in contact. Find f. (Ans: +50 cm)



★★★ *Sign convention + formula + unit: teeno likhe to numerical ke poore marks pakke, Nitesh Bhaiya ki guarantee!*



MEME BREAK
Numericals ki practice skip karke seedha kal ka ***Boss Level****? Phir paper dekh ke Tom wala yahi chehra banega!*

5 PRACTICE MCQS

1. The SI unit of power of a lens is: NCERT ★★★
(a) metre (b) dioptre
(c) watt (d) joule

2. The lens formula is: NCERT ★★★
(a) $1/v + 1/u = 1/f$ (b) $v - u = f$
(c) $1/v - 1/u = 1/f$ (d) $1/u - 1/v = 1/f$

3. A lens has power −2.5 D. It is a: ★★★
(a) concave lens, $f = -40$ cm (b) convex lens, $f = +40$ cm
(c) concave lens, $f = -25$ cm (d) convex lens, $f = +25$ cm

4. A lens gives magnification +0.5. The image is: ★★☆
(a) real, inverted, diminished (b) real, inverted, enlarged
(c) virtual, erect, enlarged (d) virtual, erect, diminished

5. For a real object placed in front of a lens, u is always: ★★☆

(a) positive
(b) negative
(c) zero
(d) positive for a convex lens only

6. Lenses of power +3 D and −1 D are kept in contact. The net power is: ★★☆

(a) +4 D
(b) −2 D
(c) +2 D
(d) +3 D

7. An object is 40 cm from a convex lens of focal length 20 cm. The image is: ★★☆

(a) 40 cm behind the lens
(b) 20 cm behind the lens
(c) 40 cm in front of the lens
(d) at infinity

8. A lens forms a virtual, erect image 3 times the size of the object. Its magnification is: ★★☆

(a) −3
(b) −1/3
(c) +1/3
(d) +3

9. The power of a convex lens of focal length 25 cm is: NCERT ★★★

(a) +0.25 D
(b) +4 D
(c) −4 D
(d) +25 D

10. **Assertion (A):** The power of a concave lens is negative.
Reason (R): The focal length of a concave lens is taken as negative in the New Cartesian sign convention. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Define 1 dioptre. Find the focal length of a lens of power −2.0 D. What type of lens is it? NCERT **(2 marks)**
2. Find the power of a concave lens of focal length 2 m. NCERT **(1 mark)**
3. A 4 cm tall object is placed 30 cm from a convex lens of focal length 20 cm. Find the position, size and nature of the image. **(3 marks)**
4. An object is placed 20 cm from a concave lens of focal length 20 cm. Find the image distance and magnification. **(3 marks)**
5. Two thin lenses of power +3.5 D and −2.5 D are placed in contact. Find the power and focal length of the combination. NCERT **(2 marks)**

LEVEL 26 COMPLETE +125 XP earned. Next: Level 27



BUJHO *Samajh gaya! Pehle signs, phir formula, aur power ke liye f metre mein.*

PAHELI *Shabaash! Lenses ka chapter ab tumhari jeb mein.*

Answer key: MCQs

Q	Ans	Why
1	(b)	1 D = 1 m^{-1}.
2	(c)	Lens: minus sign between $1/v$ and $1/u$.
3	(a)	$f = 1/(-2.5) = -0.4$ m.
4	(d)	+ means erect and virtual; 0.5 means smaller.
5	(b)	It is measured against the incident light.
6	(c)	$P = 3 + (-1) = +2$ D.
7	(a)	$1/v = 1/20 - 1/40 = 1/40$.
8	(d)	Erect means +, enlarged 3 times.
9	(b)	100/25 = +4 D.
10	(a)	$P = 1/f$ and f is negative.

Model answers: short and long answer

1 **Dioptre and −2.0 D**
- 1 D is the power of a lens of focal length 1 m
- $f = 1/(-2.0) = -0.5$ m, so it is a **concave lens**

2 **Concave lens, f = −2 m**
- $P = 1/f = 1/(-2) =$ **−0.5 D**

3 **u = −30 cm, f = +20 cm, h = +4 cm**
- $1/v = 1/20 - 1/30 = 1/60$, so $v =$ **+60 cm**
- $m = 60/(-30) = -2$, $h' =$ **−8 cm**: real, inverted, enlarged

4 **u = −20 cm, f = −20 cm**
- $1/v = 1/(-20) + 1/(-20) = -1/10$, so $v =$ **−10 cm**
- $m = (-10)/(-20) =$ **+0.5**: virtual, erect, half size

Day 26 checklist: tick before you sleep

- ☐ I can state the New Cartesian sign convention for lenses
- ☐ I can write the lens formula and the magnification formula
- ☐ I can solve a lens numerical with correct signs
- ☐ I can find power in dioptre and tell the lens type from its sign
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2 ke numericals phir se karo.





NITESH SIR SAYS
Numerical mein marks steps ke milte hain: data with signs, formula, substitution, unit. Ek bhi step mat chhodo. ***Kal milte hai Day 27, Boss Level pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Light chapter board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 27 LEVEL 27

PHYSICS | CHAPTER 5 | BOSS LEVEL

CHAPTER 5 BOSS LEVEL

Goal: the whole Light chapter in 75 minutes.



by Nitesh Prakash

01 Revision Map | 02 MCQs | 03 Short Answers | 04 Long Answers | 05 Answer Key | +150 XP today | Level 27 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 5 REVISION MAP

Chapter 5 mind map: revise in 5 minutes

REFLECTION
Angle i = angle r; incident ray, normal, reflected ray in one plane. Plane mirror: virtual, erect, same size, laterally inverted

MIRRORS
R = 2f. Concave: converging, f negative. Convex: diverging, f positive, always virtual, erect, diminished

MIRROR FORMULA ★★★
1/v + 1/u = 1/f; m = h′/h = −v/u. m negative: real and inverted; m positive: virtual and erect

LIGHT

REFRACTION
Snell: sin i / sin r = constant. n = c/v. Rarer to denser: bends towards the normal

LENSES
Convex: converging, f positive. Concave: diverging, f negative, always virtual, erect, diminished

LENS FORMULA & POWER ★★★
1/v − 1/u = 1/f; m = h′/h = v/u. P = 1/f (f in metres), unit dioptre (D). Convex P positive, concave P negative

New Cartesian sign convention: the scorecard

Measure every distance from the pole P (mirror) or optical centre O (lens). Along the incident light = positive; against it = negative.

Quantity	Sign
u (real object)	always negative
f concave mirror, concave lens	negative
f convex mirror, convex lens	positive
Height above the axis	positive
Real image (mirror)	v negative
Real image (lens)	v positive

Image cases: concave mirror

Object at	Image at	Nature, size
Infinity	F	real, inverted, point-sized
Beyond C	Between F and C	real, inverted, diminished
C	C	real, inverted, same size
Between C and F	Beyond C	real, inverted, enlarged
F	Infinity	real, inverted, highly enlarged
Between P and F	Behind the mirror	virtual, erect, enlarged



BUJHO *Boss Level? Mirror ka formula yaad hai, lens ka bhi same hi hoga na?*

PAHELI *Nahi! Mirror: 1/v + 1/u. Lens: 1/v − 1/u. Ek sign ki galti aur poora numerical gaya.*



BOSS LEVEL RULES

- ✔ **Time:** 75 minutes. **Maximum marks:** 40. Close the book and set a timer.
- ✔ **A:** 14 MCQs × 1 = 14. **B:** 5 short answers = 12. **C:** 2 long answers × 5 = 10. **D:** 1 case-based question = 4.
- ✔ Write Given (with signs), Formula, Substitution and Answer with unit in every numerical. Draw ray diagrams with a pencil and arrows.

Mirror-lens ka har doubt? Sign convention ka flex tape lagao, sab fix!



2 SECTION A: MCQS

1. The image formed by a plane mirror is: ★★★

(a) real, inverted, same size (b) virtual, erect, same size
(c) virtual, inverted, magnified (d) real, erect, diminished

2. The radius of curvature of a spherical mirror is 20 cm. Its focal length is: ★★☆

(a) 40 cm (b) 20 cm
(c) 10 cm (d) 5 cm

3. A spherical mirror and a thin spherical lens each have a focal length of −15 cm. The mirror and the lens are likely to be: NCERT ★★★

(a) both concave (b) both convex
(c) mirror concave, lens convex (d) mirror convex, lens concave

4. No matter how far you stand from a mirror, your image always appears erect. The mirror is: NCERT ★★☆

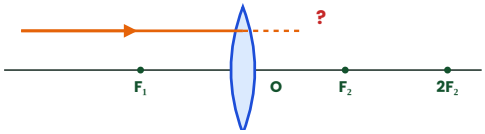
(a) only plane (b) only concave
(c) only convex (d) either plane or convex

5. When light enters glass from air, it: ★★★
(a) speeds up and bends away from the normal
(b) slows down and bends towards the normal
(c) speeds up and bends towards the normal
(d) slows down and does not bend

6. The refractive index of diamond is 2.42. This means: ★★☆
(a) light travels 2.42 times faster in diamond
(b) speed of light in diamond is 1/2.42 of its speed in vacuum
(c) diamond reflects 2.42 times more light
(d) diamond absorbs light

7. An object is at $2F_1$ of a convex lens. Where is the image? ★★★
(a) F_2, point-sized
(b) between F_2 and $2F_2$
(c) $2F_2$, same size
(d) beyond $2F_2$, enlarged

8. A ray parallel to the principal axis falls on a convex lens as shown. After refraction it will: ★★★
(a) pass through F_2
(b) pass through $2F_2$
(c) go undeviated
(d) appear to come from F_1



9. The SI unit of the power of a lens is: ★★☆
(a) metre
(b) dioptre
(c) watt
(d) candela

10. To get a virtual, erect and enlarged image from a convex lens, the object must be placed: ★★★
(a) at $2F_1$
(b) between F_1 and $2F_1$
(c) between F_1 and O
(d) at infinity

Must-know formulas: write them daily

Mirror	Lens
$1/v + 1/u = 1/f$	$1/v - 1/u = 1/f$
$m = h'/h = -v/u$	$m = h'/h = v/u$
$f = R/2$	$P = 1/f$ (f in m), 1 D = 1 m^{-1}
$n = c/v = 3 \times 10^8 / v$	$n_{21} = n_2/n_1 = v_1/v_2$

How to attack Sections B, C and D

- ✔ 2 marks = two clear points; 3 marks = three points, or a full numerical with steps.
- ✔ Numericals: Given with signs, formula, substitution, answer with unit and the nature of the image.
- ✔ Ray diagrams: two standard rays, arrows on every ray, dashed virtual rays, labels P, F, C or O, F_1, F_2.



Light ka nature samajhne gaye the, MCQs ne dimaag ka bulb jala diya!

11. Light travels fastest in which medium? (n: water 1.33, kerosene 1.44, crown glass 1.52, diamond 2.42) ★★☆
(a) water
(b) kerosene
(c) crown glass
(d) diamond

12. A mirror gives m = +0.4 for every position of the object. The mirror is: ★★☆
(a) plane
(b) concave
(c) convex
(d) either concave or plane

13. **Assertion (A):** A concave mirror can form a virtual image. **Reason (R):** When the object is between P and F, the reflected rays diverge and appear to meet behind the mirror. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** The power of a concave lens is negative. **Reason (R):** A concave lens is a converging lens. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Image cases: convex lens

Object at	Image at (convex lens)
Infinity	F_2: real, inverted, point-sized
Beyond $2F_1$	Between F_2 and $2F_2$: diminished
$2F_1$	$2F_2$: real, inverted, same size
Between F_1 and $2F_1$	Beyond $2F_2$: real, enlarged
F_1	Infinity: highly enlarged
Between F_1 and O	Same side: virtual, erect, enlarged

3 SECTION B: SHORT ANSWERS

15 State the two laws of refraction. What is meant by the absolute refractive index of a medium? **(2 marks)**

16 Draw a ray diagram to show the image formed by a concave mirror when the object is placed between C and F. State the nature of the image. **(2 marks)**

17 The rear-view mirror of a bus is convex with radius of curvature 3 m. A car is 6 m behind it. Find the position, nature and magnification of the image. **(3 marks)**

18 An object is placed 10 cm from a concave lens of focal length 15 cm. Find the image distance and magnification. State the nature of the image. **(3 marks)**

19 Give two uses each of concave and convex mirrors with one reason for each. **(2 marks)**

4 SECTION C: LONG ANSWERS

20 (A) A 4 cm tall object is placed 30 cm in front of a concave mirror of focal length 20 cm. Find the position, nature and size of the image. (B) Draw the ray diagram. **(5 marks)**

21 (A) An object 3 cm high is placed 20 cm from a convex lens of focal length 15 cm. Find the position, size and nature of the image. (B) Draw the ray diagram. (C) Find the power of the lens. **(5 marks)**

5 SECTION D: CASE-BASED QUESTION

Case: the refractive index chart (4 marks)

Riya measured the absolute refractive index of some media in her school lab:

Medium	Air	Water	Kerosene	Crown glass	Diamond
n	1.0003	1.33	1.44	1.52	2.42

22 Which medium is optically the densest? **(1)**

23 Find the speed of light in crown glass ($c = 3 \times 10^8$ m/s). **(1)**

24 A ray goes from water into kerosene. Does it bend towards or away from the normal? **(1)**

25 Find the refractive index of diamond with respect to water. **(1)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Plane mirror: virtual, erect, same size, laterally inverted.
2	(c)	f = R/2 = 20/2 = 10 cm.
3	(a)	Negative f: concave mirror and concave lens.
4	(d)	Plane and convex mirrors always give erect images.
5	(b)	Glass is optically denser: speed falls, ray bends towards the normal.
6	(b)	n = c/v, so v = c/2.42.
7	(c)	Object at $2F_1$ gives a same-size image at $2F_2$.
8	(a)	A ray parallel to the axis passes through F_2.
9	(b)	1 D = 1 m^{-1}.
10	(c)	Object between F_1 and O: virtual, erect, enlarged.
11	(a)	Lowest n means highest speed: water.
12	(c)	m positive and less than 1 for every position: convex mirror.
13	(a)	R correctly explains the virtual image.
14	(c)	A is true; R is false: a concave lens is diverging.

Model answers: Section C

20 Concave mirror numerical

Given: h = +4 cm, u = −30 cm, f = −20 cm
Formula: 1/v + 1/u = 1/f, so 1/v = 1/f − 1/u
1/v = −1/20 + 1/30 = (−3 + 2)/60 = −1/60
v = −60 cm: 60 cm in front of the mirror
m = −v/u = −(−60)/(−30) = **−2**; h′ = m × h = **−8 cm**
Image: real, inverted, enlarged (twice). (B) Ray diagram: as in answer 16 (object between C and F).

Model answers: Section D

- (22.1) Diamond (highest n = 2.42).
- (22.2) v = c/n = (3 × 10⁸)/1.52 ≈ **1.97 × 10⁸ m/s**.
- (22.3) Towards the normal: kerosene (1.44) is optically denser than water (1.33).
- (22.4) n(diamond, water) = 2.42/1.33 ≈ **1.82**.

Sign convention skip kiya? Clown makeup complete! Har answer ka sign check karo.



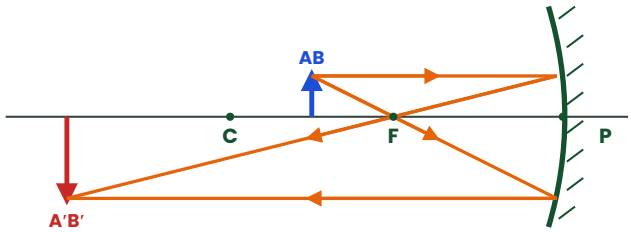
Model answers: Section B

15 Laws of refraction

- (1) The incident ray, the refracted ray and the normal at the point of incidence lie in the same plane.
- (2) Snell's law: for a given pair of media and a given colour, sin i / sin r = constant (the refractive index of the second medium with respect to the first).
- Absolute refractive index: n = speed of light in vacuum / speed of light in the medium = c/v.

16 Concave mirror, object between C and F

- Image: beyond C, real, inverted and enlarged.



Ray 1: parallel to the axis, reflects through F. Ray 2: through F, reflects parallel to the axis.

Model answers: Section B (numericals)

17 Convex rear-view mirror

Given: R = +3 m, so f = R/2 = +1.5 m; u = −6 m
Formula: 1/v + 1/u = 1/f, so 1/v = 1/f − 1/u
1/v = 1/1.5 − 1/(−6) = 4/6 + 1/6 = 5/6
v = +1.2 m (behind the mirror)
m = −v/u = −(1.2)/(−6) = **+0.2**
Image: virtual, erect, diminished (one-fifth the size).

18 Concave lens

Given: f = −15 cm, u = −10 cm
Formula: 1/v − 1/u = 1/f, so 1/v = 1/f + 1/u
1/v = −1/15 − 1/10 = (−2 − 3)/30 = −1/6
v = −6 cm (same side as the object)
m = v/u = (−6)/(−10) = **+0.6**
Image: virtual, erect, diminished.

19 Uses of spherical mirrors

- Concave: torches and headlights (bulb at F gives a strong parallel beam); shaving and dentist's mirrors (enlarged, erect image when close); solar furnaces (concentrate sunlight).
- Convex: rear-view mirrors of vehicles: always erect image and a much wider field of view.

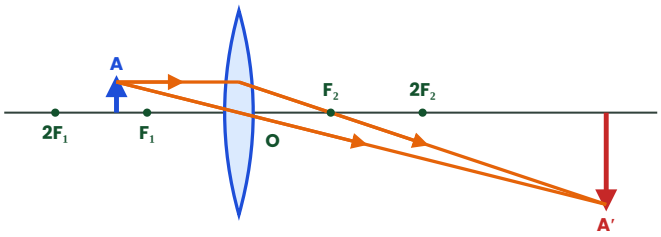
Light: kabhi wave, kabhi particle. Board exam mein bas sahi ray diagram banao, observer khush!



Model answers: Section C (continued)

21 Convex lens numerical

Given: h = +3 cm, u = −20 cm, f = +15 cm
Formula: 1/v − 1/u = 1/f, so 1/v = 1/f + 1/u
1/v = 1/15 − 1/20 = (4 − 3)/60 = 1/60
v = +60 cm: on the other side of the lens
m = v/u = 60/(−20) = **−3**; h′ = m × h = **−9 cm**
Image: real, inverted, enlarged (three times).
(C) P = 1/f = 1/(+0.15 m) = **+6.67 D**



(B) Object between F_1 and $2F_1$: image beyond $2F_2$. Ray 1 parallel, then through F_2; ray 2 through O, undeviated.

Most asked in Chapter 5: Eduspark PYQ analysis

Topic	Day	Rating
Mirror formula numericals	22	★★★
Lens formula and power numericals	25	★★★
Ray diagrams: concave mirror, convex lens	22, 24	★★★
Refractive index, n = c/v	23	★★★
Uses of concave and convex mirrors	21	★★☆
Laws of refraction, Snell's law	23	★★☆
Image by concave lens and convex mirror	22, 24	★★☆

Common mistakes that cost marks

× WRONG	✓ RIGHT
u written as +30 cm	u is always negative for a real object: −30 cm
Lens formula 1/v + 1/u = 1/f	Lens: 1/v − 1/u = 1/f; mirror: 1/v + 1/u = 1/f
Mirror m = v/u	Mirror m = −v/u; lens m = v/u
P = 1/f with f in cm	Convert f to metres first: P = 100/f(cm)
No arrows on rays	Every ray needs an arrow; virtual rays are dashed

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 5 board and sample-paper questions.



BUJHO *Convex mirror ki image screen pe kyun nahi aati? Main toh try karta rahunga!*

PAHELI *Kyunki wo virtual hai, rays actually milti hi nahi. Screen pe sirf real image aati hai.*



Day 27 checklist: tick before you sleep

- [ ] I revised the mind map, the sign table and all formulas
- [ ] I attempted all 14 MCQs without looking at the answer key
- [ ] I solved every numerical with Given, Formula, Substitution and Answer
- [ ] I drew both ray diagrams neatly from memory with arrows and labels
- [ ] I scored myself honestly and revised the days of my wrong answers

My score: ____ / 5

5/5 = Jalwa hai! Light chapter ka boss ab hamara hai!



LEVEL 27 COMPLETE **CHAPTER 5 COMPLETE**

+150 XP earned. Next: Level 28



BUJHO *Samajh gaya! Mirror mein −v/u, lens mein v/u. Sign convention hi asli hero hai!*

PAHELI *Boss defeated! Kal se Chapter 6: hamari apni aankh ka camera.*

Bonus from class notes: solved example

B Concave mirror: where to place the screen?

A 7 cm tall object is 27 cm in front of a concave mirror of focal length 18 cm. Where should the screen be placed for a sharp image? Find the image size.

u = −27 cm, f = −18 cm, h = +7 cm
1/v = 1/f − 1/u = −1/18 + 1/27 = (−3 + 2)/54 = −1/54
v = −54 cm: screen 54 cm in front of the mirror
m = −v/u = −(−54)/(−27) = −2; **h′ = −14 cm** (real, inverted, enlarged)

Bonus rapid-fire round: quick answers

Question	Answer
Speed of light in vacuum	3×10^8 m/s
Mirror used by dentists	Concave
Lens in a magnifying glass	Convex
Unit of refractive index	None (a ratio)
Ray through the optical centre	Goes undeviated
Power of a lens with f = −25 cm	−4 D (concave)



NITESH SIR SAYS
Chapter 5 khatam! Galat numericals dobara solve karo, sign ke saath. ***Consistency hi jalwa hai!***

PHYSICS | CHAPTER 5 COMPLETE ASTRA SERIES

CHAPTER 5 SWAHAA!

Light: Reflection and Refraction | Days 21–27



BUJHO *Mirror, lens, sign convention, numericals sab clear! Ab concave aur convex mein kabhi confuse nahi hounga.*

PAHELI *Shabaash, Bujho! Light ka chapter Swahaa. Ab chalo aankh ki duniya mein, Human Eye wait kar raha hai.*

CHAPTER XP EARNED
+900 XP

LEVELS CLEARED
7 levels (Days 21–27)

BOOK PROGRESS
Level 27 / 60 3400 XP total

WHAT YOU HAVE MASTERED

- ✔ Laws of reflection, plane mirror images
- ✔ Mirror formula, magnification, sign convention
- ✔ Convex and concave lenses, image formation
- ✔ Concave and convex mirrors, ray diagrams
- ✔ Refraction, Snell's law, refractive index
- ✔ Lens formula, magnification, power of a lens

NEXT CHAPTER Day 28: The Human Eye and the Colourful World *Light cleared!*

DAY 28 · LEVEL 28

PHYSICS | CHAPTER 6 | PART 1 · ASTRA SERIES

THE HUMAN EYE

Goal: know every part of the eye, then fix its defects.



by Nitesh Prakash

01 Eye Structure | 02 Accommodation | 03 Near, Far Point | 04 Defects | 05 Numericals | +125 XP today | Level 28 / 60

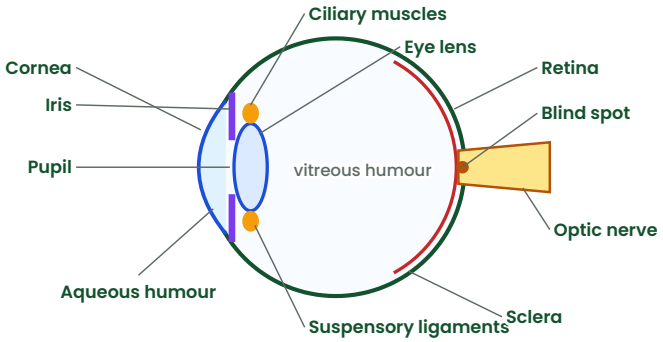
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 THE HUMAN EYE: OUR NATURAL CAMERA

DEFINITION

- **Human eye:** a natural optical instrument that lets us see objects around us. Its lens system forms a **real, inverted** image on the light-sensitive retina.
- The brain processes the signals so we see the object **erect**.

Structure of the human eye (labelled diagram)



Draw it with at least 8 labels: cornea, iris, pupil, eye lens, ciliary muscles, retina, optic nerve, blind spot.

Retina: rods and cones (from class notes)

Rod cells	Cone cells
Sensitive to **dim light**	Sensitive to **bright light**
Side (peripheral) vision	Colour and sharp central vision
No colour information	Detect colours

The retina turns the image into electrical impulses; the optic nerve takes them to the brain.

2 POWER OF ACCOMMODATION

DEFINITION

- **Power of accommodation:** the ability of the eye lens to adjust its focal length so that both near and far objects are focused on the retina.

Looking at	Ciliary muscles	Eye lens	Focal length
Distant object	relaxed	thin	increases
Nearby object	contract	thick	decreases

Range of vision of a normal eye: from 25 cm to infinity.

Parts of the eye and their functions

Part	Function
Sclera	Tough white outer coat; protects the eye
Cornea	Transparent front bulge of the sclera; light enters here and **most refraction** happens at its outer surface
Iris	Dark muscular ring behind the cornea; gives the eye its colour; controls the size of the pupil
Pupil	Opening in the iris; controls the light entering: **small in bright light, wide in dim light**
Eye lens	Crystalline, flexible convex lens; fine-tunes the focal length to focus the image on the retina
Ciliary muscles	Hold the lens and change its curvature, so its focal length changes
Aqueous humour	Clear fluid between the cornea and the lens
Vitreous humour	Jelly behind the lens; gives nutrients and keeps the shape of the eyeball
Retina	Light-sensitive screen with rods and cones; forms the image and makes electrical signals
Optic nerve	Carries the signals from the retina to the brain
Blind spot	Where the optic nerve leaves the retina; no rods or cones, so no vision



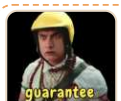
BUJHO *Retina pe image ulti banti hai, toh duniya ulti kyun nahi dikhti?*

PAHELI *Kyunki brain signals ko process karke image ko seedha kar deta hai. Aankh camera hai, brain editor!*



Near point and far point

- ✔ **Near point** (least distance of distinct vision): the closest point seen clearly without strain. Normal young adult: **25 cm**.
- ✔ **Far point:** the farthest point seen clearly. Normal eye: **infinity**.
- ✔ Closer than 25 cm, the lens cannot become thicker any more, so the object looks blurred.



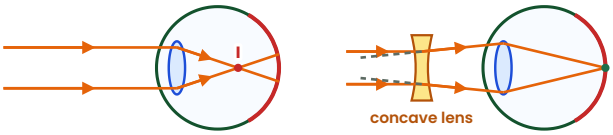
★★★ *Near point 25 cm, far point infinity, accommodation ka meaning: ye teen line har saal aati hain, Nitesh Bhaiya ki guarantee!*

3 DEFECTS OF VISION AND THEIR CORRECTION

A defect of vision (refractive defect) means the eye cannot focus the image exactly on the retina. Spectacle lenses fix it.

1. Myopia (near-sightedness) ★★★

- ✔ **Sees:** near objects clearly, but distant objects look blurred.
- ✔ **Far point** comes nearer than infinity (a few metres or less).
- ✔ **Image** of a distant object forms **in front of the retina**.
- ✔ **Causes:** (i) excessive curvature of the eye lens; (ii) elongation of the eyeball.
- ✔ **Correction:** a **concave (diverging) lens** of suitable power.



(a) Myopic eye: image in front of retina (b) Corrected: image on retina

Class notes say "excessive curvature of the cornea"; NCERT says excessive curvature of the **eye lens**. Write the NCERT line in the exam.



MEME BREAK

*Myopia mein concave, hypermetropia mein convex. Button galat dabaya toh **marks bhi blur**!*

NCERT quick Q&A

> **Q.** Why can a normal eye not see clearly an object closer than 25 cm? **A.** The ciliary muscles cannot make the lens thicker beyond a limit, so its focal length cannot decrease any further.

4 NUMERICALS ON DEFECTS OF VISION

CLASS TRICK: NUMERICAL HACK (NITESH SIR)

- ✔ **Step 1:** Find the case of defect (myopia or hypermetropia).
- ✔ **Step 2:** Find the correction lens of that defect.
- ✔ **Step 3:** If myopia: u = ∞, v = given in question. If hypermetropia: u = 25 cm, v = given in question.
- ✔ **Step 4:** Jitna bhi value aaye usme pehle negative sign laga do (such as u = −∞, u = −25 cm).
- ✔ Yaad rakho: 1/∞ = 0. Then use 1/f = 1/v − 1/u and P = 100/f (f in cm).

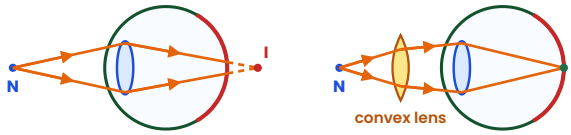
Solved Example 1 (class notes): myopia

A person cannot see objects beyond 40 cm clearly. Name the defect and find the power of the correcting lens.

Defect: myopia (far point = 40 cm), so a concave lens is needed.
Given: u = −∞, v = −40 cm (the lens must form the image of a distant object at the far point)
1/f = 1/v − 1/u = −1/40 − 0 = −1/40, so **f = −40 cm**
P = 100/f = 100/(−40) = **−2.5 D** (concave lens)

2. Hypermetropia (far-sightedness) ★★★

- ✔ **Sees:** distant objects clearly, but near objects look blurred.
- ✔ **Near point** moves farther than 25 cm.
- ✔ **Image** of a near object forms **behind the retina**.
- ✔ **Causes:** (i) focal length of the eye lens is too long; (ii) the eyeball is too small.
- ✔ **Correction:** a **convex (converging) lens** of suitable power.



(a) Hypermetropia: image behind retina (b) Corrected: image on retina

3. Presbyopia (old-age defect)

- ✔ **Sees:** near objects with difficulty; reading and writing become uncomfortable.
- ✔ **Cause:** with age the **ciliary muscles weaken** and the **eye lens loses flexibility**, so the near point recedes.
- ✔ A person may suffer from **both myopia and hypermetropia**. Class notes say near and far both blur; NCERT: mainly near vision, both only in some people.
- ✔ **Correction: bifocal lenses**: upper part concave (distant vision), lower part convex (reading).

4. Cataract

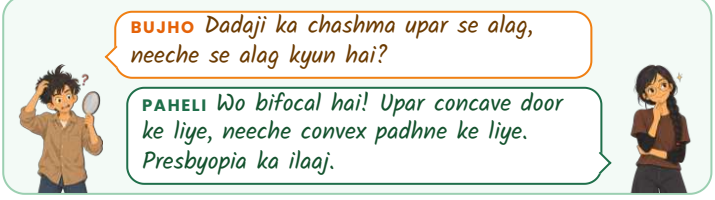
- ✔ The crystalline lens of older people often becomes **milky and cloudy**.
- ✔ Seeing through it is like looking through a frosty or fogged-up window; vision is partly or fully lost.
- ✔ **Correction:** vision is restored by **cataract surgery**.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Astigmatism (from class notes): the eye cannot focus horizontal and vertical lines at the same time, so it sees clearly in only one plane at a time. Cause: an irregularly shaped cornea or a distorted eye lens. Correction: a **cylindrical lens**.

Defects at a glance

Defect	Image	Lens
Myopia	in front of retina	concave
Hypermetropia	behind retina	convex
Presbyopia	near point recedes	bifocal
Cataract	cloudy lens	surgery



Solved Example 2 (class notes): hypermetropia

An old person can read a newspaper only when it is 80 cm away. Find the power of the lens needed to read at 25 cm.

Defect: hypermetropia (near point = 80 cm), so a convex lens is needed.
Given: u = −25 cm, v = −80 cm (image of the paper must be at his near point)
1/f = 1/v − 1/u = −1/80 + 1/25 = (−5 + 16)/400 = 11/400
f = +36.4 cm; P = 100/f = 1100/400 = **+2.75 D** (convex lens)

Check your notes: 1100/400 = 2.75, not 2.25. The correct answer is +2.75 D.

Solved Examples 3 and 4 (NCERT)

3. The far point of a myopic person is 80 cm in front of the eye. Find the nature and power of the lens needed.

u = −∞, v = −80 cm; 1/f = 1/v − 1/u = −1/80, so **f = −80 cm**
P = 100/(−80) = **−1.25 D**: a concave lens

4. The near point of a hypermetropic eye is 1 m. Find the power of the lens needed (normal near point 25 cm).

u = −25 cm, v = −100 cm
1/f = 1/v − 1/u = −1/100 + 1/25 = 3/100, so **f = +33.3 cm**
P = 100/f = 300/100 = **+3.0 D**: a convex lens

Board dhundhla, teacher dhundhli, sab dhundhla... "This is fine." Nahi bhai, eye test karao aur concave lens lagao!



Common mistakes that cost marks

× WRONG	✓ RIGHT
Myopia: image behind retina	Myopia: in front; hypermetropia: behind
Iris controls light directly	Iris controls the pupil size; the pupil lets light in
Hypermetropia: u = −∞	Hypermetropia: u = −25 cm; myopia: u = −∞
Near point of normal eye: 25 m	Near point: 25 cm; far point: infinity

EXAM TIPS: SCORE FULL MARKS

- ✓ Defect questions: write **three things**: where the image forms, two causes, and the correcting lens with a ray diagram.
- ✓ In the eye diagram, the retina is at the back and the blind spot is where the optic nerve leaves.
- ✓ Power answer: always give the sign and the lens type, e.g. −2.5 D, concave.

REMEMBER IN ONE LINE

- ✓ Myopia: image in front, concave. Hypermetropia: image behind, convex.

5 PRACTICE MCQS

1. The least distance of distinct vision for a normal young adult is about: NCERT ★★★
(a) 25 m (b) 2.5 cm
(c) 25 cm (d) 2.5 m

2. The change in focal length of the eye lens is caused by the action of the: NCERT ★★★
(a) pupil (b) retina
(c) ciliary muscles (d) iris

3. The human eye forms the image of an object at its: NCERT ★★☆
(a) cornea (b) iris
(c) pupil (d) retina

4. The eye focuses on objects at different distances by adjusting the focal length of its lens. This is due to: NCERT ★★★
(a) presbyopia (b) accommodation
(c) near-sightedness (d) far-sightedness

5. The size of the pupil is controlled by the: ★★☆
(a) cornea (b) iris
(c) retina (d) optic nerve

6. A student sees far objects clearly but not a book held close. Defect and lens: ★★★
(a) myopia, concave (b) myopia, convex
(c) hypermetropia, convex (d) hypermetropia, concave

7. The cells of the retina that respond to colour are the: ★★☆
(a) rods (b) cones
(c) ciliary cells (d) iris cells

8. A myopic person has a far point of 2 m. The power of the correcting lens is: ★★★
(a) +2 D (b) −2 D
(c) +0.5 D (d) −0.5 D

9. **Assertion (A):** A myopic person uses spectacles with a concave lens.
Reason (R): A concave lens diverges the rays from a distant object so that the image forms on the retina. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The far point of a normal eye is at infinity.
Reason (R): The eye lens is thickest when we look at distant objects. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

6 SHORT AND LONG ANSWER QUESTIONS

1. Draw a labelled diagram of the human eye. Give the functions of the iris and the pupil. **(3 marks)**
2. A student cannot read the blackboard from the last row. Name the defect, give two causes, and draw ray diagrams of the defect and its correction. **(5 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Near point of a normal eye: 25 cm.
2	(c)	Ciliary muscles change the curvature of the lens.
3	(d)	The image forms on the retina.
4	(b)	Adjusting focal length = accommodation.
5	(b)	The iris controls the size of the pupil.
6	(c)	Near point has moved away: convex lens.
7	(b)	Cones detect colour; rods work in dim light.
8	(d)	f = −2 m, so P = 1/f = −0.5 D.
9	(a)	R correctly explains A.
10	(c)	A true; R false: the lens is thinnest for distant objects.

Model answers: short and long questions

11 Eye diagram, iris and pupil

- Diagram: as in Section 1, with 8 labels.
- Iris: controls the size of the pupil. Pupil: lets light in; it becomes small in bright light and wide in dim light.

12 Blackboard from the last row: myopia

- Image of a distant object forms in front of the retina.
- Causes: (i) excessive curvature of the eye lens; (ii) elongation of the eyeball.
- Correction: concave lens of suitable power. Diagrams: as in Section 3.

NCERT quick Q&A (continued)

- **Q.** What happens to the image distance in the eye when the object moves farther away? **A.** It does not change: the image stays on the retina; only the focal length of the eye lens changes.
- **Q.** Why do older people need bifocal lenses? **A.** Presbyopia: weak ciliary muscles and a stiff lens, so they may need correction for both near and far vision.

Day 28 hit-list: Eduspark PYQ analysis

Topic	Rating
Myopia: causes, correction, ray diagram	★★★
Hypermetropia: causes, correction, ray diagram	★★★
Power of accommodation, near and far point	★★★
Labelled diagram of the eye	★★☆
Presbyopia and cataract	★★☆
Power of correcting lens numericals	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 6 board and sample-paper questions.

Did you know? Eye donation (NCERT)

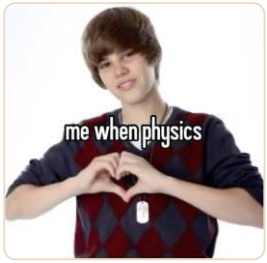
- ✔ People of any age, and even those who wear spectacles or had cataract surgery, can donate their eyes.
- ✔ Eyes must be removed within 4 to 6 hours after death.
- ✔ One pair of eyes can give vision to up to four people with corneal blindness.

Day 28 checklist: tick before you sleep

- ☐ I can draw and label the human eye with 8 parts
- ☐ I can explain accommodation, near point and far point
- ☐ I can draw the ray diagrams of myopia and hypermetropia and their correction
- ☐ I solved all four numericals with the Class Trick steps
- ☐ I know presbyopia, cataract and bifocal lenses

My score: ____ / 5

Physics ka eye chapter samajh aaya toh dil se pyaar ho gaya. 5/5 = Jalwa!



LEVEL 28 COMPLETE +125 XP earned. Next: Level 29



BUJHO *Samajh gaya! Myopia mein concave, hypermetropia mein convex, aur 1/∞ = 0!*

PAHELI *Shabaash! Kal Level 29: prism, rainbow, blue sky aur Chapter 6 ka Boss.*

Try yourself: two quick numericals

1. A myopic person's far point is 2.5 m. Find the power of the lens needed. **[Answer: −0.4 D, concave]**
2. A person's near point is 50 cm. Find the power of the lens needed to read at 25 cm. **[Answer: +2 D, convex]**

Bonus rapid-fire round: quick answers

Question	Answer
Where does most refraction happen?	Outer surface of the cornea
Part that gives the eye its colour	Iris
Screen of the eye	Retina
Point with no vision	Blind spot
Lens for presbyopia	Bifocal
Normal range of vision	25 cm to infinity

Coming up: Day 29

The Colourful World: prism, dispersion, rainbow, twinkling stars, scattering of light, and the Chapter 6 Boss Level.



NITESH SIR SAYS

Aankh ka diagram roz ek baar banao, aur numericals mein pehle defect pehchano. ***Kal milte hai Day 29 pe!***

DAY 29 · LEVEL 29

PHYSICS | CHAPTER 6 | PART 2 · ASTRA SERIES

COLOURFUL WORLD + BOSS

Goal: prism, rainbow, stars and blue sky, then the boss.



by Nitesh Prakash

01 Prism | 02 Dispersion | 03 Rainbow | 04 Atmospheric Refraction | 05 Scattering | +175 XP today | Level 29 / 60

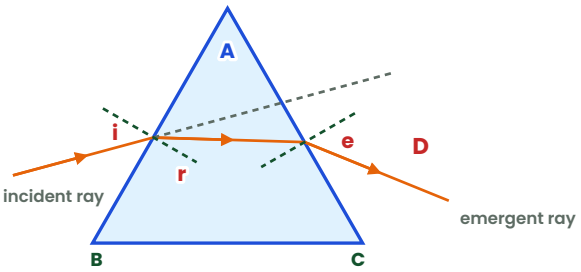
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 REFRACTION THROUGH A GLASS PRISM

DEFINITION

- **Prism:** a transparent block with two triangular bases and three rectangular side faces. The angle between its two refracting faces is the **angle of the prism (A)**.
- **Angle of deviation (D):** the angle between the incident ray produced forward and the emergent ray.

Refraction of light through a glass prism



i = angle of incidence, r = angle of refraction, e = angle of emergence, A = angle of the prism, D = angle of deviation.

KEY POINT

- ✔ Unlike a glass slab, the two refracting faces of a prism are not parallel, so the emergent ray is **not parallel** to the incident ray: it bends towards the **base** of the prism.

2 DISPERSION OF WHITE LIGHT

DEFINITION

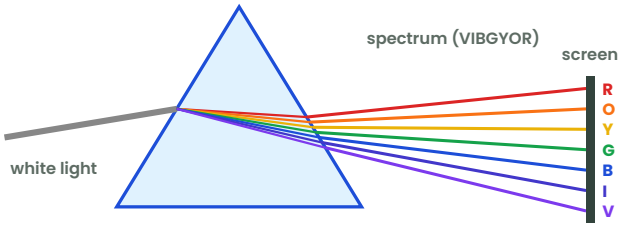
- **Dispersion:** the splitting of white light into its seven component colours.
- **Spectrum:** the band of coloured components of a light beam: **VIBGYOR** (violet, indigo, blue, green, yellow, orange, red).

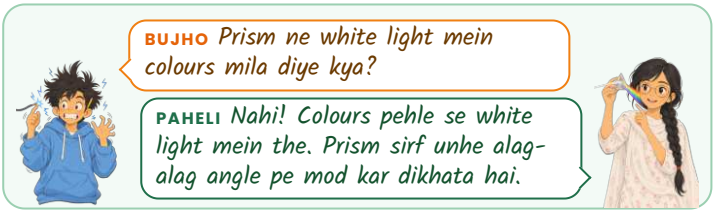


Tesla aur Einstein baad mein aaye, Newton ne pehle hi prism se dikha diya: white light = 7 colours!

Why does dispersion happen? (class notes) ★★★

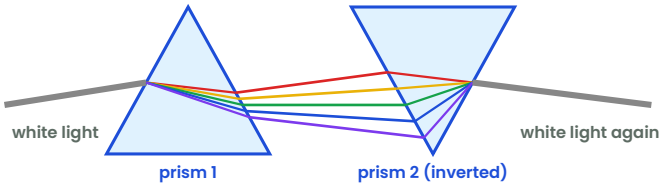
- ✔ All colours travel at the same speed in vacuum, but at **different speeds in glass**, so each colour bends through a different angle.
- ✔ **Red**: longest wavelength, travels fastest in glass, **bends the least** (top of the spectrum).
- ✔ **Violet**: shortest wavelength, travels slowest, **bends the most** (bottom of the spectrum).





Newton's experiment: recombination of the spectrum

- ✔ Newton placed a second, identical prism **upside down** in the path of the spectrum.
- ✔ All seven colours recombined and **white light** came out of the other side.
- ✔ So white light is made of seven colours (class notes: "How can we say white light has 7 colours?").



★★★ *Dispersion ka reason (red sabse kam mudta, violet sabse zyada) aur VIBGYOR: Nitesh Bhaiya ki guarantee!*

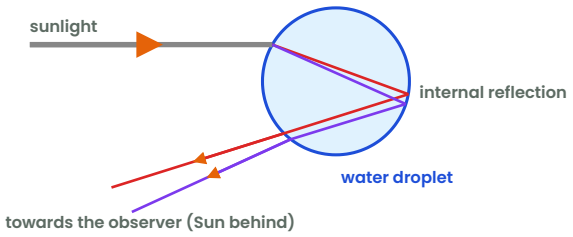
Roses are red kyunki red ki wavelength sabse lambi hai. Violet sabse zyada mudta hai, poetry mein bhi!

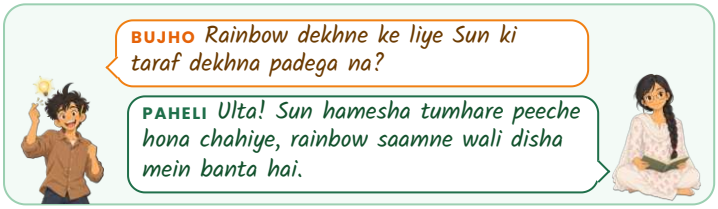


3 RAINBOW: A NATURAL SPECTRUM

How a rainbow forms ★★★

- A rainbow is a natural spectrum seen in the sky after a rain shower, caused by **dispersion of sunlight by tiny water droplets**.
- Each droplet acts like a small prism: it **refracts and disperses** the sunlight, **reflects it internally**, and **refracts it again** as it comes out.
- A rainbow is always seen in the direction **opposite to the Sun** (the Sun is behind you).





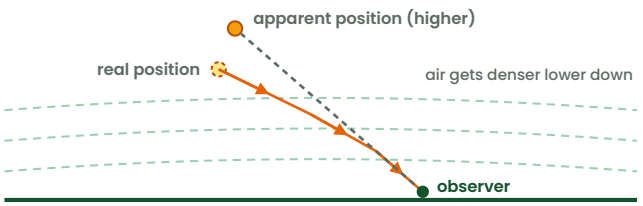
4 ATMOSPHERIC REFRACTION

DEFINITION

- **Atmospheric refraction:** refraction of light by the Earth's atmosphere, whose air layers have different optical densities.
- Example: objects seen through hot air above a fire seem to waver, because the hot air is optically rarer and keeps changing.

Why do stars twinkle? (class notes) ★★★

- Starlight bends towards the normal as it enters denser and denser air, so the star appears **slightly higher** than its real position.
- Stars are far away and act as **point sources**. The atmosphere keeps changing, so the apparent position keeps shifting and the light entering the eye **flickers**: brighter, then fainter. This is twinkling.



★★★ *Twinkling of stars aur planets kyun nahi twinkle karte: har saal ka 3-mark question, Nitesh Bhaiya ki guarantee!*

Why don't planets twinkle?

- Planets are much closer, so they act as **extended sources** (a collection of many point sources).
- The changes in light from all these points **average out to zero**, so there is no twinkling.

Planet ki light steady, star ki light flicker. Point source vs extended source, bas itna yaad rakho!



Advanced sunrise and delayed sunset

- Because of atmospheric refraction, we see the Sun about **2 minutes before** the actual sunrise and about **2 minutes after** the actual sunset.
- Actual sunrise: when the Sun actually crosses the horizon.
- The same effect makes the Sun's disc look **flattened** at sunrise and sunset.

5 SCATTERING OF LIGHT

DEFINITION

- **Scattering of light:** tiny particles (air molecules, dust, droplets) take in light and send it out again in all directions.
- **Tyndall effect:** scattering of light by colloidal particles, which makes the path of a beam of light visible.
- Examples: sunlight entering a smoke-filled room through a small hole; sunlight through the canopy of a dense forest, where tiny water droplets in the mist scatter it.

The colour depends on particle size

- **Very fine particles** scatter mainly blue light (short wavelength).
- **Larger particles** scatter longer wavelengths too; if large enough, the scattered light looks **white** (clouds).

Why is the clear sky blue? ★★★

- Air molecules and fine particles are **smaller than the wavelength** of visible light, so they scatter **blue** (shorter wavelength) much more than **red**.
- Red light has a wavelength about **1.8 times** that of blue light. The scattered blue light enters our eyes, so the sky looks blue.
- **Sky for astronauts:** very high up there is almost no air to scatter light, so the sky looks **dark**.

Sky blue hai kyunki blue hai? Nahi, kyunki short wavelength wala blue sabse zyada scatter hota hai!



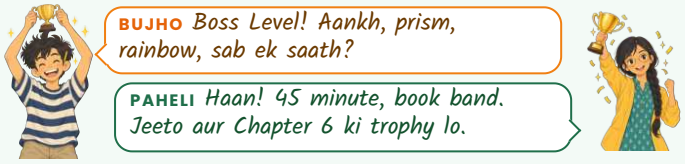
Why are danger signals red?

- ✔ Red light has the longest wavelength, so it is **scattered the least** by fog and smoke.
- ✔ So it travels farthest and is seen clearly from a distance.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Reddish Sun at sunrise and sunset (from class notes): near the horizon, sunlight crosses a thick layer of air, so most blue light is scattered away and mostly red reaches us. **Deep sea looks blue** (class notes): water absorbs red light, and the light scattered back from the water is mainly blue. **Law:** class notes write scattering $\propto 1/\lambda$; for fine particles the correct relation is $\propto 1/\lambda^4$ (Rayleigh scattering). NCERT gives no formula: just remember shorter wavelength, more scattering.

6 CHAPTER 6 BOSS LEVEL



BOSS LEVEL RULES

- ✔ **Time:** 45 minutes. **Maximum marks:** 31. Close the book and set a timer.
- ✔ **A:** 11 MCQs × 1 = 11. **B:** 4 short answers = 11. **C:** 1 long answer = 5. **D:** 1 case = 4.

1. Splitting of white light into its colours is called: ★★★
(a) reflection (b) refraction
(c) dispersion (d) scattering

2. In the spectrum formed by a glass prism, the colour that deviates the least is: ★★★
(a) red (b) yellow
(c) blue (d) violet

3. The clear sky looks blue because: ★★★
(a) air absorbs blue light
(b) fine particles scatter blue light more
(c) the sea reflects blue light
(d) blue has the longest wavelength

4. The twinkling of stars is due to: ★★★
(a) dispersion (b) atmospheric refraction
(c) scattering (d) reflection

5. A rainbow is always seen: ★★☆
(a) towards the Sun (b) opposite to the Sun
(c) only at noon (d) only at night

6. The path of a light beam through a colloid is visible due to: ★★☆
(a) dispersion (b) Tyndall effect
(c) refraction (d) reflection

Tip: in every "why" answer, name the phenomenon first, then give the reason.

7. Danger signal lights are red because red light: ★★★
(a) is scattered the least (b) is scattered the most
(c) bends the most (d) travels the slowest

8. Bifocal lenses are used to correct: ★★☆
(a) myopia (b) hypermetropia
(c) cataract (d) presbyopia

9. The far point of a myopic eye is 1 m. The power of the correcting lens is: ★★☆
(a) +1 D (b) −1 D
(c) +0.5 D (d) −2 D

10. **Assertion (A):** Planets do not twinkle.
Reason (R): Planets are close and act as extended sources, so the changes in their light average out. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

11. **Assertion (A):** The sky appears dark to astronauts at very high altitudes.
Reason (R): Scattering of light is greater at higher altitudes. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

7 SHORT ANSWERS (SECTION B)

12 Draw a labelled diagram of refraction of light through a glass prism. Mark the angles i, e, A and D. **(3 marks)**

13 Why does a star appear slightly higher than its real position? Why do planets not twinkle? **(3 marks)**

14 What is the Tyndall effect? Give two examples from daily life. **(2 marks)**

15 Explain with a labelled diagram how a rainbow is formed. **(3 marks)**

8 LONG ANSWER (SECTION C)

16 (A) What is myopia? Give two causes and show its correction with a ray diagram. (B) A person cannot see beyond 1.5 m clearly. Find the nature and power of the lens needed. **(5 marks)**

9 CASE-BASED (SECTION D)

Case 17: a morning at the hill station (4 marks)

In the morning mist, Aarav sees bright sunbeams streaming through the trees. At noon the sky is deep blue. At night the stars twinkle, but Jupiter shines steadily.

1. Name the effect that makes the sunbeams visible. **(1)**
2. Why is the noon sky blue? **(1)**
3. Why does Jupiter not twinkle? **(1)**
4. Name the phenomenon that makes stars twinkle. **(1)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Splitting of white light = dispersion.
2	(a)	Red has the longest wavelength: least deviation.
3	(b)	Fine particles scatter short-wavelength blue light more.
4	(b)	Changing air layers bend starlight: atmospheric refraction.
5	(b)	The Sun must be behind the observer.
6	(b)	Colloidal particles scatter light: Tyndall effect.
7	(a)	Red is scattered least, so it is seen from far.
8	(d)	Bifocal lenses correct presbyopia.
9	(b)	f = −1 m, so P = 1/f = −1 D.
10	(a)	R correctly explains A.
11	(c)	A true; R false: high up there is almost no air to scatter light.

Model answers: Sections B, C and D

12 Prism diagram

- Draw as in Section 1: incident ray, refracted ray, emergent ray, normals at both faces; mark i, e, A, and D between the incident ray produced and the emergent ray.

13 Star higher, planets steady

- Starlight bends towards the normal as it passes into denser air layers, so the star appears higher than it is.
- Planets are extended sources; the changes in light from their many points average out, so they do not twinkle.

14 Tyndall effect

- Scattering of light by colloidal particles, which makes the path of a light beam visible.
- Examples: a sunbeam in a smoky or dusty room; sunlight through the canopy of a misty forest.

15 Rainbow

- Water droplets act as tiny prisms: refraction and dispersion on entry, internal reflection, refraction again on exit. Seen opposite to the Sun. Diagram: as in Section 3.

16 Myopia (5 marks)

- (A) Distant objects look blurred; image forms in front of the retina. Causes: excessive curvature of the eye lens; elongation of the eyeball. Correction: concave lens (ray diagram as on Day 28).

(B) u = −∞, v = −1.5 m (far point)
1/f = 1/v − 1/u = −1/1.5, so f = −1.5 m
P = 1/f = **−0.67 D**: a concave lens

17 Case: hill station

- (1) Tyndall effect. (2) Air molecules scatter blue light (shorter wavelength) much more than red. (3) Jupiter is a planet: an extended source, so the changes average out. (4) Atmospheric refraction.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 6 board and sample-paper questions.

Score guide: 27+ Jalwa, 20 to 26 good, below 20: revise Days 28 and 29 once more.

Most asked in Chapter 6: Eduspark PYQ analysis

Topic	Day	Rating
Myopia and hypermetropia with diagrams	28	★★★
Twinkling of stars, planets do not twinkle	29	★★★
Dispersion and rainbow formation	29	★★★
Blue sky, danger signals, Tyndall effect	29	★★★
Power of accommodation, near and far point	28	★★☆
Advanced sunrise and delayed sunset	29	★★☆

Day 29 checklist: tick before you sleep

- ☐ I can draw the prism diagram and mark i, e, A and D
- ☐ I can explain dispersion, recombination and the rainbow
- ☐ I can explain twinkling of stars and advanced sunrise
- ☐ I can explain the blue sky, red danger signals and the Tyndall effect
- ☐ I attempted the Boss Level honestly and checked my score

My score: ____ / 5

5/5 = Chapter 6 ka boss khatam! Hooded Kermit ki mat suno.



LEVEL 29 COMPLETE **CHAPTER 6 COMPLETE**

+175 XP earned. Next: Level 30



BUJHO *Samajh gaya! Red sabse kam mudta, blue sabse zyada scatter hota, aur stars point source hain!*

PAHELI *Boss defeated! Kal se Chapter 7: Electricity, current ka khel.*



BUJHO *Sunrise se pehle hi Sun dikh gaya, ye jaadu hai kya?*

PAHELI *Jaadu nahi, atmospheric refraction! Sun horizon ke neeche hota hai, phir bhi 2 minute pehle dikh jaata hai.*



Coming up: Day 30

Electricity begins: electric charge, current, potential difference and how a circuit works.



NITESH SIR SAYS

Aankh aur aasmaan dono samajh aa gaye! Galat answers ka chapter dobara padho. ***Consistency hi jalwa hai!***

PHYSICS | CHAPTER 6 COMPLETE ASTRA SERIES

CHAPTER 6 SWAHAA!

The Human Eye and the Colourful World | Days 28–29



BUJHO *Myopia, hypermetropia, prism, rainbow, blue sky, sab samajh aa gaya! Ab chashma bhi physics lagta hai.*

PAHELI *Bilkul sahi, Bujho! Chapter 6 Swahaa. Next stop: Electricity, current aur circuits ki duniya.*

CHAPTER XP EARNED
+300 XP

LEVELS CLEARED
2 levels (Days 28–29)

BOOK PROGRESS
Level 29 / 60 3700 XP total

WHAT YOU HAVE MASTERED

- ✔ Structure of the human eye and its parts
- ✔ Myopia, hypermetropia, presbyopia, cataract
- ✔ Rainbow and atmospheric refraction
- ✔ Power of accommodation, near and far point
- ✔ Refraction through a prism, dispersion
- ✔ Scattering of light, Tyndall effect, blue sky

NEXT CHAPTER Day 30: Electricity

Human Eye cleared!

DAY 30 · LEVEL 30

PHYSICS | CHAPTER 7 | PART 1 · ASTRA SERIES

CURRENT & POTENTIAL DIFFERENCE

Goal: master charge, current, volt and circuit symbols today.



by Nitesh Prakash

01 Charge | 02 Current | 03 Circuit | 04 Potential Difference | 05 Symbols

+100 XP today | Level 30 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 ELECTRIC CHARGE

DEFINITION

- **Electric charge:** the property of matter that produces **electric and magnetic effects**
- Two kinds: **positive** (carried by protons) and **negative** (carried by electrons)
- Charge of one electron, **$e = 1.6 \times 10^{-19}$ C** (negative)
- SI unit: **coulomb (C)**; symbol Q or q; charge is a **scalar**

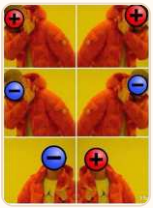
Particle	Charge	Where it sits
Electron	-1.6×10^{-19} C	outside the nucleus, free to move in metals
Proton	$+1.6 \times 10^{-19}$ C	inside the nucleus, fixed

How a body gets charged

- ✔ A body that **loses electrons** becomes **positively** charged
- ✔ A body that **gains electrons** becomes **negatively** charged
- ✔ Only electrons move; protons stay locked in the nucleus
- ✔ Like charges repel; unlike charges attract

CLASS TRICK: WHO TURNS POSITIVE?

"Dene wala hamesha bada hota hai." Jo electron **deta** hai (loses) wo **positive** ban jata hai; jo leta hai wo negative.



MEME BREAK

*Plus-plus? Nahi. Minus-minus? Nahi. **Plus-minus? Haan!** Unlike charges hi ek doosre ko attract karte hain.*

EXTRA KNOWLEDGE (HIGH LEVEL) properties of charge, from class notes

- **Additivity:** total charge = algebraic sum, e.g. +3 C and −1 C give +2 C
- **Conservation:** charge is neither created nor destroyed, only transferred
- **Invariance:** charge does not change with speed (you will see this in Class 12)
- **Quantisation:** Q = ne, where n is a whole number; no body has a charge smaller than e

2 ELECTRIC CURRENT ★★★ MOST ASKED

DEFINITION

- **Electric current:** the **rate of flow of charge** through any cross-section of a conductor
- **I = Q / t** (Q = net charge, t = time in seconds)
- SI unit: **ampere (A)**, named after André-Marie Ampère; 1 A = 1 C/s

One ampere and smaller units

- **1 ampere:** current when **1 coulomb** of charge flows through a cross-section in **1 second**
- 1 mA (milliampere) = 10^{-3} A; 1 μA (microampere) = 10^{-6} A
- Current is measured by an **ammeter**, always joined **in series**
- A **galvanometer** detects small currents and shows their direction



★★★ *Q = It wale sawal mein minute ko second aur mA ko A mein badalna mat bhoolna. Nitesh Bhaiya ki guarantee!*

Solved Example 1: Electrons in one coulomb ★★★

Given	Q = 1 C, $e = 1.6 \times 10^{-19}$ C
Formula	Q = ne, so n = Q / e
Solve	$n = 1 / (1.6 \times 10^{-19}) = 0.625 \times 10^{19}$
Answer	**$n = 6.25 \times 10^{18}$ electrons** make up 1 C

Solved Example 2: Charge from current ★★★

Given	I = 0.5 A, t = 10 min = 600 s (NCERT)
Formula	Q = I × t
Solve	Q = 0.5 × 600
Answer	**Q = 300 C** flows through the filament

Class question: how do metals conduct?

Metals have a large number of **free electrons** that are loosely held. When a cell is connected, these electrons drift through the metal and carry charge, so metals are good conductors.

REMEMBER: UNITS FIRST

Convert time to **seconds** and current to **amperes** before using Q = It. 2 min = 120 s; 250 mA = 0.25 A.

3 DIRECTION OF CURRENT ★★☆

	Conventional current	Electron flow
Outside the cell	**+ terminal → − terminal**	− terminal → + terminal
Fixed by	agreed before electrons were discovered	actual motion of free electrons
Used in	all circuit diagrams and answers	explaining conduction in metals



MEME BREAK
*Pata tha current + se − chalta hai, phir bhi arrow ulta bana diya? **Diagram ke marks gaye!** Conventional current hamesha + se −, electrons − se +.*

Why do electrons move at all?

- Without a cell, free electrons move **randomly** in all directions, so there is **no net flow** and no current
- A cell sets up an **electric force** across the wire; the electrons now **drift** towards the + terminal
- This steady drift of charge is the electric current

BUJHO *Electron − se + ki taraf jaata hai, toh current bhi − se + hona chahiye na?*

PAHELI *Current ki direction electrons discover hone se pehle tay ho gayi thi: **+ se −**. Electrons ulta chalte hain. Diagram mein hamesha conventional current hi dikhao.*

4 ELECTRIC CIRCUIT

DEFINITION

- **Electric circuit:** a **continuous and closed path** of an electric current
- Parts: a **source** (cell or battery), **connecting wires**, a **switch or key** and a **device** such as a bulb

Closed vs open circuit

- ✔ **Closed circuit:** key ON, path complete, current flows, bulb glows
- ✔ **Open circuit:** key OFF or wire broken, path incomplete, **no current**
- ✔ The switch only completes or breaks the path; it does not make charge

Solved Example 5: Counting electrons ★★☆

Given	I = 1.5 A, t = 2 min = 120 s, e = 1.6×10^{-19} C
Step 1	Q = It = 1.5 × 120 = 180 C
Step 2	n = Q / e = 180 / (1.6×10^{-19})
Answer	**n = 1.125 × 10^{21} electrons** pass in 2 min

5 POTENTIAL DIFFERENCE ★★★ MOST ASKED

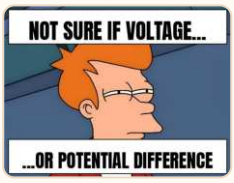
DEFINITION

- **Potential difference (V)** between two points: **work done** to move a **unit positive charge** from one point to the other
- **V = W / Q**; SI unit **volt (V)** = joule per coulomb (J/C)
- Another name for potential difference is **voltage**

One volt and the voltmeter

- **1 volt:** PD between two points when **1 joule** of work is done to move **1 coulomb** of charge between them
- Measured by a **voltmeter**, joined **in parallel** across the two points; it has a **high resistance**
- A cell keeps up the PD by the **chemical reaction** inside it; this PD drives the current
- Like water flows from a higher level to a lower level, current flows from **higher to lower potential**

CLASS TRICK: VOLTAGE = POTENTIAL DIFFERENCE
"Gangadhar hi Shaktiman hain, aur Shaktiman hi Gangadhar hain!" Voltage hi potential difference hai, aur potential difference hi voltage. Naam do, cheez ek.



MEME BREAK
*Fry confuse mat ho: **voltage aur potential difference ek hi cheez** hain.*

Solved Example 3: Work done on a charge ★★★

Given	Q = 2 C, V = 12 V (NCERT)
Formula	V = W / Q, so W = V × Q
Solve	W = 12 × 2
Answer	**W = 24 J** of work is done

Solved Example 4: Energy given to each coulomb ★★☆

Given	Battery of 6 V; Q = 1 C (NCERT)
Formula	W = V × Q
Solve	W = 6 × 1
Answer	**6 J** of energy is given to every coulomb

EXTRA KNOWLEDGE (HIGH LEVEL) beyond the Class 10 syllabus
Electric potential at a point: the work done to bring a **unit positive charge** from **infinity** to that point. Its unit is also the volt. The difference of the potentials of two points is the potential difference you use in Class 10.

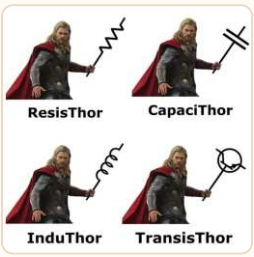
REMEMBER: THREE FORMS
V = W / Q, so W = V × Q and Q = W / V.

EXAM TIP: VOLTMETER AND AMMETER
Ammeter: **series**, very low resistance. Voltmeter: **parallel**, very high resistance.

6 CIRCUIT SYMBOLS ★★★ MOST ASKED

Symbols you must draw neatly

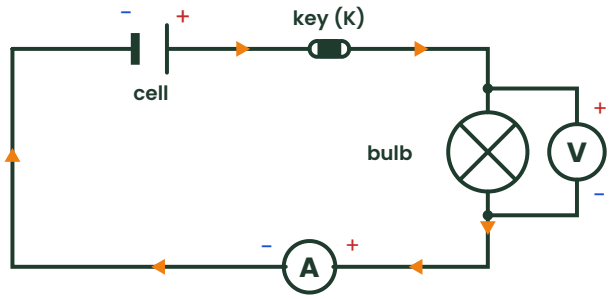
Electric cell (long plate +)	Battery: combination of cells
Plug key or switch, open (OFF)	Plug key or switch, closed (ON)
Wire joint (wires connected)	Wires crossing without joining
Electric bulb	Resistor of resistance R
Variable resistance (rheostat)	Ammeter (in series)
Voltmeter (in parallel)	Galvanometer (detects small current)



MEME BREAK

Thor ka hammer bhi symbol se pehchana jata hai! Resistor zig-zag, cell lambi-chhoti line: ***symbol galat toh diagram ke marks gaye.***

Circuit diagram: measuring I and V for a bulb ★★★



conventional current: + terminal → circuit → − terminal

Ammeter in series with the bulb; voltmeter across (in parallel with) the bulb; long plate of the cell = + terminal.

Ammeter vs voltmeter

	Ammeter	Voltmeter
Measures	current (A)	potential difference (V)
Joined	**in series**	**in parallel**
Resistance	very low	very high
+ terminal	towards the + terminal of the cell	

TRY YOURSELF: QUICK NUMERICALS

(i) 2 A flows for 5 min. Find Q. (ii) 96 C flows in 1 min. Find I.
(iii) 10 J moves 2 C. Find V. (iv) Write 3.2 mA in amperes.

Answers are on page 4, with the answer key.

× WRONG	✓ RIGHT
Ammeter in parallel with the bulb	ammeter always in series
Current flows from − to + in the circuit	conventional current: + to −
Q = It with t in minutes	convert t to seconds first
1 V = 1 C/J	1 V = 1 J/C (joule per coulomb)



BUJHO *Ammeter ko bhi bulb ke parallel laga doon, voltmeter ki tarah?*

PAHELI *Nahi! Ammeter ka resistance bahut kam hota hai. Parallel lagaya toh saara current usi mein bhaagega.* ***Ammeter series mein, voltmeter parallel mein.***





★★★ ***Circuit diagram mein ammeter series, voltmeter parallel aur + terminal sahi taraf: ye 2 marks pakke hain, Nitesh Bhaiya ki guarantee!***

Day 30 mind map: revise in 2 minutes

CHARGE
e = 1.6 × 10^{-19} C; lose e^- → +, gain e^- → −

CURRENT ★★★
I = Q/t; 1 A = 1 C/s; ammeter in series

CURRENT & POTENTIAL DIFFERENCE

VOLTAGE ★★★
V = W/Q; 1 V = 1 J/C; voltmeter in parallel

DIRECTION
current + to −; electrons − to +

CIRCUIT
closed path: source + wires + key + device; 6.25 × 10^{18} electrons = 1 C



BUJHO *Battery current banati hai na, isliye current khatam ho jata hai?*

PAHELI *Battery* ***potential difference*** *banati hai, chemical reaction se. Wahi PD charges ko dhakka deta hai aur current behta hai. Reaction ka saman khatam, toh cell khatam.*



Day 30 hit-list: Eduspark PYQ analysis

Topic	Rating
Define current and 1 ampere; Q = It numericals	★★★
Define potential difference and 1 volt; W = VQ	★★★
Circuit diagram with ammeter and voltmeter	★★★
Electrons in 1 C (6.25 × 10^{18})	★★☆
Symbols: cell, key, rheostat, wires crossing	★★☆
Direction of current vs electron flow	★☆☆

7 PRACTICE MCQS

1. The SI unit of electric charge is: ★★☆
(a) ampere (b) volt
(c) coulomb (d) ohm

2. One ampere is equal to: ★★★
(a) 1 J/C (b) 1 C/s
(c) 1 C × s (d) 1 J/s

3. The number of electrons that make up 1 C of charge is: ★★★
(a) 1.6×10^{19} (b) 6.25×10^{18}
(c) 1.6×10^{-19} (d) 6.25×10^{-18}

4. An ammeter is always connected in a circuit: ★★★
(a) in series (b) in parallel
(c) across the cell (d) either way

5. The unit joule per coulomb is the same as: ★★☆
(a) ampere (b) watt
(c) ohm (d) volt

6. The work done in moving 2 C of charge across a potential difference of 6 V is: ★★☆
(a) 3 J (b) 8 J
(c) 12 J (d) 4 J

7. A body has a charge of $+3.2 \times 10^{-19}$ C. It has: ★☆☆
(a) gained 2 electrons (b) lost 2 electrons
(c) lost 3 electrons (d) gained 3 protons

8. A current of 500 mA, expressed in amperes, is: ★☆☆
(a) 5 A (b) 0.05 A
(c) 0.5 A (d) 50 A

9. **Assertion (A):** A body that loses electrons becomes positively charged.
Reason (R): An electron carries a negative charge. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Electric current in a metal wire is due to the flow of electrons.
Reason (R): Conventional current flows from the negative to the positive terminal outside the cell. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Define electric current and state its SI unit. A current of 0.5 A is drawn by a bulb for 10 minutes. Find the charge that flows. **(3 marks)**
2. Define potential difference and 1 volt. How much work is done in moving a charge of 2 C across two points with a potential difference of 12 V? **(3 marks)**
3. Draw a circuit with a cell, a plug key, a bulb and an ammeter, and show a voltmeter that measures the potential difference across the bulb. **(3 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Charge is measured in coulomb.
2	(b)	I = Q/t, so 1 A = 1 C/s.
3	(b)	$n = 1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$.
4	(a)	Series, so the full current passes through it.
5	(d)	V = W/Q, so 1 V = 1 J/C.
6	(c)	W = VQ = 6 × 2 = 12 J.
7	(b)	$n = 3.2 \times 10^{-19} / 1.6 \times 10^{-19} = 2$, lost.
8	(c)	500 mA = 500×10^{-3} A = 0.5 A.
9	(a)	Losing negative electrons leaves extra positive charge.
10	(c)	A is true; conventional current flows + to −.

Model answers: short and long answer

1 Electric current
Rate of flow of charge, I = Q/t; SI unit ampere. Q = It = 0.5 × 600 s = **300 C**.

2 Potential difference
Work done to move a unit positive charge from one point to another, V = W/Q. 1 V = 1 J/C. W = VQ = 12 × 2 = **24 J**.

3 Circuit diagram
As on page 3: ammeter in series, voltmeter across the bulb, + terminals towards the cell's + terminal.

Try-yourself answers: (i) 600 C (ii) 1.6 A (iii) 5 V (iv) 3.2×10^{-3} A

Day 30 checklist: tick before you sleep

- ☐ I can define charge, current and potential difference with their SI units
- ☐ I can use Q = It and W = VQ with correct units
- ☐ I know 1 C = 6.25×10^{18} electrons and e = 1.6×10^{-19} C
- ☐ I can draw all circuit symbols and a circuit with ammeter and voltmeter
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 30 COMPLETE +100 XP earned. Next: Level 31



BUJHO *Samajh gaya! Charge behta hai toh current, aur use dhakka deta hai potential difference.*

PAHELI *Shabaash! Kal Level 31: Ohm's law aur resistance.*



NITESH SIR SAYS
Bas do formula: Q = It aur V = W/Q. Par units pe dhyan do: minute ko second, mA ko A. ***Kal milte hai Day 31 pe!***

DAY 31 · LEVEL 31

PHYSICS | CHAPTER 7 | PART 2 · ASTRA SERIES

OHM'S LAW & RESISTIVITY

Goal: use V = IR and R = ρl/A with full confidence.



by Nitesh Prakash

01 Ohm's Law | 02 V–I Graph | 03 Resistance | 04 Resistivity | 05 Materials | +100 XP today | Level 31 / 60

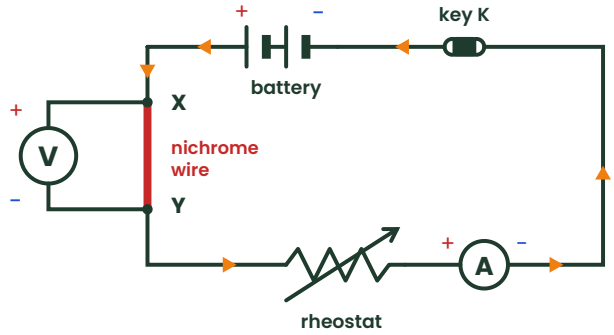
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 OHM'S LAW ★★★ MOST ASKED

DEFINITION

- **Ohm's law:** the potential difference V across a metallic conductor is **directly proportional** to the current I through it, provided its **temperature remains the same**
- V ∝ I, so **V = IR**, where R is a constant called the **resistance**
- Stated by the German physicist **Georg Simon Ohm** in 1827

Activity: verifying Ohm's law ★★☆



- Close the key; note the ammeter (I) and voltmeter (V) readings
- Change the current with the rheostat or by adding cells; take 4 to 5 readings
- V/I comes out the **same** every time: that constant is R of the wire



MEME BREAK

"OHMMMM..." Resistance ke monks bhi Ohm ka naam japte hain! Unit yaad rakho: ***ohm (Ω).***

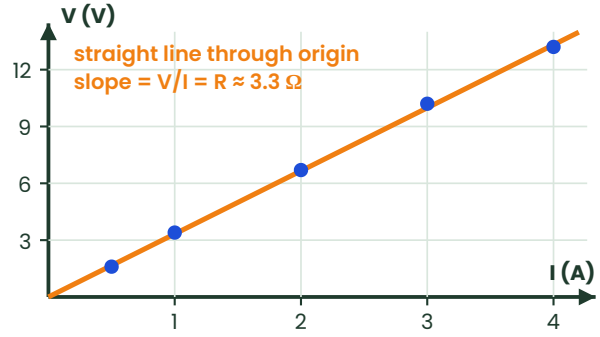
BUJHO *Ohm's law toh har cheez pe lagta hoga, bulb ke filament pe bhi?*



PAHELI *Ohm's law tab tak sahi hai jab tak* ***temperature same*** *rahe. Filament bahut garam hota hai, uska resistance badh jata hai, isliye uska V–I graph poori tarah seedha nahi rehta.*



V–I graph: a straight line ★★★



I (A)	0.5	1.0	2.0	3.0	4.0
V (V)	1.6	3.4	6.7	10.2	13.2
V/I (Ω)	3.2	3.4	3.35	3.4	3.3

Data from the NCERT exercise (nichrome wire). The graph is a straight line through the origin, so V ∝ I.

2 RESISTANCE ★★★ MOST ASKED

DEFINITION

- **Resistance:** the property of a conductor to **oppose the flow of current** through it
- **R = V / I**; SI unit **ohm (Ω)**; measured by an **ohmmeter**
- **1 ohm:** resistance of a conductor when a PD of **1 V** across it drives a current of **1 A**; 1 Ω = 1 V/A

What R tells you

- ✔ For a given V, current is **inversely proportional** to R: double R, half the current
- ✔ A **rheostat** (variable resistance) changes R to control the current
- ✔ Good conductors have low R; insulators have very high R

Solved Example 1: Current through a bulb ★★★

Given	R = 1200 Ω, V = 220 V (NCERT)
Formula	I = V / R
Solve	I = 220 / 1200 = 0.183 A
Answer	**I ≈ 0.18 A** flows through the filament

3 RESISTANCE FACTORS ★★★ MOST ASKED

Factor	Relation	Example
Length l	R ∝ l	double the length → R doubles
Thickness (area A)	R ∝ 1/A	double the area → R halves
Temperature	R rises with temperature (metals)	a hot filament has more R
Nature of material	set by ρ	alloy (nichrome) > its metals

Why each factor works

- **Longer wire:** electrons travel a longer path and collide more, so R increases
- **Thicker wire:** more room for electrons to flow, so R decreases; current flows more easily through a thick wire
- **Hotter wire:** atoms vibrate more and block electrons, so R of a metal increases
- **Alloys:** have higher resistance than the pure metals they are made of, e.g. nichrome



★★★ *Ohm's law ka statement "temperature same rahe" ke saath likhna, aur V–I graph seedhi line origin se. Nitesh Bhaiya ki guarantee!*



BUJHO *Mota taar ho ya patla, current toh same hi behta hoga na?*

PAHELI *Nahi! Mote taar ka area zyada, toh resistance kam. Isliye same voltage pe **mote taar mein current zyada** behta hai. R ∝ 1/A yaad rakho.*





MEME BREAK

*Resistance lambai-motai se badalta hai, par **resistivity sirf material aur temperature** pe depend karti hai. Lambai se ρ badalna = Patrick logic!*

Solved Example 2: Resistance of a wire (class notes) ★★☆

Given	l = 2 m, A = 1.55×10^{-6} m², ρ = 2.8×10^{-8} Ω m
Formula	R = ρl / A
Solve	R = $(2.8 \times 10^{-8} \times 2) / (1.55 \times 10^{-6}) = 3.61 \times 10^{-2}$
Answer	**R ≈ 0.036 Ω**

EXAM SHORTCUT: STRETCHED WIRE

If a wire is stretched to **n times** its length (same volume), its resistance becomes **n^2 times**. Double the length → 4R.

4 RESISTIVITY ★★★ MOST ASKED

Derivation: R = ρl/A ★★★

- R ∝ l (length) and R ∝ 1/A (area)
- Combining: R ∝ l/A, so **R = ρ l / A**
- ρ (rho) is a constant for the material at a given temperature: the **resistivity**
- Put l = 1 m and A = 1 m²: then **ρ = R**

DEFINITION

- **Resistivity (ρ):** the resistance of a conductor of **unit length** and **unit area of cross-section**
- ρ = RA / l; SI unit **ohm metre (Ω m)**
- **1 Ω m:** resistivity of a material when a conductor of it, 1 m long with 1 m² cross-section, has a resistance of 1 Ω

Resistivity: the key facts

- ✔ Depends only on the **material** and the **temperature**, not on length or thickness
- ✔ Metals and alloys: very low ρ, 10^{-8} to 10^{-6} Ω m; insulators: 10^{12} to 10^{17} Ω m
- ✔ Resistivity of an alloy is generally **higher** than that of its constituent metals

Solved Example 3: Length and radius both doubled (class notes) ★★★

Given	l′ = 2l, r′ = 2r, so A′ = π(2r)² = 4A
Formula	R = ρl/A
Solve	R′ = ρ(2l)/(4A) = ½ × ρl/A
Answer	**R′ = R/2**: the resistance becomes half

Solved Example 4: Wire stretched to double length (class notes) ★★★

Given	R = 4 Ω; volume stays the same, so l′ = 2l gives A′ = A/2
Formula	R = ρl/A
Solve	R′ = ρ(2l)/(A/2) = 4 × ρl/A = 4R
Answer	**R′ = 4 × 4 = 16 Ω**

Solved Example 5: Finding resistivity ★★★

Given	l = 1 m, R = 26 Ω, d = 0.3 mm, so r = 0.15×10^{-3} m (NCERT)
Area	A = πr² = $3.14 \times (0.15 \times 10^{-3})^2 = 7.07 \times 10^{-8}$ m²
Solve	ρ = RA / l = $26 \times 7.07 \times 10^{-8} / 1$
Answer	**ρ ≈ 1.84×10^{-6} Ω m** (manganese, from the table)

Solved Example 6: Length needed for 10 Ω ★★☆

Given	R = 10 Ω, d = 0.01 cm, so r = 5×10^{-5} m; ρ = 50×10^{-8} Ω m (NCERT)
Area	A = πr² = 7.85×10^{-9} m²
Solve	l = RA / ρ = $10 \times 7.85 \times 10^{-9} / (5 \times 10^{-7})$
Answer	**l ≈ 0.157 m**; if the diameter is doubled, R becomes one-fourth

5 CONDUCTORS, ALLOYS AND INSULATORS ★★☆

Resistivity of some materials (NCERT table) ★★☆

	Material	ρ (Ω m) at 20 °C
Conductors	Silver	1.60×10^{-8}
	Copper	1.62×10^{-8}
	Aluminium	2.63×10^{-8}
	Tungsten	5.20×10^{-8}
	Nickel	6.84×10^{-8}
	Iron	10.0×10^{-8}
	Chromium	12.9×10^{-8}
	Mercury	94.0×10^{-8}
	Manganese	1.84×10^{-6}
Alloys	Constantan (Cu, Ni)	49×10^{-6}
	Manganin (Cu, Mn, Ni)	44×10^{-6}
	Nichrome (Ni, Cr, Mn, Fe)	100×10^{-6}
Insulators	Glass	10^{10} to 10^{14}
	Hard rubber	10^{13} to 10^{16}
	Ebonite	10^{15} to 10^{17}
	Diamond	10^{12} to 10^{13}
	Paper (dry)	10^{12}

Choosing the right material

- **Heating elements** (iron, toaster, heater): **nichrome**, because alloys have high resistivity and do not oxidise (burn) easily at high temperature
- **Bulb filament: tungsten**, as it has a very high melting point (3380 °C)
- **Connecting and transmission wires: copper and aluminium**, because of their very low resistivity
- **Best conductor:** silver (lowest ρ), but it is too costly for wiring



MEME BREAK

*Heater coil ke liye copper bola? Seedha window ke bahar! **Nichrome** chahiye: high resistivity, aur garam hokar jalta nahi.*

Solved Example 7: Heater at two voltages ★★★

Given	I = 4 A at V = 60 V (NCERT)
Step 1	R = V / I = 60 / 4 = 15 Ω
Step 2	At 120 V: I = V / R = 120 / 15
Answer	**I = 8 A**: double the voltage, double the current

CLASS QUESTION: IRON OR MERCURY?

Iron (ρ = 10.0×10^{-8} Ω m) conducts better than mercury (94.0×10^{-8} Ω m).

× WRONG	✓ RIGHT
Resistivity changes with length	ρ depends only on material and temperature
Unit of resistivity: Ω/m	unit is Ω m (ohm metre)
Thick wire has more resistance	thick wire: larger A, less R
Ohm's law holds at any temperature	only when temperature stays same

Solved Example 8: Unknown resistor ★★☆

Given	V = 12 V, I = 2.5 mA = 2.5×10^{-3} A (NCERT)
Formula	R = V / I
Solve	R = 12 / (2.5×10^{-3})
Answer	**R = 4800 Ω = 4.8 kΩ**

Exam tips

- ✔ Draw the V–I graph through the **origin**, with V on the y-axis
- ✔ In R = ρl/A, convert mm and cm to **metres**; A = πr^2 uses the **radius**
- ✔ For a stretched wire, remember the area also changes

Day 31 mind map: revise in 2 minutes

OHM'S LAW ★★★
V ∝ I at constant temperature; V = IR

RESISTANCE
R = V/I; ohm (Ω); 1 Ω = 1 V/A

OHM'S LAW & RESISTIVITY

FACTORS ★★★
R ∝ l, R ∝ 1/A, rises with temperature, depends on material

RESISTIVITY
R = ρl/A; Ω m; only material and temperature

MATERIALS
nichrome for heaters, tungsten for filaments, copper and aluminium for wires

TRY YOURSELF: QUICK NUMERICALS

(i) V = 6 V, R = 3 Ω. Find I.
(ii) I = 0.5 A, R = 20 Ω. Find V.
(iii) Length halved: new R?
(iv) Area tripled: new R?

Answers are on page 4, with the answer key.

Short and long answer questions

1. State Ohm's law. Draw the circuit used to verify it and sketch the V–I graph you expect. **(5 marks)**
2. Name the factors on which the resistance of a conductor depends. How does the resistance change when both the length and the radius of a wire are doubled? **(3 marks)**
3. Why are the coils of electric toasters and irons made of an alloy rather than a pure metal? **(2 marks)**

REMEMBER

Silver: best conductor. Tungsten melts at 3380 °C.

6 PRACTICE MCQS



MCQ TIME
Pehle khud solve karo, phir answer key dekho. ***Seedha key dekhna mana hai!***

1. The SI unit of resistance is: ★★☆
(a) volt (b) ampere
(c) ohm (d) ohm metre

2. The V–I graph of a metallic wire at constant temperature is: ★★★
(a) a curve (b) a straight line through the origin
(c) parallel to the I-axis (d) a parabola

3. If the length of a wire is doubled, its resistance becomes: ★★★
(a) half (b) double
(c) four times (d) unchanged

4. The resistivity of a wire depends on: ★★★
(a) its length (b) its area of cross-section
(c) its material (d) the current through it

5. The SI unit of resistivity is: ★★☆
(a) Ω (b) Ω/m
(c) Ω m (d) Ω m^2

6. A wire of resistance 4 Ω is stretched to twice its length. Its new resistance is: ★★★
(a) 8 Ω (b) 2 Ω
(c) 16 Ω (d) 4 Ω

7. A current of 2.5 mA flows when a 12 V battery is connected across a resistor. Its resistance is: ★★☆
(a) 30 Ω (b) 480 Ω
(c) 4800 Ω (d) 0.2 Ω

8. Which material is used for the filament of an electric bulb? ★☆☆
(a) nichrome (b) tungsten
(c) copper (d) aluminium

9. **Assertion (A):** The heating elements of electric irons are made of an alloy.
Reason (R): Alloys have high resistivity and do not oxidise readily at high temperatures. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Resistance of a conductor increases when its area of cross-section is increased.
Reason (R): Resistance is inversely proportional to the area of cross-section. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	R = V/I is measured in ohm.
2	(b)	V ∝ I, so a straight line through the origin.
3	(b)	R ∝ l.
4	(c)	ρ depends on material and temperature only.
5	(c)	ρ = RA/l gives Ω m^2/m = Ω m.
6	(c)	l doubles, A halves: R × 4 = 16 Ω.
7	(c)	R = 12/(2.5 × 10^{-3}) = 4800 Ω.
8	(b)	Tungsten has a very high melting point.
9	(a)	R is exactly why alloys are chosen.
10	(d)	A is false (R decreases); R is true.

Model answers: short and long answer

1 Ohm's law
At constant temperature, V across a metallic wire is directly proportional to I through it: V = IR. Circuit as on page 1 (battery, key, ammeter, rheostat in series with wire XY, voltmeter across XY); the V–I graph is a straight line through the origin.

2 Factors
Length, area of cross-section, temperature and material. A becomes 4A, so R′ = ρ(2l)/(4A) = **R/2**.

3 Alloys in heaters
Alloys have a high resistivity and do not oxidise (burn) readily at high temperature.

Try-yourself answers: (i) 2 A (ii) 10 V (iii) R/2 (iv) R/3

Day 31 checklist: tick before you sleep

- ☐ I can state Ohm's law and draw its circuit and V–I graph
- ☐ I can define resistance, 1 ohm, resistivity and 1 ohm metre
- ☐ I know how length, area, temperature and material change R
- ☐ I can solve stretched-wire and R = ρl/A numericals
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 31 COMPLETE +100 XP earned. Next: Level 32



BUJHO *Samajh gaya! Lamba taar zyada R, mota taar kam R, aur ρ sirf material ka.*

PAHELI *Shabaash! Kal Level 32: resistors in series.*



NITESH SIR SAYS
V = IR aur R = ρl/A, dono formula se 3 se 5 marks pakke. Stretch wale sawal mein area badalna mat bhoolna. ***Kal milte hai Day 32 pe!***

DAY 32 LEVEL 32

PHYSICS | CHAPTER 7 | PART 3 ASTRA SERIES

RESISTORS IN SERIES

Goal: same current, shared voltage, added resistance.



by Nitesh Prakash

01 Series Basics | 02 Same Current | 03 Derivation | 04 Numericals | 05 Drawbacks | +100 XP today | Level 32 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

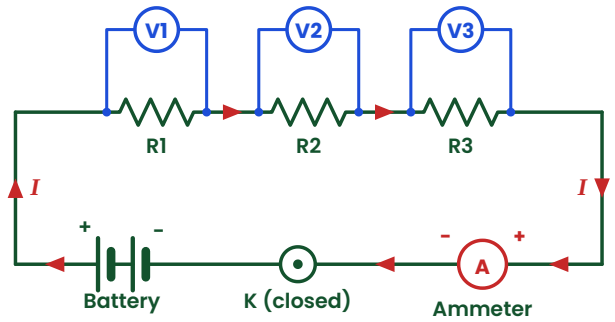
1 SERIES COMBINATION OF RESISTORS

Recap (Day 31): Ohm's law, **V = IR**. Today we join two or more resistors and find one **equivalent resistance** that can replace them all.

DEFINITION

- **Series combination:** resistors joined **end to end**, one after the other, so that charge has **only one path** through them
- The **same current** flows through each resistor when a potential difference is applied across the combination
- **Equivalent (resultant) resistance R_s:** the single resistance that draws the same current from the same battery

Circuit: three resistors in series ★★★



Ammeter in series (anywhere in the loop); each voltmeter in parallel across its resistor. Red arrows: conventional current, + to −.

Where series is used

- ✔ A **fuse** is joined in series with the appliance it protects, so all the current passes through it
- ✔ An **ammeter** is always joined in series
- ✔ **Decorative lights**: small bulbs share the mains voltage; one fused bulb darkens the string

Activity: is the current the same everywhere?

1. Join resistors of 1 Ω, 2 Ω and 3 Ω in series with a battery, a plug key and an ammeter
2. Note the ammeter reading, then shift the ammeter between resistors: **same reading**
3. Put a voltmeter across all three, then across each one: **$V = V_1 + V_2 + V_3$**

2 KEY FACTS OF A SERIES CIRCUIT

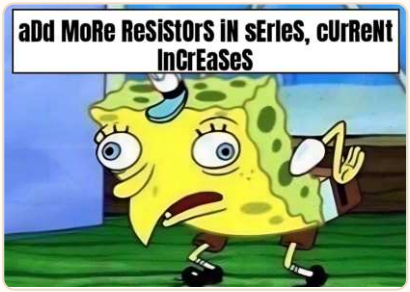
★★★ MOST ASKED

Learn these five lines

- ✔ **Current is the same** in every resistor: $I_1 = I_2 = I_3 = I$. The ammeter reads the same anywhere in the loop.
- ✔ **Potential difference divides:** $V = V_1 + V_2 + V_3$
- ✔ **Resistances add:** $R_s = R_1 + R_2 + R_3$, so R_s is **greater than the largest** single resistance
- ✔ **n equal resistors** R in series: $R_s = nR$
- ✔ **$V \propto R$** (same I): $V_1 : V_2 : V_3 = R_1 : R_2 : R_3$. The bigger resistor takes the bigger share of voltage.

CLASS TRICK: K³B (CLASS KEY POINT)

"Series mein current same rehta hai, voltage divide hota hai." Notes wala derivation: $V = V_1 + V_2 + V_3 \rightarrow IR_s = IR_1 + IR_2 + IR_3 \rightarrow \mathbf{R_s = R_1 + R_2 + R_3}$. Current same hai, isliye I cancel ho jaata hai.



MEME BREAK

*"Series mein resistor badhao, current badhega"? Bilkul nahi! R_s badha toh current **ghatega**: $I = V / R_s$.*



BUJHO *Pehla resistor thoda current kha jaata hoga na? Last wale tak kam pahunchega.*

PAHELI *Bilkul nahi! Current use nahi hota. Series mein ek hi raasta hai, toh har resistor se **same current** guzarta hai. Ammeter kahin bhi lagao, reading same.*





BUJHO *Toh zyada resistor series mein lagaun toh bulb aur tez jalega?*

PAHELI *Ulta! Rs badhta hai, $I = V / R_s$ ghatta hai, isliye bulb **dimmer**. Har resistor ek naya speed breaker.*



3 DERIVATION OF SERIES RESISTANCE

★★★ MOST ASKED

Let resistors R_1, R_2, R_3 be joined in series between points A and B, with a battery of potential difference V. Let I be the current.

Step 1: $V = V_1 + V_2 + V_3$
total potential difference is shared by the three resistors ...(i)

Step 2: $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$
Ohm's law for each resistor; the current I is the same ...(ii)

Step 3: for the equivalent resistor, $V = IR_s$
one resistor R_s drawing the same current I ...(iii)

Step 4: $IR_s = IR_1 + IR_2 + IR_3$
put (ii) and (iii) in (i), then divide both sides by I

$$R_s = R_1 + R_2 + R_3$$

for n resistors: $R_s = R_1 + R_2 + \ldots + R_n$



★★★ *3 marks ka derivation: diagram banao, V = V1 + V2 + V3 likho, I cancel karo. Nitesh Bhaiya ki guarantee!*

SHORTCUT: VOLTAGE SHARE

Same I flows, so $V_1 = V \times R_1 / R_s$. Example: 5 Ω and 15 Ω on 12 V: $V_1 = 12 \times 5/20 = 3$ V, $V_2 = 9$ V.

Series at a glance

Quantity	In a series circuit
Current	Same through each resistor
Potential difference	Divides; largest across the largest R
Total resistance	$R_s = R_1 + R_2 + \ldots$, more than the largest
Add one more resistor	R_s rises, current falls
One part breaks	Circuit opens; current becomes zero

4 COMMON MISTAKES

× WRONG	✓ RIGHT
Current gets used up as it passes each resistor	The **same current** flows through every resistor
Each resistor gets the full battery voltage	Voltage **divides**: $V = V_1 + V_2 + V_3$
Adding a resistor in series raises the current	R_s rises, so **I = V / R_s falls**
Voltmeter drawn in series with a resistor	Voltmeter goes **in parallel**, ammeter in series



GRU KA PLAN

*Bulbs series mein lagaye, Rs badha, current ghata, aur bulbs **dim ho gaye**. Plan ka last step dekh ke Gru bhi hairan!*

5 SOLVED EXAMPLES

★★★ MOST ASKED



★★★ *Given, Formula, Substitution, Answer + unit: ye 4 steps likho, poore marks pakke. Nitesh Bhaiya ki guarantee!*

Solved Example 1: Lamp and conductor ★★★

Q. An electric lamp of 20 Ω and a conductor of 4 Ω are joined in series to a 6 V battery. Find the total resistance, the current and the potential difference across each. NCERT

Given: $R_1 = 20$ Ω, $R_2 = 4$ Ω, V = 6 V

- $R_s = R_1 + R_2 = 20 + 4 = \mathbf{24\ \Omega}$
- $I = V / R_s = 6 / 24 = \mathbf{0.25\ A}$
- $V_{lamp} = IR_1 = 0.25 \times 20 = \mathbf{5\ V}$; $V_{conductor} = 0.25 \times 4 = \mathbf{1\ V}$

Answer: 24 Ω, 0.25 A, 5 V and 1 V. Check: 5 + 1 = 6 V ✔

Solved Example 2: Three resistors ★★★

Q. Resistors of 5 Ω, 10 Ω and 15 Ω are connected in series across a 12 V supply. Find the current and the voltage across each.

Given: $R_1 = 5$ Ω, $R_2 = 10$ Ω, $R_3 = 15$ Ω, V = 12 V

- $R_s = 5 + 10 + 15 = \mathbf{30\ \Omega}$
- $I = V / R_s = 12 / 30 = \mathbf{0.4\ A}$
- $V_1 = 0.4 \times 5 = \mathbf{2\ V}$, $V_2 = 0.4 \times 10 = \mathbf{4\ V}$, $V_3 = 0.4 \times 15 = \mathbf{6\ V}$

Answer: 0.4 A; 2 V, 4 V, 6 V (ratio 1 : 2 : 3, like R). Total 12 V ✔

Solved Example 3: Five resistors ★★★

Q. A 9 V battery is connected in series with resistors of 0.2 Ω, 0.3 Ω, 0.4 Ω, 0.5 Ω and 12 Ω. How much current flows through the 12 Ω resistor? NCERT

Given: V = 9 V; five resistors in series

- $R_s = 0.2 + 0.3 + 0.4 + 0.5 + 12 = \mathbf{13.4\ \Omega}$
- $I = V / R_s = 9 / 13.4 = 0.6716$ A ≈ **0.67 A**
- In series the current is the same everywhere, so the 12 Ω resistor also carries 0.67 A

Answer: ≈ 0.67 A through the 12 Ω resistor

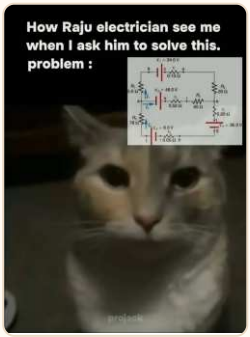
Solved Example 4: Work backwards from one voltage ★★☆

Q. A 4 Ω resistor and a resistor R are in series with a 12 V battery. The voltage across the 4 Ω resistor is 4 V. Find the current and R.

Given: $R_1 = 4$ Ω, $V_1 = 4$ V, V = 12 V

- $I = V_1 / R_1 = 4 / 4 = \mathbf{1\ A}$ (same through R)
- $V_R = V - V_1 = 12 - 4 = \mathbf{8\ V}$
- $R = V_R / I = 8 / 1 = \mathbf{8\ \Omega}$

Answer: I = 1 A, R = 8 Ω



MEME BREAK

*Circuit dekh ke Raju electrician ka bhi yahi haal! Darr mat, series mein bas **R jodo, I nikalo, V = IR lagao**. Done.*

Solved Example 5: Find the unknown resistor ★★★

Q. A 5 Ω resistor and an unknown resistor R are joined in series across 6 V. The current is 0.4 A. Find R.

Given: $R_1 = 5\ \Omega$, V = 6 V, I = 0.4 A

- $R_s = V / I = 6 / 0.4 =$ **15 Ω**
- $R_s = R_1 + R$, so $R = 15 - 5 =$ **10 Ω**

Answer: R = 10 Ω

Solved Example 6: Ammeter and voltmeter readings ★★★

Q. A battery of three cells of 2 V each, resistors of 5 Ω, 8 Ω and 12 Ω and a plug key are all in series. Find the ammeter reading and the voltmeter reading across the 12 Ω resistor. NCERT

Given: $V = 3 \times 2 = 6$ V; $R_1 = 5\ \Omega$, $R_2 = 8\ \Omega$, $R_3 = 12\ \Omega$

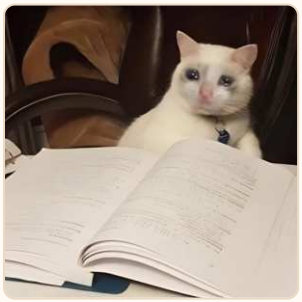
- $R_s = 5 + 8 + 12 =$ **25 Ω**
- $I = V / R_s = 6 / 25 =$ **0.24 A** (ammeter)
- $V_3 = IR_3 = 0.24 \times 12 =$ **2.88 V** (voltmeter)

Answer: Ammeter 0.24 A; voltmeter 2.88 V

6 DISADVANTAGES OF SERIES CIRCUITS ★★☆

Why homes are not wired in series

- **One fault stops all:** if one bulb fuses, the only path breaks and every appliance goes off
- **No separate control:** appliances cannot be switched on and off independently
- **Same current for all:** a bulb, fan and heater need very different currents, but series forces one current on all
- **High total resistance:** R_s is large, so the current is small and devices do not work at full power



SERIES KA DUKH

Ek bulb fuse hua aur poori series lights andhere mein. Ek gaya, sab gaye!

Day 32 mind map: revise in 2 minutes

CURRENT ★★★
Same in every resistor; ammeter reads the same anywhere

VOLTAGE ★★★
Divides: $V = V_1 + V_2 + V_3$; $V \propto R$

RESISTORS IN SERIES

RESISTANCE ★★★
$R_s = R_1 + R_2 + R_3$, more than the largest

EQUAL ONES
n resistors of R each: $R_s = nR$

DRAWBACKS
One fault stops all; no separate switching; same current forced on all; homes use parallel (Day 33)

7 PRACTICE MCQS

1. In a series combination of resistors, the quantity that is the same for every resistor is: ★★★

(a) potential difference (b) current
(c) resistance (d) heat produced

2. Resistors of 2 Ω, 3 Ω and 5 Ω are joined in series. The equivalent resistance is: ★★★

(a) 0.97 Ω (b) 5 Ω
(c) 10 Ω (d) 30 Ω

3. Equivalent resistance of n identical resistors, each R, joined in series is: ★★☆

(a) R/n (b) nR
(c) n^2R (d) R

4. 4 Ω and 8 Ω resistors in series are connected to 12 V. The potential difference across the 8 Ω resistor is: ★★★

(a) 4 V (b) 6 V
(c) 8 V (d) 12 V

5. The equivalent resistance of resistors in series is always: ★★☆

(a) less than the smallest (b) equal to the average
(c) greater than the largest (d) equal to the smallest

6. Ammeters placed at different points of a series circuit will read: ★★☆

(a) more near + terminal (b) the same current
(c) zero in the middle (d) less near − terminal

7. In a string of bulbs joined in series, one bulb fuses. The other bulbs will: ★★☆

(a) glow brighter (b) all go off
(c) glow dimmer (d) keep glowing

8. Resistors of 1 Ω, 2 Ω and 3 Ω are in series with a 6 V battery. The current is: ★★★

(a) 0.5 A (b) 1 A
(c) 2 A (d) 6 A

9. **Assertion (A):** Series arrangement is not used for domestic circuits.
Reason (R): In series, if one appliance fails the circuit breaks and all appliances stop working. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** When resistors are joined in series, the potential difference across each resistor is the same.
Reason (R): The same current flows through each resistor in a series combination. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

EXAM TIPS

- Write the unit with every answer: Ω, A or V
- Self-check: $V_1 + V_2 + V_3$ must equal the battery voltage

Day 32 hit-list: Eduspark PYQ analysis

Topic	Rating
Derivation of $R_s = R_1 + R_2 + R_3$	★★★
Numericals: R_s, current, voltage across each	★★★
Same current, divided voltage	★★★
Why series is not used at home	★★☆
n equal resistors: nR	★★☆
Voltage ratio $V \propto R$	★☆☆

Case-based question (4 marks)

Resistors of 2 Ω, 4 Ω and 6 Ω are joined in series with a 6 V battery and a key.

1. Find the equivalent resistance. **(1)**
2. Find the current in the circuit. **(1)**
3. Find the potential difference across the 6 Ω resistor. **(1)**
4. What happens to the current if the 4 Ω resistor breaks? **(1)**

Short and long answer questions

1. Derive the expression for the equivalent resistance of three resistors R_1, R_2 and R_3 joined in series. Draw the circuit diagram. **(3 marks)**
2. Why is the series arrangement not used in domestic wiring? Give two reasons. **(2 marks)**
3. A 4 Ω and a 6 Ω resistor are joined in series with a 5 V battery. Find (i) the equivalent resistance, (ii) the current and (iii) the potential difference across each resistor. **(3 marks)**

TRY YOURSELF

1. 2 Ω, 3 Ω and 7 Ω are in series across 6 V. Find the current.
2. How many 4 Ω resistors in series give 20 Ω?
3. 10 Ω and R in series across 9 V draw 0.3 A. Find R.
4. Find the ratio of the voltages across 3 Ω and 9 Ω resistors joined in series.

Answers are just below this box.

Try-yourself answers

(1) 12 Ω, so 0.5 A
(2) 5 resistors
(3) R_s = 30 Ω, R = 20 Ω
(4) 1 : 3



NITESH SIR SAYS

K³B yaad rakho: current same, voltage divide. Baaki sab V = IR se nikal jaata hai. ***Kal milte hai Day 33 pe!***

Answer key: MCQs

Q	Ans	Why
1	(b)	One path, so one current.
2	(c)	2 + 3 + 5 = 10 Ω.
3	(b)	R added n times.
4	(c)	I = 12/12 = 1 A; V = 1 × 8 = 8 V.
5	(c)	Adding only increases the total.
6	(b)	Current is the same everywhere.
7	(b)	The only path is broken.
8	(b)	I = 6/6 = 1 A.
9	(a)	R correctly explains A.
10	(d)	V divides ($V \propto R$); R is true.

Model answers: short and long answer

1 Derivation

Draw R_1, R_2, R_3 in series with battery, key and ammeter. Same current I; $V = V_1 + V_2 + V_3$. By Ohm's law $IR_s = IR_1 + IR_2 + IR_3$, so $\mathbf{R_s = R_1 + R_2 + R_3}$.

2 Not for homes

(i) If one appliance fails, the circuit breaks and all stop.
(ii) Appliances cannot be switched separately and need different currents.

3 Numerical

R_s = 4 + 6 = **10 Ω**; I = 5/10 = **0.5 A**; V_4 = 0.5 × 4 = **2 V**, V_6 = 0.5 × 6 = **3 V**.

Case-based answers

(1) 2 + 4 + 6 = 12 Ω
(2) I = 6/12 = 0.5 A
(3) V = 0.5 × 6 = 3 V
(4) Circuit opens; I = 0

Day 32 checklist: tick before you sleep

- ☐ I can define a series combination and draw its circuit
- ☐ I know: current same, voltage divides (K³B)
- ☐ I can derive $R_s = R_1 + R_2 + R_3$
- ☐ I can find R_s, I and each V in a numerical
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Ek concept chhoota toh domino ki tarah sab girega, page 1–2 phir se padho.



LEVEL 32 COMPLETE +100 XP earned. Next: Level 33



BUJHO *Samajh gaya! Current same, voltage divide, resistance add.*

PAHELI *Shabaash! Kal Level 33: parallel combination of resistors.*

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Electricity board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 33 · LEVEL 33

PHYSICS | CHAPTER 7 | PART 4 · ASTRA SERIES

PARALLEL & MIXED CIRCUITS

Goal: find R_p and solve any series-parallel circuit.



by Nitesh Prakash

01 Parallel Rules | 02 Derivation | 03 Series vs Parallel | 04 Mixed Circuits | 05 Home Wiring | +100 XP today | Level 33 / 60

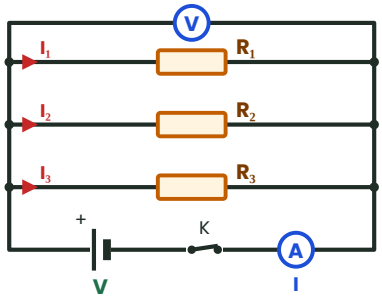
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 RESISTORS IN PARALLEL ★★★ MOST ASKED

DEFINITION

- **Parallel combination:** resistors joined **between the same two points**, so each one is connected directly across the source
- The **potential difference across every resistor is the same** (equal to V of the source)
- The **total current divides** among the branches: $I = I_1 + I_2 + I_3$

Circuit: three resistors in parallel



Ammeter A (in series) reads the total current I; voltmeter V (in parallel) reads the same p.d. across R_1, R_2 and R_3.

CLASS TRICK: SERIES VS PARALLEL IN ONE LINE

Parallel: Voltage remains same, current is divided inversely proportional to the resistance.
Series: Current remains same, voltage divides itself on all the resistors.

Activity: what the ammeters show

- ✔ Put an ammeter in each branch: I_1, I_2, I_3 come out different.
- ✔ Their sum equals the main-line reading I.
- ✔ Smallest resistor carries the **largest** current ($I = V/R$, same V).

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Junction rule (Kirchhoff's current law): the total current entering a junction equals the total current leaving it, because charge is neither created nor destroyed at the junction. That is exactly why $I = I_1 + I_2 + I_3$. Named in Class 12; not asked by name in Class 10.

2 THE PARALLEL FORMULA ★★★ MOST ASKED

Step by step (write all steps in the exam) ★★★

STEP 1 Total current = sum of branch currents: $I = I_1 + I_2 + I_3$

STEP 2 Same V across each: by Ohm's law $I_1 = V/R_1$, $I_2 = V/R_2$, $I_3 = V/R_3$

STEP 3 For the equivalent resistor R_p: $I = V/R_p$

STEP 4 So $V/R_p = V/R_1 + V/R_2 + V/R_3$; cancel V

$$\mathbf{1/R_p = 1/R_1 + 1/R_2 + 1/R_3}$$

The reciprocal of the equivalent resistance is the sum of the reciprocals

Shortcuts that save time ★★☆

Case	Equivalent resistance
Two resistors	$\mathbf{R_p = R_1R_2 / (R_1 + R_2)}$ (product over sum)
n equal resistors R	$\mathbf{R_p = R/n}$; in series it would be nR
Any parallel set	R_p is **less than the smallest** resistor

Quick check: 3 Ω ‖ 6 Ω = 18/9 = 2 Ω. For n equal resistors, $R_s : R_p = nR : R/n = \mathbf{n^2 : 1}$.



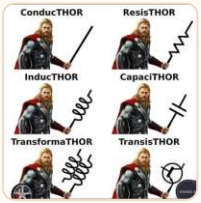
BUJHO *Resistor jodne se resistance badhna chahiye na? Parallel mein kam kaise ho gaya?*

PAHELI *Har naya branch current ko ek extra raasta deta hai. Zyada raaste matlab current ko kam rukawat, isliye R_p sabse chhote se bhi chhota.*



REMEMBER: ADDING ANY BRANCH LOWERS R_p

Even a 1 MΩ resistor added in parallel makes R_p a little smaller. Removing a branch makes R_p larger and the main current smaller.



MEME BREAK

*ResisTHOR kitne bhi lagao, **parallel** mein total resistance sabse chhote wale se bhi kam hi rahega!*

★★★ *Derivation of the parallel formula with circuit diagram: 3 marks ka favourite sawaal. Nitesh Bhaiya ki guarantee!*

3 SERIES VS PARALLEL ★★★ MOST ASKED

Compare before you calculate

Feature	Series	Parallel
Current	same in every resistor	divides; $I = I_1 + I_2 + \ldots$
Potential difference	divides; $V = V_1 + V_2 + \ldots$	same across every resistor
Equivalent R	$R_s = R_1 + R_2 + \ldots$	$1/R_p = 1/R_1 + 1/R_2 + \ldots$
Value	more than the largest	less than the smallest
One device fails	whole circuit stops	others keep working



MEME BREAK

*Yoda ki seekh: "Parallel mein jodo, **R_p sabse chhote se bhi kam** aayega, hmm." Bada aaya to galti hai!*

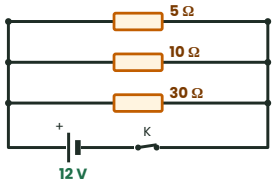
4 MIXED CIRCUITS: THE METHOD ★★★ MOST ASKED

Four steps for any series-parallel circuit

1. Spot the **parallel group** (resistors between the same two junctions) and replace it by R_p.
2. Add the series parts: $R_{total} = R_{series} + R_p$.
3. Main current $I = V / R_{total}$.
4. Work back: p.d. across each part = $I \times R$, then each branch current = V/R.

Solved Example 1: three in parallel ★★★

Q. Resistors of 5 Ω, 10 Ω and 30 Ω are connected in parallel to a 12 V battery. Find R_p, the current in each resistor and the total current.



GIVEN $R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$, $R_3 = 30\ \Omega$, $V = 12$ V

FORMULA $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$; $I = V/R$

SOLVE $1/R_p = 6/30 + 3/30 + 1/30 = 10/30$, so $\mathbf{R_p = 3\ \Omega}$

CURRENTS $I_1 = 12/5 = \mathbf{2.4\ A}$, $I_2 = 12/10 = \mathbf{1.2\ A}$, $I_3 = 12/30 = \mathbf{0.4\ A}$

ANSWER $I = 12/3 = \mathbf{4\ A} = 2.4 + 1.2 + 0.4$ (check)

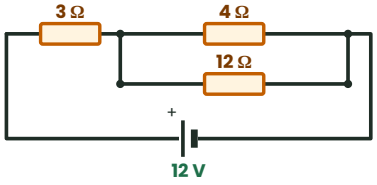
TRY YOURSELF: MIXED CIRCUITS

1. Two 4 Ω resistors in parallel, in series with a 3 Ω resistor, across 10 V. Find the main current.
2. Find R_p of 2 Ω, 4 Ω and 4 Ω in parallel.
3. What is the highest and lowest resistance from four 4 Ω resistors?

Answers are on page 4, below the model answers.

Solved Example 2: series + parallel ★★★

Q. A 3 Ω resistor is in series with a parallel pair of 4 Ω and 12 Ω, across a 12 V battery. Find the total resistance, the main current and the current in the 4 Ω resistor.



STEP 1 $R_p = (4 \times 12)/(4 + 12) = 48/16 = 3\ \Omega$

STEP 2 $R_{total} = 3 + 3 = \mathbf{6\ \Omega}$

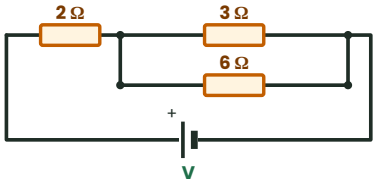
STEP 3 $I = V/R = 12/6 = \mathbf{2\ A}$

STEP 4 p.d. across 3 Ω = $2 \times 3 = 6$ V, so across the pair = $12 - 6 = 6$ V

ANSWER $I_{4\Omega} = 6/4 = \mathbf{1.5\ A}$; $I_{12\Omega} = 6/12 = 0.5$ A (sum 2 A)

Solved Example 3: get 4 Ω and 1 Ω ★★★

Q. How will you connect three resistors of 2 Ω, 3 Ω and 6 Ω to get a total resistance of (a) 4 Ω (b) 1 Ω? NCERT



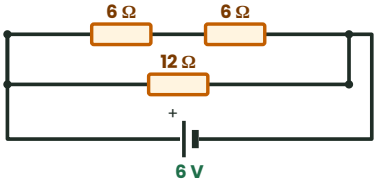
(A) 3 Ω ‖ 6 Ω = 18/9 = 2 Ω; then in series with 2 Ω: 2 + 2 = **4 Ω** (circuit above)

(B) All three in parallel: 1/R = 1/2 + 1/3 + 1/6 = 6/6, so **R = 1 Ω**

NOTE Maximum possible = 11 Ω (all series); minimum = 1 Ω (all parallel)

Solved Example 4: branch with two resistors ★★☆

Q. Two 6 Ω resistors in series form one branch; a 12 Ω resistor forms a second branch in parallel with it. The combination is connected to a 6 V battery. Find the total current and the current in each branch.



STEP 1 Branch 1 = 6 + 6 = 12 Ω; branch 2 = 12 Ω

STEP 2 $R_p = 12/2 = \mathbf{6\ \Omega}$ (two equal resistors: R/n)

STEP 3 $I = 6/6 = \mathbf{1\ A}$

ANSWER Each branch has 6 V across it, so each carries $6/12 = \mathbf{0.5\ A}$

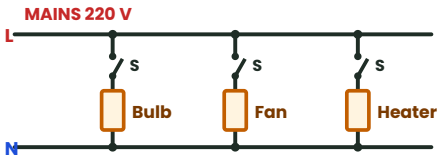
EXAM TIP: SHOW THE REDRAW

Write "3 Ω ‖ 6 Ω = 2 Ω" beside a redrawn circuit before the next step. Units on every line: Ω, A, V.

★★★ *Mixed circuit numericals 3 marks ke pakke hain: pehle parallel ko ek resistor banao, phir series jodo. Nitesh Bhaiya ki guarantee!*

5 PARALLEL WIRING AT HOME ★★★ MOST ASKED

Home circuit: every appliance in parallel



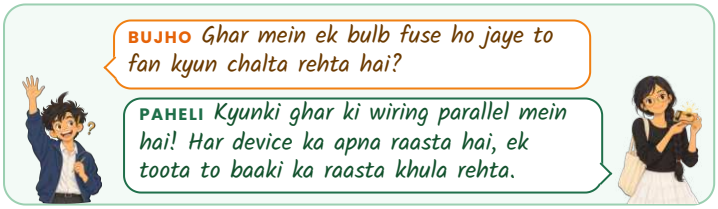
L = live wire, N = neutral wire, S = switch. Each appliance sits between L and N, so each gets the full 220 V.

Why not series for domestic circuits?

1. If one bulb fuses or one device is switched off, the circuit breaks and **every device stops**.
2. The **same current** flows through all, but a bulb and a heater need very different currents.
3. Resistances add up, so the total resistance is large and the current is too small to run the devices.

Advantages of parallel connection ★★★

- ✔ Each appliance gets the **full supply voltage** (220 V).
- ✔ If one appliance fails, the **others keep working**; each has its own switch.
- ✔ Each appliance draws the **current it needs** ($I = V/R$), according to its own resistance.
- ✔ Total resistance is low, so the circuit can run many devices together.



Class questions (from Nitesh Sir's notes)

Q1. Judge the equivalent resistance when 1 Ω and 10^6 Ω are connected in parallel.
Ans. R_p is less than the smallest, so it is **less than 1 Ω** (about 0.999999 Ω).
Q2. Why does the main current increase when one more bulb is switched on at home?
Ans. A new parallel branch lowers R_p; with V fixed, $I = V/R_p$ increases.

Common mistakes that cost marks

× WRONG	✔ RIGHT
$1/R_p = 1/2 + 1/3$, so $R_p = 5/6$ Ω	$1/R_p = 5/6$, so R_p = **6/5 Ω**: invert at the end
Same current in all parallel branches	Same **p.d.**; currents differ
Voltmeter in series	Voltmeter in parallel, ammeter in series

6 PRACTICE MCQS

Mixed circuit dekh ke rona aa gaya? Method follow karo, 10 MCQs khud solve karo!



1. Three resistors of 6 Ω each are connected in parallel. The equivalent resistance is: ★★★
(a) 18 Ω (b) 2 Ω
(c) 3 Ω (d) 6 Ω

2. In a parallel combination, the quantity that is the same for every resistor is: ★★★
(a) current (b) potential difference
(c) resistance (d) charge

3. The equivalent resistance of 1 Ω and 10^6 Ω in parallel is: NCERT ★★☆
(a) more than 10^6 Ω (b) between 1 Ω and 10^6 Ω
(c) less than 1 Ω (d) exactly 10^6 Ω

4. 4 Ω and 12 Ω resistors in parallel give an equivalent resistance of: ★★☆
(a) 16 Ω (b) 8 Ω
(c) 3 Ω (d) 0.33 Ω

5. A wire of resistance R is cut into five equal parts, which are joined in parallel to give R′. The ratio R/R′ is: NCERT ★★★
(a) 1/25 (b) 1/5
(c) 5 (d) 25

6. Domestic appliances are connected in parallel because: ★★☆
(a) the same current flows through each (b) each gets the same voltage and works independently
(c) total resistance increases (d) it saves wire

7. How many 176 Ω resistors in parallel are needed to draw 5 A from a 220 V line? NCERT ★★☆
(a) 2 (b) 4
(c) 5 (d) 44

8. The lowest total resistance from coils of 4 Ω, 8 Ω, 12 Ω and 24 Ω is: NCERT ★★☆
(a) 48 Ω (b) 4 Ω
(c) 2 Ω (d) 1 Ω

9. **Assertion (A):** When more resistors are added in parallel, the total current drawn from the source increases.
Reason (R): The equivalent resistance of a parallel combination decreases as more resistors are added. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

10. **Assertion (A):** Bulbs in a house are connected in series.
Reason (R): In a series circuit, every bulb carries the same current. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Short and long answer questions

1. Derive the expression for the equivalent resistance of three resistors connected in parallel. **(3 marks)**
2. Give two disadvantages of series connection and two advantages of parallel connection for domestic appliances. **(3 marks)**
3. An electric lamp of 100 Ω, a toaster of 50 Ω and a water filter of 500 Ω are connected in parallel to a 220 V source. What is the resistance of an electric iron that draws as much current as all three together, and what current flows through it? **(3 marks)** NCERT

Answer key: MCQs

Q	Ans	Why
1	(b)	R/n = 6/3 = 2 Ω.
2	(b)	Every branch is across the same two points.
3	(c)	R_p is less than the smallest resistor.
4	(c)	48/16 = 3 Ω.
5	(d)	Each part R/5; five in parallel: R/25, so R/R′ = 25.
6	(b)	Full 220 V each; one off, others work.
7	(b)	R = 220/5 = 44 Ω; n = 176/44 = 4.
8	(c)	1/R = (6 + 3 + 2 + 1)/24 = 12/24, so 2 Ω.
9	(a)	Lower R_p with fixed V means larger I.
10	(d)	Home bulbs are in parallel; R itself is true.

Model answers: short and long answer

1 Derivation of the parallel formula

$I = I_1 + I_2 + I_3$; same V across each, so $I_1 = V/R_1$, $I_2 = V/R_2$, $I_3 = V/R_3$. For R_p, $I = V/R_p$. Hence $V/R_p = V/R_1 + V/R_2 + V/R_3$, giving $\mathbf{1/R_p = 1/R_1 + 1/R_2 + 1/R_3}$. Draw the circuit with ammeter in series and voltmeter across the combination.

2 Series vs parallel at home

Series: one fault stops all; same current for devices that need different currents. Parallel: each gets 220 V; devices work independently, each has its own switch and draws the current it needs. Also, the total resistance stays low.

3 Lamp, toaster and filter

GIVEN $R_1 = 100\ \Omega$, $R_2 = 50\ \Omega$, $R_3 = 500\ \Omega$, V = 220 V

SOLVE $1/R = 1/100 + 1/50 + 1/500 = (5 + 10 + 1)/500 = 16/500$

R $R = 500/16 = \mathbf{31.25\ \Omega}$ (the iron must have this resistance)

I $I = 220/31.25 = \mathbf{7.04\ A}$

Try yourself answers: (1) 4 Ω ‖ 4 Ω = 2 Ω, total 5 Ω, I = 2 A (2) 1/R = 1/2 + 1/4 + 1/4 = 1, R_p = 1 Ω (3) highest 16 Ω, lowest 1 Ω

FORMULA RECAP

Parallel: $I = I_1 + I_2 + I_3$, V same, $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$. Mixed: $R_{total} = R_{series} + R_p$, then $I = V/R_{total}$.

Coming up: Day 34

Heating effect of current: Joule's law $H = I^2Rt$, bulbs, heaters and fuses, electric power and the kilowatt-hour.

Day 33 mind map: revise in 2 minutes

SAME V
Every branch is across the same two points

CURRENT DIVIDES
$I = I_1 + I_2 + I_3$; small R takes big I

RESISTORS IN PARALLEL

FORMULA ★★★
$1/R_p = \Sigma\ 1/R$; two: $R_1R_2/(R_1+R_2)$; n equal: R/n

HOME WIRING
Parallel: full 220 V, independent switches

MIXED CIRCUITS
Parallel group → R_p, add series parts, find I = V/R, then work back to each branch

Day 33 hit-list: Eduspark PYQ analysis

Topic	Rating
Derivation of $1/R_p$ with circuit	★★★
Mixed series-parallel numericals	★★★
Why domestic wiring is parallel	★★★
Highest / lowest resistance combinations	★★☆
Cut wire, parts in parallel (R/n^2)	★★☆

REVISION PLAN FOR TONIGHT

Redo Solved Examples 2 and 4 without looking, then attempt the Try yourself box. Time yourself: 4 minutes per numerical.

Day 33 checklist: tick before you sleep

- ☐ I can state the rules for current and p.d. in a parallel circuit
- ☐ I can derive the formula for R_p with a circuit diagram
- ☐ I can solve a mixed series-parallel numerical step by step
- ☐ I can explain why homes use parallel wiring
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

LEVEL 33 COMPLETE +100 XP earned. Next: Level 34



BUJHO Samajh gaya! Parallel mein voltage same, current bant jaata hai!

PAHELI Shabaash! Kal Level 34: heating effect, power aur bijli ka bill.



NITESH SIR SAYS
Circuit pehle redraw karo, phir parallel group ko ek resistor banao. **Diagram saaf to answer saaf!**



DAY 34 · LEVEL 34

PHYSICS | CHAPTER 7 | PART 5 · ASTRA SERIES

HEATING EFFECT & ELECTRIC POWER

Goal: use $H = I^2Rt$, $P = VI$ and the kWh with confidence.



by Nitesh Prakash

01 Joule's Law | 02 Bulb & Heater | 03 Fuse | 04 Electric Power | 05 kWh & Bill | +100 XP today | Level 34 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 THREE EFFECTS OF CURRENT ★★☆

Electric current can produce

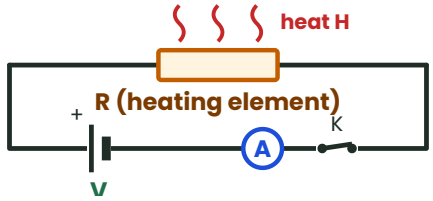
Effect	What happens	Used in
Heating	conductor gets hot	heater, iron, bulb, fuse
Magnetic	wire behaves like a magnet	electromagnet, bell (Day 36)
Chemical	solution decomposes	electroplating, refining

2 HEATING EFFECT OF CURRENT ★★★ MOST ASKED

DEFINITION

- **Heating effect:** when a current flows through a resistor, part of the electrical energy is **converted into heat**
- The source keeps doing work to push charge through the resistance; this work appears as **heat energy**
- Purely resistive circuit: **all** the energy from the source is dissipated as heat

Circuit: current through a heating element



More current, more resistance or more time means more heat. Nichrome coil gets red-hot; copper wires stay cool.

Energy chain in a heater

Cell / mains chemical or electrical energy → **Moving charges** work done = VQ → **Resistor** heat energy H

CLASS NOTE: THREE EFFECTS OF CURRENT

Current ke teen effect: Heating (nichrome), Magnetic, Chemical. Aaj ka hero: heating effect!

REMEMBER: WHY CONNECTING WIRES STAY COOL

Copper connecting wires have very low resistance, so I^2Rt is tiny there. The heat appears in the high-resistance element.

3 JOULE'S LAW OF HEATING ★★★ MOST ASKED

JOULE'S LAW

- The heat produced in a resistor is directly proportional to (i) the **square of the current**, I^2 (ii) the **resistance**, R (iii) the **time** for which current flows, t

Derivation of $H = I^2Rt$ ★★★

STEP 1 Charge Q moves through p.d. V: work done W = VQ
STEP 2 Charge in time t: Q = It, so W = VIt
STEP 3 This work is converted to heat: H = VIt
STEP 4 By Ohm's law V = IR, so H = (IR)It

$$H = I^2Rt = VIt = V^2t/R$$

H in joule (J), I in ampere, R in ohm, t in second

Read the law like a graph ★★☆

- ✔ Double I → heat becomes **4 times** ($H \propto I^2$).
- ✔ Double R (same I) → heat doubles.
- ✔ Double t → heat doubles.



BUJHO *Heater ka coil laal ho jata hai, par usse judi copper ki taar thandi kyun rehti?*

PAHELI *Same current, par coil ka resistance bahut zyada hai. $H = I^2Rt$, to garmi wahin banti jahan R bada hai.*



Solved Example 1: heat from charge ★★☆

Q. Compute the heat generated while transferring 96000 C of charge in one hour through a potential difference of 50 V. NCERT

GIVEN Q = 96000 C, V = 50 V, t = 1 h
FORMULA H = VQ (work done on the charge)
ANSWER $H = 50 \times 96000 = \mathbf{4.8 \times 10^6\ J}$

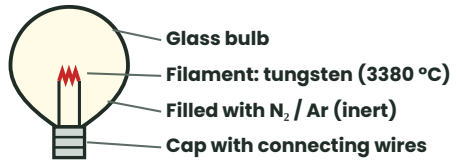
lly = I love you
lmy = I miss you
Brb = Be right back
pV = nRT
F = ma
KE = (1/2)mv^2
p1v1 = p2v2
E = mc^2

MEME BREAK

*Formula list mein ek aur pyaar jod lo: **$H = I^2Rt$**. Current double, garmi chaar guna!*

4 USES OF THE HEATING EFFECT ★★★ MOST ASKED

Electric bulb ★★★



- ✔ Filament made of **tungsten**: very high melting point (3380 °C), glows white-hot without melting.
- ✔ Bulb filled with chemically inactive **nitrogen and argon** to prolong the life of the filament.
- ✔ Most of the energy is given out as **heat**; only a small part as light.

Heater, iron, toaster, kettle

- ✔ Heating element is a coil of **nichrome** (alloy of nickel and chromium).
- ✔ The coil sits in grooves of a **china clay (porcelain)** plate, which is a heat-resistant insulator.

Class question: why only nichrome for the heating coil? ★★★

Ans. (i) Nichrome has **high resistivity**, so more heat is produced ($H = I^2Rt$). (ii) It has a **high melting point**. (iii) It **does not oxidise (burn) readily** at high temperature, so it can stay red-hot for long.

Electric fuse ★★★

- ✔ A short wire of an alloy with **low melting point** (aluminium, copper, iron, lead), connected **in series** with the device.
- ✔ If the current exceeds the safe value, the wire heats up, **melts and breaks the circuit**, protecting the appliance.
- ✔ Common ratings: 1 A, 2 A, 3 A, 5 A, 10 A.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Modern homes use a **miniature circuit breaker (MCB)**: it switches off automatically on overload and can be reset by a switch, so no fuse wire needs replacing.



BUJHO *Fuse ki jagah moti copper ki taar laga dein to kya hoga?*

PAHELI *Wo pighlegi hi nahi! Zyada current mein appliance jal jayega ya aag lag sakti hai. Sahi rating ka fuse hi lagao.*



Solved Example 2: resistance of a bulb ★★☆

Q. Find the resistance of the filament of a 100 W, 220 V bulb. Which has more resistance: this bulb or a 60 W, 220 V bulb?

FORMULA $P = V^2/R$, so $R = V^2/P$

100 W $R = 220^2/100 = 48400/100 = \mathbf{484\ \Omega}$

60 W $R = 48400/60 = \mathbf{806.7\ \Omega}$: the 60 W bulb has more resistance

Solved Example 3: choosing a fuse ★★★

Q. An electric iron of 1 kW is used on a 220 V supply. Which fuse should be used: 2 A, 5 A or 10 A?

GIVEN P = 1000 W, V = 220 V

FORMULA $I = P/V$

SOLVE $I = 1000/220 = \mathbf{4.54\ A}$

ANSWER Use the **5 A** fuse: just above the normal current

Solved Example 4: heat in an iron ★★★

Q. An electric iron of resistance 20 Ω takes a current of 5 A. Calculate the heat developed in 30 s. NCERT

GIVEN R = 20 Ω, I = 5 A, t = 30 s

FORMULA $H = I^2Rt$

SOLVE $H = 5^2 \times 20 \times 30 = 25 \times 600$

ANSWER $H = \mathbf{15000\ J = 1.5 \times 10^4\ J}$

Power ratings you will meet (typical values)

Appliance	Power	Current at 220 V
LED bulb	10 W	about 0.05 A
Ceiling fan	75 W	about 0.34 A
Electric iron	1000 W	about 4.5 A
Geyser	2000 W	about 9.1 A

High-power appliances draw large currents, so they need thicker wires and higher fuse ratings.

Undesirable heating

- ✔ Heat in wires, motors and computers wastes energy and can damage insulation.
- ✔ Thick copper wires and cooling fans keep this loss low.

Common mistakes that cost marks

× WRONG	✔ RIGHT
kWh is a unit of power	kWh is a unit of **energy**
Time in minutes put directly in kWh	Convert to **hours** first
Higher wattage bulb has higher R	$R = V^2/P$: higher P, **lower** R

FUSE SAFETY

Always replace a blown fuse with one of the **same rating**. Never use a thicker wire or a nail in its place.



MEME BREAK

*Joule aur kWh ki dosti: ek hi energy, bas **3.6 × 10⁶** ka farak. Unit badlo, answer nahi!*

★★★ *Tungsten filament, nichrome coil, fuse in series: har saal ek na ek pakka aata hai. Nitesh Bhaiya ki guarantee!*

5 ELECTRIC POWER ★★★ MOST ASKED

DEFINITION

- **Electric power:** the rate at which electrical energy is consumed (or work is done) in a circuit, P = W/t
- SI unit: **watt (W). 1 W** is the power consumed by a device that carries **1 A** when operated at a p.d. of **1 V**
- 1 kilowatt (kW) = 1000 W

Derivation: three forms of P ★★★

STEP 1 P = W/t and W = VQ, so P = VQ/t
STEP 2 Q/t = I, so **P = VI**
STEP 3 Put V = IR: **P = I^2R**
STEP 4 Put I = V/R: **P = V^2/R**

$$P = VI = I^2R = V^2/R$$

Same current (series): use I^2R. Same voltage (parallel, home): use V^2/R

Solved Example 5: power of a bulb ★★★

Q. An electric bulb connected to a 220 V generator draws a current of 0.50 A. What is its power and its resistance? NCERT

GIVEN V = 220 V, I = 0.50 A
POWER P = VI = 220 × 0.50 = **110 W**
RESISTANCE R = V/I = 220/0.50 = **440 Ω**

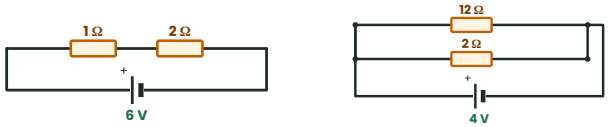
Solved Example 6: heater on the mains ★★☆

Q. An electric heater of resistance 8 Ω draws 15 A from the service mains for 2 hours. Calculate the rate at which heat is developed in the heater. NCERT

FORMULA Rate of heat = power, P = I^2R
ANSWER P = 15^2 × 8 = 225 × 8 = **1800 W = 1800 J/s**

Solved Example 7: compare the power in the 2 Ω resistor ★★★

Q. Compare the power used in the 2 Ω resistor in: (i) a 6 V battery in series with 1 Ω and 2 Ω (ii) a 4 V battery in parallel with 12 Ω and 2 Ω. NCERT



(I) I = 6/(1 + 2) = 2 A; P = I^2R = 2^2 × 2 = **8 W**
(II) V across 2 Ω = 4 V; P = V^2/R = 16/2 = **8 W**
ANSWER Power is the **same, 8 W**, in both cases



BUJHO *100 W ka bulb aur 60 W ka bulb ghar mein lagaye, to zyada resistance kiska hoga?*

PAHELI *R = V^2/P. Same 220 V par kam power wale ka resistance zyada: 60 W bulb ka R bada hai.*



REMEMBER: RATED BULB ON LOWER VOLTAGE

R stays fixed, so P ∝ V^2. A 220 V, 100 W bulb on 110 V gives $(1/2)^2$ × 100 = **25 W**.

6 ELECTRICAL ENERGY ★★★ MOST ASKED

DEFINITION

- **Electrical energy** used by a device = power × time: **E = P × t**
- SI unit: joule (J). Practical unit: watt-hour (Wh)
- Commercial unit: **kilowatt-hour (kWh)**, called **1 unit** on the electricity bill

How big is 1 kWh? ★★★

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = \mathbf{3.6 \times 10^6 \text{ J}}$$

Energy used by a 1000 W device running for 1 hour

Bill = units (kWh) × rate per unit. Units = (watts × hours) / 1000.



MEME BREAK
*Bill pe **1 unit** likha hai? Wo 1 joule nahi, poore **3.6 × 10^6 J** hai!*

Solved Example 8: refrigerator bill ★★★

Q. An electric refrigerator rated 400 W operates 8 hours a day. What is the cost of the energy to operate it for 30 days at ₹ 3.00 per kWh? NCERT

ENERGY E = 400 W × 8 h × 30 = 96000 Wh = **96 kWh**
COST 96 × 3.00 = **₹ 288**

Solved Example 9: which uses more energy ★★☆

Q. Which uses more energy: a 250 W TV set in 1 hour, or a 1200 W toaster in 10 minutes? NCERT

TV E = 250 W × 1 h = **250 Wh**
TOASTER E = 1200 W × (10/60) h = **200 Wh**
ANSWER The **TV** uses more energy (250 Wh > 200 Wh)

Exam tips

- ✔ Write H = I^2Rt with units: J, A, Ω, s.
- ✔ For "why tungsten / why nichrome", give two reasons each.
- ✔ Fuse: say "in series" and "low melting point" every time.

Short and long answer questions

1. State Joule's law of heating and derive H = I^2Rt. **(3 marks)**
2. An electric motor takes 5 A from a 220 V line. Find its power and the energy consumed in 2 h. **(2 marks)** NCERT
3. Two lamps, 100 W and 60 W, are connected in parallel to a 220 V supply. What current is drawn from the line? **(2 marks)** NCERT
4. Why is tungsten used for bulb filaments, and why are heater coils made of an alloy? **(2 marks)**

7 PRACTICE MCQS

Ek side: "bijli ka bill joule mein?"
Doosri side: "units = kWh", mast view.
Ab MCQs khud solve karo!



1. The heat produced in a resistor is given by: ★★★

(a) IRt (b) I^2Rt
(c) IR^2t (d) I^2R/t

2. The filament of an electric bulb is made of tungsten because it has: ★★★

(a) low resistivity (b) a very high melting point
(c) a low melting point (d) low cost

3. A fuse wire is connected: ★★★

(a) in parallel with the appliance (b) in series with the appliance
(c) between live and earth (d) only in the neutral wire

4. 1 kWh is equal to: ★★★

(a) 3.6×10^3 J (b) 3.6×10^5 J
(c) 3.6×10^6 J (d) 36×10^6 J

5. If the current through a resistor is doubled, the heat produced in the same time becomes: ★★☆

(a) double (b) half
(c) four times (d) unchanged

6. Which of the following does not represent electrical power in a circuit? NCERT ★★☆

(a) I^2R (b) IR^2
(c) VI (d) V^2/R

7. A bulb rated 220 V, 100 W is operated on 110 V. The power consumed is: NCERT ★★☆

(a) 100 W (b) 75 W
(c) 50 W (d) 25 W

8. **Assertion (A):** Nichrome is used in the heating elements of electric heaters.
Reason (R): Nichrome has high resistivity and does not oxidise readily at high temperature. ★★☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

9. **Assertion (A):** The commercial unit of electrical energy is the watt.
Reason (R): 1 kWh = 3.6×10^6 J. ★☆☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

TRY YOURSELF: BILLS AND POWER

1. Five 60 W bulbs run 6 h a day and a 1 kW iron runs 1 h a day. Find the units used in 30 days and the bill at ₹ 5 per unit.
2. Find the resistance of a 60 W, 220 V bulb.

Answers: 84 units, ₹ 420; $R = V^2/P = 48400/60 = 806.7\ \Omega$.

Answer key: MCQs

Q	Ans	Why
1	(b)	Joule's law: $H = I^2Rt$.
2	(b)	Melting point 3380 °C; glows without melting.
3	(b)	In series, so it breaks the whole current path.
4	(c)	1000 W × 3600 s.
5	(c)	$H \propto I^2$: $2^2 = 4$.
6	(b)	IR^2 has wrong units.
7	(d)	R fixed; $P \propto V^2$: half V gives P/4 = 25 W.
8	(a)	R gives the exact reason for A.
9	(d)	Unit of energy is kWh; watt is power.

Model answers: numericals

2 Motor
POWER $P = VI = 220 \times 5 =$ **1100 W**
ENERGY $E = 1.1\ kW \times 2\ h =$ **2.2 kWh** $= 2.2 \times 3.6 \times 10^6 =$ **7.92 × 10^6 J**

3 Two lamps in parallel
TOTAL P 100 + 60 = 160 W
CURRENT $I = P/V = 160/220 =$ **0.73 A**

Day 34 hit-list: Eduspark PYQ analysis

Topic	Rating
Joule's law with derivation	★★★
kWh to J and electricity bill	★★★
Tungsten, nichrome, fuse reasons	★★★

Day 34 checklist: tick before you sleep

- ☐ I can state Joule's law and derive $H = I^2Rt$
- ☐ I can explain the bulb, heater and fuse using the heating effect
- ☐ I can use $P = VI = I^2R = V^2/R$
- ☐ I can convert kWh to joules and calculate a bill
- ☐ I solved all 9 MCQs and checked them with the key

My score: ____ / 5

LEVEL 34 COMPLETE +100 XP earned. Next: Level 35



BUJHO *Samajh gaya! Garmi I^2Rt, power VI, aur bill kWh mein!*

PAHELI *Shabaash! Kal Level 35: Chapter 7 ka Boss Level test.*



NITESH SIR SAYS
Har numerical mein units likho aur kWh ko joule mein badalna mat bhoolo. ***Unit sahi, marks pakke!***

DAY 35 · LEVEL 35

PHYSICS | CHAPTER 7 | BOSS LEVEL

CHAPTER 7 BOSS LEVEL

Goal: test all of Electricity in 90 minutes.



by Nitesh Prakash

01 Revision Map · 02 MCQs · 03 Short Answers · 04 Case & Long · 05 Answer Key · +200 XP today | Level 35 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 7 REVISION MAP ★★★ MOST ASKED

Chapter 7 mind map: revise in 5 minutes

CURRENT & P.D.
I = Q/t (A); V = W/Q (V); 1 C = 6.25 × 10^{18} electrons

OHM'S LAW
V = IR at constant temperature; V–I graph a straight line

ELECTRICITY

RESISTANCE ★★★
R = ρl/A; ρ depends on material and temperature

COMBINATIONS
$R_s = R_1 + R_2 + \ldots$; $1/R_p = 1/R_1 + 1/R_2 + \ldots$

HEAT, POWER, ENERGY
$H = I^2Rt$; $P = VI = I^2R = V^2/R$; 1 kWh = 3.6 × 10^6 J

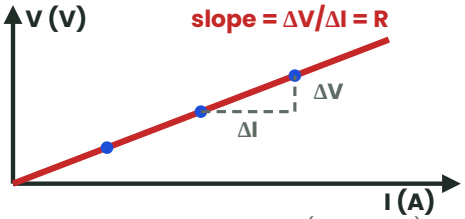
Must-know formulas: write them daily ★★★

Quantity	Formula	SI unit
Current	I = Q/t	ampere (A)
Potential difference	V = W/Q	volt (V)
Resistance	R = V/I = ρl/A	ohm (Ω)
Resistivity	ρ = RA/l	ohm metre (Ω m)
Heat	$H = I^2Rt$	joule (J)
Power	P = VI	watt (W)
Energy	E = Pt	J or kWh

CLASS TRICK: VOLTAGE = POTENTIAL DIFFERENCE

"Gangadhar hi Shaktiman hain or Shaktiman hi Gangadhar hain": potential difference aur voltage ek hi cheez ke do naam hain.

V–I graph you must draw ★★☆



A straight line through the origin shows V ∝ I (Ohm's law). A steeper line means a larger resistance.

How to attack the Boss: 90 minutes

1. Section A (20 MCQs, 1 mark each): **20 min**
2. Section B (5 × 2 marks): **12 min**
3. Section C (4 × 3 marks): **20 min**
4. Section D (2 cases × 4 marks): **14 min**
5. Section E (2 × 5 marks): **18 min**; final check of units: 6 min

Total 60 marks. Write Given, Formula, Substitution and Answer with unit in every numerical.



BOSS INTRO

*Ohm, Volta, Ampere, Joule: saare scientists dekh rahe hain. Aaj **Boss Level** mein unka naam roshan karo!*

2 SECTION A: MCQS ★★★ MOST ASKED

1. The number of electrons that make up a charge of 1 C is: ★★★
(a) 1.6 × 10^{-19} (b) 6.25 × 10^{18}
(c) 6.25 × 10^{19} (d) 1.6 × 10^{19}

2. 1 volt is equal to: ★★★
(a) 1 J/C (b) 1 C/J
(c) 1 J × 1 C (d) 1 A/s

3. An ammeter is always connected: ★★☆
(a) in series (b) in parallel
(c) across the cell (d) across the resistor

4. The resistivity of a conductor depends on: ★★★
(a) its length (b) its area of cross-section
(c) its material and temperature (d) its shape

5. A wire of resistance R is stretched to twice its length (volume unchanged). Its new resistance is: ★★★
(a) R/2 (b) 2R
(c) 4R (d) R

6. The slope of a V–I graph (V on the y-axis) of an ohmic conductor gives: ★★☆
(a) conductance (b) resistance
(c) resistivity (d) power

7. Using three 3 Ω resistors, which value can NOT be obtained? ★★☆

(a) 9 Ω (b) 1 Ω
(c) 4.5 Ω (d) 6 Ω

8. A charge of 150 C flows through a wire in 1 minute. The current is: ★★☆

(a) 150 A (b) 2.5 A
(c) 0.4 A (d) 9000 A

9. The work done in moving 2 C of charge across a p.d. of 12 V is: NCERT ★★★

(a) 6 J (b) 12 J
(c) 24 J (d) 14 J

10. A 100 W bulb glows for 10 hours. The energy used is: ★★☆

(a) 1 kWh (b) 10 kWh
(c) 0.1 kWh (d) 1000 kWh

11. If the current in a heater coil is halved, the heat produced in the same time becomes: ★★★

(a) half (b) one-fourth
(c) double (d) four times

12. Heating elements are made of alloys rather than pure metals because alloys: NCERT ★★★

(a) have low resistivity (b) melt easily
(c) have high resistivity and do not oxidise readily (d) are good conductors

13. **Assertion (A):** The resistivity of a material does not depend on the length of the wire.
Reason (R): Resistivity is a characteristic property of the material at a given temperature. ★★☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

14. **Assertion (A):** In a parallel circuit, the equivalent resistance is less than the smallest resistance.
Reason (R): In a parallel circuit, the potential difference across each resistor is different. ★★☆

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

15. The SI unit of resistivity is: ★★☆

(a) Ω (b) Ω m
(c) Ω/m (d) Ω m^2

16. The device used to change the resistance in a circuit is a: ★☆☆

(a) ammeter (b) voltmeter
(c) rheostat (d) galvanometer

17. The conventional direction of electric current is: ★★☆

(a) the direction of electron flow (b) opposite to the direction of electron flow
(c) from negative to positive terminal outside the cell (d) random

MCQ TIP
Read all four options before you tick.

18. Wire A has twice the length and half the area of cross-section of wire B (same material). $R_A : R_B$ is: ★★★

(a) 1 : 1 (b) 2 : 1
(c) 4 : 1 (d) 1 : 4

19. A 60 W and a 100 W bulb (both 220 V) are joined in series to 220 V. Which glows brighter? ★★☆

(a) 100 W bulb (b) 60 W bulb
(c) both equally (d) neither glows

20. A fuse wire should have: ★★☆

(a) high melting point (b) low melting point
(c) very low resistance (d) high melting point and low resistance

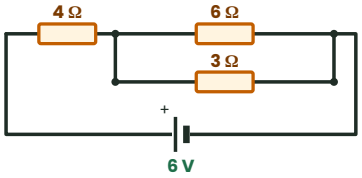
3 SECTIONS B AND C ★★★ MOST ASKED

Section B: Short Answers (2 marks each)

1. Define 1 volt. State Ohm's law.
2. Why are copper and aluminium wires used for electricity transmission?
3. Find the equivalent resistance of 10 Ω and 15 Ω (a) in series (b) in parallel.
4. Why is a series arrangement not used for domestic circuits? Give two reasons.
5. Draw a schematic diagram of a circuit with a battery of three 2 V cells, resistors of 5 Ω, 8 Ω and 12 Ω and a plug key, all in series. Find the current. NCERT

Section C: Short Answers (3 marks each)

1. The resistance of a metal wire of length 1 m is 26 Ω at 20 °C. If the diameter of the wire is 0.3 mm, what is the resistivity of the metal? NCERT
2. An electric iron consumes energy at 840 W when heating is at the maximum and 360 W when at the minimum, on 220 V. Find the current and resistance in each case. NCERT
3. A hot plate has two coils A and B of 24 Ω each. Find the current drawn from a 220 V supply when they are used (a) separately (b) in series (c) in parallel. NCERT
4. In the circuit below, find (a) the total resistance (b) the main current (c) the current in the 3 Ω resistor.



Resistivity aur electric iron wale numericals: pakka aayenge, isliye do baar practice karo!





BUJHO Section D mein units kaise nikalun? Watt ko seedha 30 se guna kar doon?

PAHELI Watt × hour = Wh; 1000 se bhaag do to kWh, phir din aur rate se guna karo!



4 SECTIONS D AND E ★★★ MOST ASKED

Section D, Case 1: Riya's home (4 marks)

Riya's home runs on a 220 V supply. Every day she uses 4 LED bulbs of 10 W for 5 h, 2 fans of 75 W for 10 h and a refrigerator of 200 W for 24 h. The electricity board charges ₹ 6 per unit.

1. How are these appliances connected, and why? **(1)**
2. Find the total energy used per day in kWh. **(1)**
3. Find the electricity bill for 30 days. **(1)**
4. What current flows in the main line when all of them run together? **(1)**

Case 2: choosing the right material (4 marks)

Resistivity (Ω m) at 20 °C: silver 1.60×10^{-8}, copper 1.62×10^{-8}, aluminium 2.63×10^{-8}, tungsten 5.20×10^{-8}, nichrome 100×10^{-6}, glass 10^{10} to 10^{14}.

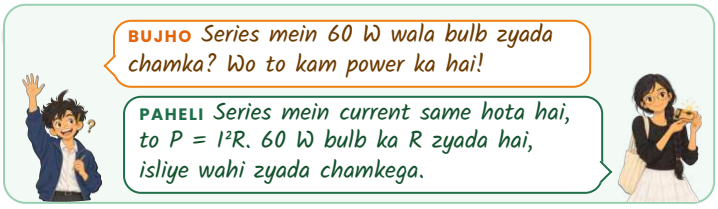
1. Which is the best conductor in the list? **(1)**
2. Which material is used in heating elements and which in bulb filaments? **(1)**
3. Why is copper used for wiring rather than silver? **(1)**
4. A wire's length and area are both doubled. What happens to its resistance and resistivity? **(1)**

Section E: Long Answers (5 marks each)

1. (a) Derive the expression for the equivalent resistance of three resistors R_1, R_2 and R_3 connected in series. **(3)** (b) Resistors of 5 Ω, 10 Ω and 15 Ω are connected in series to a 30 V battery. Find the current and the p.d. across each resistor. **(2)**
2. (a) Define electric power and derive $P = I^2R$ and $P = V^2/R$. **(2)** (b) A 1.5 kW geyser runs 2 h every day. Find the energy used in 30 days and the cost at ₹ 7 per unit. Which fuse (5 A or 10 A) suits it on 220 V? **(3)**

Common mistakes that cost marks

× WRONG	✓ RIGHT
Diameter used as radius in A = πr^2	Halve the diameter first
mm^2 left as it is	Convert to m^2: 1 mm = 10^{-3} m
Rated power used at a different voltage	R stays fixed; find R = V^2/P first
Unit missing in final answer	Every answer needs Ω, A, V, W or J



5 ANSWER KEY ★★★ MOST ASKED

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	$1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$.
2	(a)	V = W/Q.
3	(a)	Same current must pass through it.
4	(c)	Not on l or A.
5	(c)	l doubles, A halves: R × 4.
6	(b)	V/I = R.
7	(d)	Possible: 9, 1, 4.5, 2 Ω.
8	(b)	I = 150/60.
9	(c)	W = VQ = 12 × 2.
10	(a)	0.1 kW × 10 h.
11	(b)	$H \propto I^2$.
12	(c)	High ρ, no oxidation.
13	(a)	R explains A.
14	(c)	R is false: p.d. is the same.
15	(b)	ρ = RA/l: Ω × m^2/m.
16	(c)	Rheostat varies R.
17	(b)	Positive to negative outside the cell.
18	(c)	2 ÷ ½ = 4.
19	(b)	Same I; higher R (60 W) gets more I^2R.
20	(b)	It must melt on overload.

Score card: how did you fight the boss?

Marks	Result
54–60	BOSS DEFEATED
42–53	STRONG WIN: fix the small slips
30–41	ALMOST THERE: revise weak days
Below 30	RETRY: re-read Days 30–34

› **Where to revise:** Q1–3, 8, 9, 16, 17 → Day 30; Q4–6, 13, 15, 18 → Day 31; Q7, 14 → Days 32–33; Q10–12, 19, 20 → Day 34.

Bonus rapid-fire round: quick answers (from class notes)

Question	Answer
1 mA and 1 μA in ampere	10^{-3} A, 10^{-6} A
Charge on one electron	1.6×10^{-19} C
Instrument to detect a small current	galvanometer
Voltmeter: series or parallel?	parallel, high resistance
Ammeter resistance	very low
Resistance of a conductor depends on	l, A, material, temperature

Boss Level tracker: write your percentage

Ch 5 (Day 26)	Ch 6 (Day 29)	Ch 7 (Day 35)
____ %	____ %	____ %

Model answers: Sections B and C

B1 Volt and Ohm's law

1 V: p.d. when 1 J of work moves 1 C of charge (1 V = 1 J/C). Ohm's law: at constant temperature, V ∝ I, so V = IR. **(1 + 1)**

B2 Copper and aluminium

Very low resistivity, so little energy is lost as heat; they are also good, cheap conductors. **(2)**

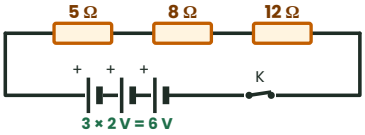
B3 10 Ω and 15 Ω

Series: 10 + 15 = **25 Ω**. Parallel: (10 × 15)/25 = **6 Ω**. **(1 + 1)**

B4 No series at home

One fault stops every device; all devices get the same current though they need different currents. **(1 + 1)**

B5 Series circuit



R = 5 + 8 + 12 = 25 Ω; I = 6/25 = **0.24 A**. **(1 diagram + 1)**

C1 Resistivity

GIVEN R = 26 Ω, l = 1 m, d = 0.3 mm, so r = 0.15×10^{-3} m

AREA A = $\pi r^2 = 3.14 \times (0.15 \times 10^{-3})^2 = 7.07 \times 10^{-8}$ m²

ANSWER ρ = RA/l = $26 \times 7.07 \times 10^{-8}$ = **1.84×10^{-6} Ω m**

C2 Electric iron

MAX I = P/V = 840/220 = **3.82 A**; R = V^2/P = 48400/840 = **57.6 Ω**

MIN I = 360/220 = **1.64 A**; R = 48400/360 = **134.4 Ω**

C3 Hot plate coils

(A) I = 220/24 = **9.17 A** (each coil alone)

(B) R = 48 Ω; I = 220/48 = **4.58 A**

(C) R = 12 Ω; I = 220/12 = **18.33 A**

C4 Mixed circuit

(A) 6 Ω ‖ 3 Ω = 18/9 = 2 Ω; total = 4 + 2 = **6 Ω**

(B) I = 6/6 = **1 A**

(C) p.d. across pair = 1 × 2 = 2 V; $I_{3\Omega}$ = 2/3 = **0.67 A**

Class questions: Nitesh Sir's favourites

Q1. Why is a voltmeter connected in parallel and an ammeter in series?

Ans. A voltmeter measures p.d. between two points, so it goes across them (high resistance, draws little current). An ammeter must carry the full current, so it goes in series (very low resistance).

Q2. Why does the resistance of a metal wire increase when it gets hot?

Ans. Resistivity of metals increases with temperature, so R = ρl/A increases.

Q3. How many electrons flow per second when a current of 1 A flows?

Ans. n = I t / e = $1/(1.6 \times 10^{-19})$ = **6.25×10^{18}** electrons.

Most asked in Chapter 7: Eduspark PYQ analysis

Topic	Rating
Series and parallel numericals	★★★
Resistivity: R = ρl/A numericals	★★★
Joule's law, power, kWh bill	★★★
Ohm's law with V–I graph	★★☆
Fuse, tungsten, nichrome reasons	★★☆

Model answers: Sections D and E

D1 Riya's home

(I) In **parallel**: each gets 220 V and works independently

(II) 4 × 10 × 5 + 2 × 75 × 10 + 200 × 24 = 6500 Wh = **6.5 kWh**

(III) 6.5 × 30 = 195 units; 195 × 6 = **₹ 1170**

(IV) P = 40 + 150 + 200 = 390 W; I = 390/220 = **1.77 A**

D2 Materials

(I) **Silver** (lowest ρ)

(II) Heating: **nichrome**; filament: **tungsten**

(III) Copper is almost as good and much cheaper

(IV) R = ρl/A: 2l/2A, so **R unchanged**; ρ unchanged (same material)

E1 Series resistors

(a) Same I in each; V = $V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3$. With V = IR_s: **$R_s = R_1 + R_2 + R_3$** (diagram 1 mark). (b) R_s = 30 Ω, I = **1 A**; V = **5 V, 10 V, 15 V**.

E2 Power and geyser

(a) P = rate of using electrical energy = W/t = VI; V = IR gives P = I^2R; I = V/R gives P = V^2/R. (b) E = 1.5 × 2 × 30 = **90 kWh**; cost = 90 × 7 = **₹ 630**; I = 1500/220 = 6.82 A, so the **10 A** fuse.

Day 35 checklist: tick before you sleep

- ☐ I attempted all 20 MCQs in 20 minutes
- ☐ I wrote Given, Formula and units in every numerical
- ☐ I can solve series, parallel and mixed circuits
- ☐ I can do resistivity, power and kWh bill numericals
- ☐ I checked my paper with the key and noted weak days

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Weak days phir se padho.



LEVEL 35 COMPLETE +200 XP earned. Next: Level 36



BUJHO *Boss defeat ho gaya! Electricity ab apni dost!*

PAHELI *Shabaash! Kal Level 36: Magnetic Effects of Electric Current.*

Coming up: Day 36

Magnetic Effects of Electric Current: magnetic field, field lines and the compass, the start of Chapter 8.



NITESH SIR SAYS

Galtiyan ek page pe likho aur Sunday ko yahi test dobara do. ***Consistency hi jalwa hai!***



PHYSICS | CHAPTER 7 COMPLETE ASTRA SERIES

CHAPTER 7 SWAHAA!

Electricity | Days 30–35



BUJHO *Ohm ka law, series-parallel, I squared R t, kilowatt-hour, sab numericals solve! Ab bijli ka bill bhi main check karunga.*

PAHELI *Zabardast, Bujho! Electricity Swahaa. Ab current ka magnetic jaadu dekhte hain.*

CHAPTER XP EARNED
+700 XP

LEVELS CLEARED
6 levels (Days 30–35)

BOOK PROGRESS
Level 35 / 60 4400 XP total

WHAT YOU HAVE MASTERED

- ✔ Charge, current, potential difference
- ✔ Resistance, resistivity and its factors
- ✔ Heating effect of current, Joule's law
- ✔ Ohm's law and V–I graph
- ✔ Resistors in series and in parallel
- ✔ Electric power, energy and kWh

NEXT CHAPTER Day 36: Magnetic Effects of Electric Current — *Electricity cleared!*

DAY 36 · LEVEL 36

PHYSICS | CHAPTER 8 | PART 1 · ASTRA SERIES

MAGNETS & MAGNETIC FIELD LINES

Goal: see the invisible field around a magnet and draw it right.



by Nitesh Prakash

01 Oersted | 02 Bar Magnet | 03 Magnetic Field | 04 Field Lines | 05 Properties | +100 XP today | Level 36 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

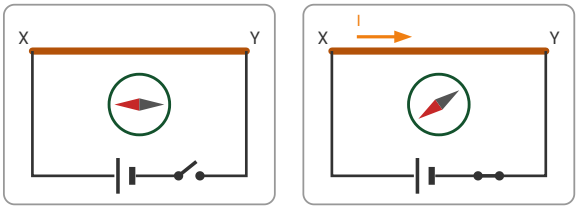
1 MAGNETS & OERSTED'S DISCOVERY

DEFINITION

- **Magnet:** an object that attracts iron, nickel and cobalt and points north–south when freely suspended
- **Natural magnet:** **lodestone** (magnetite, Fe_3O_4), a rock found in the Earth's crust
- **Man-made shapes:** bar magnet (long rectangle), horseshoe magnet (U-shaped), compass needle

Oersted's discovery (1820) ★★★

Hans Christian **Oersted** (19th century) noticed by accident that a compass needle placed near a wire **deflected** whenever current flowed. First proof that **electricity and magnetism are linked**.



Key open: needle rests along XY **Key closed: needle deflects**

- ✔ Reverse the current → the needle deflects the **opposite** way
- ✔ The unit of magnetic field strength, the **oersted**, is named after him

BUJHO *Wire mein sirf current hai, magnet toh hai hi nahi. Phir compass kyun ghooma?*

PAHELI *Kyunki **bahta hua current** khud apne chaaron taraf magnetic field banata hai. Yahi electricity aur magnetism ka rishta hai!*







MEME BREAK

*Current se compass hil gaya? Ab physics "**fysics**" nahi, full electromagnetism hai!*

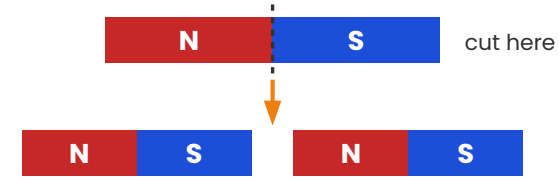
EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

The Earth is a magnet. A compass points north because the Earth behaves like a huge bar magnet whose **magnetic south pole lies near the geographic North Pole**. The north pole of the needle is attracted to it.

2 PROPERTIES OF A BAR MAGNET

Five facts to remember

1. A freely suspended bar magnet always comes to rest in the **north–south** direction. The end pointing north is the **north pole (N)**.
2. **Like poles repel**, unlike poles attract.
3. Poles always exist **in pairs** (a dipole). They **can never be separated**.
4. The attracting power is strongest at the **poles** and weakest in the middle.
5. The space around a magnet where its effect is felt is its **magnetic field**.



Each piece is a complete magnet: poles never separate

Magnetic and non-magnetic materials

Magnetic (attracted)	Non-magnetic
iron, steel, nickel, cobalt	wood, plastic, glass, copper, aluminium

BUJHO *Magnet ko beech se tod diya toh ek tukda sirf N aur doosra sirf S hoga?*

PAHELI *Nahi Bujho! Har tukda khud ek **poora magnet** ban jaata hai, naye N aur S ke saath. Akela pole kabhi nahi milta.*





How to find the N pole of an unmarked magnet

- Hang it freely by a thread; the end that settles facing **geographic north** is its N pole
- Check: that end **repels** the N pole of a compass needle

ATTRACTION IS NOT PROOF; REPULSION IS

An iron nail is attracted by both poles. Only **repulsion** proves that a body is a magnet, because it needs a like pole.

3 MAGNETIC FIELD & COMPASS NEEDLE

DEFINITION

- **Magnetic field:** the region around a magnet in which its force can be detected
- It has both **magnitude** (strength) and **direction**
- **Compass needle:** a tiny bar magnet; its north end shows the direction of the field at that point

Activity 1: iron filings show the pattern ★★☆

1. Fix a sheet of white paper on a drawing board and place a bar magnet at its centre.
2. Sprinkle iron filings evenly around it and **tap the board gently**.
3. The filings arrange themselves in a definite pattern of curves from pole to pole.

Why? Each filing becomes a tiny magnet and lines up along the field. The pattern shows the **magnetic field lines**.

Activity 2: plotting field lines with a compass

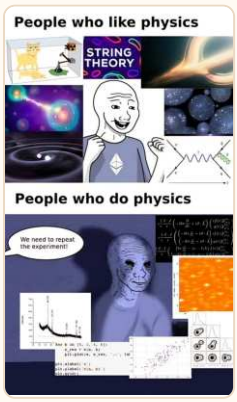
1	Place the magnet on paper and mark its outline. Put a compass near the **north pole**.
2	Mark the positions of the two ends of the needle with dots.
3	Move the compass so that its **south end** sits on the dot where the north end was. Mark again.
4	Repeat until you reach the south pole. Join the dots with a smooth curve: one field line.
5	Repeat from other points. The deflection is **largest near the poles**.

Iron filings vs compass: what each tells you

Tool	Shows the pattern?	Shows the direction?
Iron filings	Yes, the whole pattern at once	No
Compass needle	Yes, plotted point by point	**Yes**, the N end points along the field



★★★ *Field lines kabhi cross nahi karti: crossing point par compass do directions dikhayega, jo impossible hai. Nitesh Bhaiya ki guarantee!*

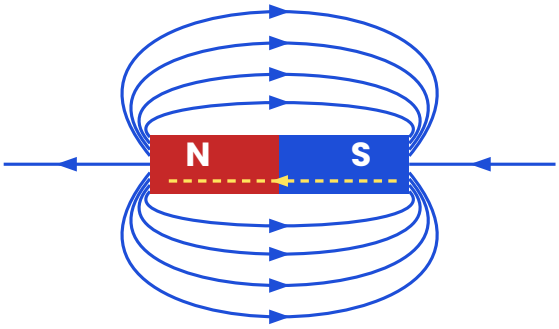


MEME BREAK
*Physics pasand karne wale: "wow, magnet!" Physics karne wale: compass ke dots **phir se plot karo**, experiment repeat!*

4 MAGNETIC FIELD LINES ★★★ MOST ASKED

DEFINITION

- **Magnetic field lines:** imaginary curves drawn to represent a magnetic field
- The **tangent** at any point gives the direction of the field there (the way a north pole would move)

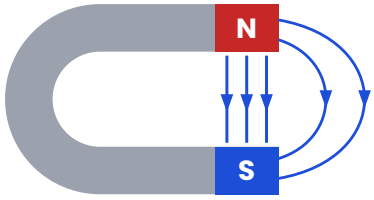


Outside: N → S | Inside the magnet: S → N (dashed)

Properties of field lines ★★★

1. Outside the magnet they **emerge from the north pole and merge at the south pole**; inside the magnet they run from S to N.
2. So they are **closed, continuous curves**.
3. **Crowded lines = stronger field.** They are closest at the poles (strongest) and far apart in the middle (weaker).
4. **No two field lines ever intersect.**

Uniform and non-uniform field



parallel, equally spaced = uniform

Horseshoe magnet: uniform field in the gap between the poles

- ✔ **Uniform field:** same strength and direction everywhere; drawn as **parallel, equally spaced** lines
- ✔ **Non-uniform field:** strength changes from point to point; lines curve and spread, as around a bar magnet



MEME BREAK
*Intezaar karte karte skeleton ban jaoge, par do field lines **kabhi nahi milengi**!*

Day 36 mind map: revise in 2 minutes

OERSTED 1820
Current in a wire deflects a compass: electricity makes magnetism

BAR MAGNET
Rests N–S; like poles repel; poles in pairs, never separate

MAGNETIC FIELD & FIELD LINES

FIELD
Region of magnetic force; has strength and direction

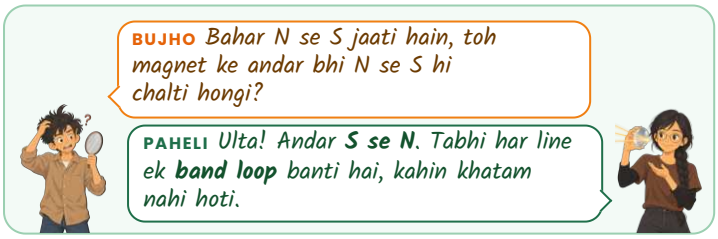
FIELD LINES ★★★
N to S outside, S to N inside; closed; never intersect

STRENGTH
Crowded lines = strong field; strongest at the poles

Why can field lines never intersect? ★★★

Class notes Q. Field lines show the **direction of the magnetic force** on a north pole. If two lines crossed, the field at that point would have **two directions**, and a compass placed there would point two ways at once. A force cannot have two directions at one point, so field lines never intersect.

× WRONG	✔ RIGHT
Field lines start at N and end at S	They are closed loops: N to S outside, S to N inside
The field is strongest in the middle	Strongest at the poles, where lines are crowded
Field lines can cross where two magnets meet	Never: the fields add to one direction
An attracted rod must be a magnet	Only repulsion proves it is a magnet





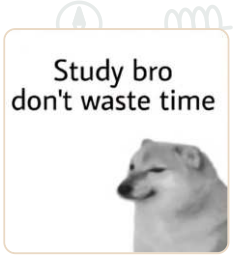
MEME BREAK
Do field lines ek dusre ko cross kar sakti hain? Bugs Bunny bole: ***No!*** *Rule yaad rakho: N se S, closed loops, no crossing.*

Exam tips

- ✔ Always put **arrows** on field lines; without arrows the diagram loses marks.
- ✔ Draw lines **closer near the poles** and wider in the middle.
- ✔ Label N and S on the magnet **before** drawing any field line.

5 PRACTICE MCQS

Study bro mode ON: phone side mein, 10 MCQs khud solve karo!



1. Oersted discovered that a current-carrying wire: ★★★
(a) attracts iron filings only at its ends
(b) deflects a nearby compass needle
(c) becomes hot and glows
(d) has no effect on a magnet

2. A freely suspended bar magnet comes to rest in the: ★★☆
(a) east–west direction
(b) north–south direction
(c) vertical direction
(d) any direction

3. Inside a bar magnet, the magnetic field lines run: ★★★
(a) from N to S
(b) from S to N
(c) in circles
(d) nowhere; there is no field inside

4. Where is the magnetic field of a bar magnet strongest? ★★☆
(a) At the middle
(b) At the poles
(c) Equal everywhere
(d) Far away from it

5. A bar magnet is broken into two equal halves at its middle. Each half: ★★★
(a) has only one pole
(b) loses all magnetism
(c) is a complete magnet with N and S
(d) has two north poles

6. The direction of the magnetic field at a point is given by: ★★☆
(a) the south end of a compass needle
(b) the north end of a compass needle
(c) iron filings only
(d) the length of the field line

7. Which property shows that two field lines cannot cross? ★★☆
(a) Field lines are closed curves
(b) Field at a point has only one direction
(c) Field lines are crowded at poles
(d) Field lines start at N

8. Lodestone is a natural magnet. It is an ore of: ★☆☆
(a) copper
(b) aluminium
(c) iron
(d) zinc

9. **Assertion (A):** Magnetic field lines never intersect each other. **Reason (R):** At the point of intersection, a compass needle would point in two directions, which is not possible. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Magnetic field lines are closed curves. **Reason (R):** Outside a magnet, field lines go from the south pole to the north pole. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Oersted, 1820: the needle deflected.
2	(b)	It points geographic north–south.
3	(b)	S to N inside, making closed loops.
4	(b)	Lines are most crowded at the poles.
5	(c)	Poles always exist in pairs.
6	(b)	North end points along the field.
7	(b)	One point, one direction of force.
8	(c)	Magnetite, Fe_3O_4, an iron ore.
9	(a)	R is the exact reason for A.
10	(c)	A true; R false: outside it is N to S.

Short and long answer questions

1. Reema placed a compass needle near a wire in a circuit and saw it deflect when the circuit was switched ON. Name the effect and the scientist who first observed it. What happens if the battery is reversed? CBSE 2025 **(2 marks)**
2. A student says that the magnetic field at X, near a pole of a bar magnet, is stronger than at Y, near its middle. Is the claim correct? Give a reason and draw the field lines with arrows. SQP 2025 **(3 marks)**
3. List four properties of magnetic field lines. Why do two field lines never intersect? **(3 marks)**

Model answers

1 Oersted's effect

- The magnetic effect of current; first seen by **Hans Christian Oersted** in 1820
- The current in the wire produces a magnetic field that turns the needle
- Reversing the battery reverses the current, so the needle deflects the **opposite way**

2 X is stronger

- Correct: field lines are **closer together** near the poles, so the field at X is stronger
- Draw closed N-to-S curves with arrows, crowded at the poles

3 Properties

- N to S outside, S to N inside; closed curves; crowded where strong; never intersect
- If they crossed, one point would have two field directions, which is impossible

Case-based question (4 marks)

Riya sprinkled iron filings around a bar magnet on a sheet of paper and tapped it. Then she placed a compass at different points.

1. Why do the filings form a pattern? **(1)**
2. Where are the filings most crowded, and what does that show? **(1)**
3. Can the filings alone show the direction of the field? What should she use? **(2)**

Day 36 checklist: tick before you sleep

- [ ] I can state Oersted's observation and what it proved
- [ ] I know five properties of a bar magnet
- [ ] I can plot field lines with a compass and explain iron filings
- [ ] I can draw bar-magnet field lines with correct arrows
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 36 COMPLETE +100 XP earned. Next: Level 37



BUJHO Samajh gaya! Field lines N se S, band loop, aur kabhi cross nahi.

PAHELI Shabaash! Kal Level 37: current wale taar ka magnetic field.

Day 36 hit-list: Eduspark PYQ analysis

Topic	Rating
Properties of magnetic field lines	★★★
Why field lines never intersect	★★★
Drawing bar-magnet field lines with arrows	★★★
Oersted's experiment	★★☆
Plotting field lines with a compass	★★☆
Poles cannot be separated	★☆☆



NITESH SIR SAYS

Magnet ko samjho toh field lines apne aap ban jaati hain: N se S, band loop, kabhi cross nahi. ***Kal milte hai Day 37 pe!***

Case-based answers

- (i) Each filing is magnetised and lines up along the field
- (ii) At the poles, where the field is strongest
- (iii) No, filings show only the pattern; a compass shows direction

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 37 · LEVEL 37

PHYSICS | CHAPTER 8 | PART 2 · ASTRA SERIES

FIELD OF A STRAIGHT WIRE & A LOOP

Goal: find the field around a current and its direction.



by Nitesh Prakash

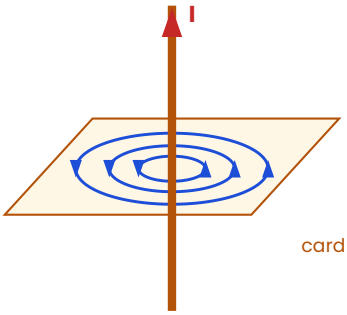
01 Straight Wire | 02 Thumb Rule | 03 B ∝ I, 1/r | 04 Circular Loop | 05 Coil Turns | +100 XP today | Level 37 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

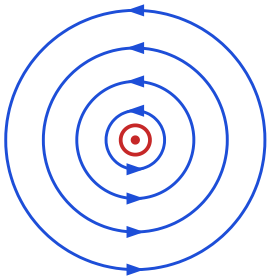
1 FIELD AROUND A STRAIGHT CONDUCTOR

Activity: wire through a cardboard ★★☆

1. Pass a thick copper wire **vertically** through the centre of a horizontal cardboard; connect it to a battery, key and rheostat.
2. Sprinkle iron filings on the cardboard, switch on the current and **tap gently**.
3. The filings form **concentric circles** centred on the wire. A compass placed on a circle shows the direction.



Current up: anticlockwise



Top view: current out of page gaps grow outward = weaker

DEFINITION

- **Pattern:** field lines around a straight wire are **concentric circles** in a plane perpendicular to the wire, centred on the wire
- **Spacing:** the circles get larger and farther apart away from the wire, because the field **fades with distance**
- **Reverse the current** → the direction of the circles reverses

SYMBOLS YOU MUST KNOW

⊙ current coming **out of** the page (you see the arrow tip); ⊗ current going **into** the page (you see the tail feathers).



BUJHO *Straight wire ka field bhi straight line mein hoga na?*

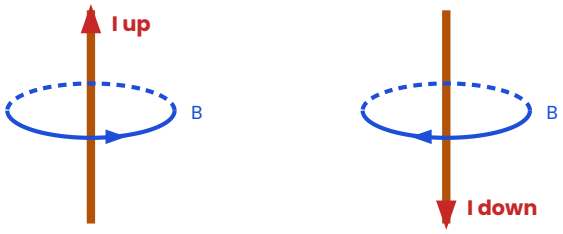
PAHELI *Nahi Bujho! Wire seedha hai, par field uske chaaron taraf **gol gol circles** mein ghoomta hai, wire ke perpendicular plane mein.*



2 RIGHT-HAND THUMB RULE ★★★ MOST ASKED

DEFINITION

- **Maxwell's right-hand thumb rule:** hold the current-carrying wire in your right hand with the **thumb along the current**; the **curled fingers** give the direction of the magnetic field lines
- It gives the **direction** of the field when the direction of the current is known



Direction at a glance ★★★

Current in a vertical wire	Field seen from above
Upwards (thumb up)	**Anticlockwise**
Downwards (thumb down)	**Clockwise**

Note on the class notes: "thumb up, fingers clockwise" is the view from **below**. Seen from **above** (the NCERT convention), current up gives an **anticlockwise** field.

Current upar ja raha hai... field clockwise hai ya anticlockwise??



MEME BREAK

*Clockwise ya anticlockwise? Mr Krabs mat bano, **right hand ka thumb** current ki taraf karo, ungliyan jawab de dengi!*



★★★ *Thumb = current, fingers = field. Ye rule har saal aata hai, Nitesh Bhaiya ki guarantee!*

3 WHAT DECIDES THE STRENGTH? ★★★

Two factors that decide B

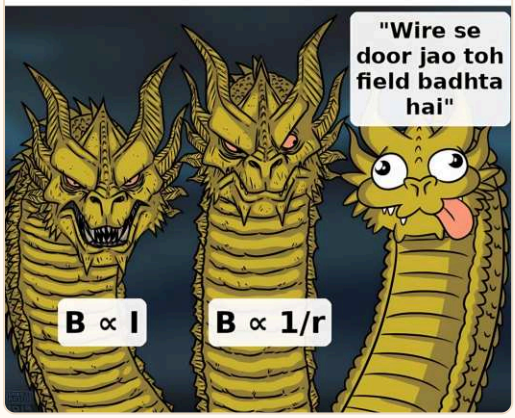
Factor	Relation	Meaning
Current, I	**B ∝ I**	double the current, double the field
Distance, r	**B ∝ 1/r**	farther from the wire, weaker field

Note: the class notes write "B = 1/r"; the correct form is **B ∝ 1/r** (proportional, not equal).

Activity: change I, then change r

- Increase the current with the **rheostat**: the compass deflection **increases**
- Keep the current fixed and move the compass **away** from the wire: the deflection **decreases**
- Conclusion: B rises with I and falls with distance r

Straight wire ke magnetic field ke rules:



MEME BREAK

*Do heads sahi: B ∝ I aur B ∝ 1/r. Teesra head galat: wire se door jao toh field **kam** hota hai, badhta nahi.*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

SI unit of magnetic field: the **tesla (T)** (from the class notes).
For a long straight wire, $B = \mu_0 I / (2\pi r)$, which shows B ∝ I and B ∝ 1/r together.
Clock-face rule for a coil: current **anticlockwise** as you look at the face → that face is a **north pole**; clockwise → south pole.

Solved Example 1: two parallel wires ★★☆

- **Q.** Two long parallel wires carry equal currents. What is the field midway between them when the currents are (a) in the same direction, (b) in opposite directions?
- (a) The two fields are equal and opposite there → **zero**
- (b) The two fields point the same way → **twice** the field of one wire

Solved Example 2: power line overhead ★★★

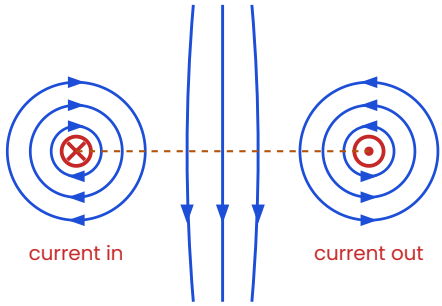
Q. A current flows through a horizontal power line from **east to west**. Find the direction of the magnetic field at a point directly below it and directly above it. NCERT

- Point the right thumb from east to west along the wire
- Fingers curl: **below** the wire the field points from **north to south**
- **Above** the wire the field points from **south to north**
- If the current flows west to east, both directions reverse

4 FIELD DUE TO A CIRCULAR LOOP

Activity: a loop through a cardboard

Bend a wire into a circular loop and pass it through a cardboard so that the cardboard cuts the loop at two points. Sprinkle iron filings and pass the current.



Section through the loop: almost straight lines at the centre

What the pattern shows ★★★

1. Near each point where the wire crosses the cardboard, the lines are **concentric circles** (each small part of the loop acts like a straight wire).
2. The circles grow **larger** as we move away from the wire.
3. At the **centre**, the arcs become so large that the lines are **almost straight** and **perpendicular to the plane** of the loop.
4. Every part of the loop makes the field at the centre point in the **same direction**, so the fields add up.

Strength at the centre of a coil

Increase	Field at centre
Current, I	increases
Number of turns, n	increases n times
Radius of the loop	decreases

Solved Example 3: change the coil ★★☆

- **Q.** A coil gives a field B at its centre. What is the field if (a) the turns are doubled, (b) the current is halved, (c) both are done together?
- (a) B ∝ n → **2B**
- (b) B ∝ I → **B/2**
- (c) 2 × 1/2 = 1 → field stays **B**

THINK OF IT THIS WAY

A loop is many tiny straight wires; inside it, all their fields point the **same way**.



BUJHO *Loop ke centre pe bhi field lines circle hongi kya?*

PAHELI *Centre pe circles itne bade ho jaate hain ki lines **lagbhag seedhi** dikhti hain, aur loop ke plane ke **perpendicular** hoti hain.*



Day 37 mind map: revise in 2 minutes

STRAIGHT WIRE
Concentric circles in a plane perpendicular to the wire

THUMB RULE ★★★
Thumb along current, curled fingers give the field

FIELD OF A CURRENT

FACTORS
B ∝ I and B ∝ 1/r for a straight wire

LOOP
Circles near the wire; almost straight at the centre

COIL
n turns give n times the field; smaller radius, stronger field

Straight wire vs circular loop

	Straight wire	Circular loop
Shape of lines	concentric circles	circles near the wire, straight at the centre
Strongest	close to the wire	near the wire; at the centre all parts add
Depends on	I, r	I, n, radius

× WRONG	✓ RIGHT
B = 1/r	B ∝ 1/r: proportional, not equal
Field lines are straight along the wire	Concentric circles around the wire
Use the left hand for the field of a wire	Right-hand thumb rule
More turns, same field at the centre	n turns give n times the field



BUJHO *Do parallel wires mein same direction mein current, toh beech mein field double ho jayega?*

PAHELI *Ulta! Beech ke point par dono wires ke fields **barabar aur opposite** hote hain, isliye total **zero**. CBSE 2023 mein yahi poocha tha.*





MEME BREAK
*Thumb rule mein left hand laga diya? Physics se nafrat ho jayegi! **Right hand** hi yaad rakho.*

Exam tips

- ✓ Always say **"viewed from above"** or **"viewed from below"** with clockwise or anticlockwise.
- ✓ Use ⊙ for current out of the page and ⊗ for current into the page.
- ✓ For a coil, mention **all three** factors: current, number of turns and radius.

5 PRACTICE MCQS

1. The magnetic field lines around a straight current-carrying wire are: ★★★

(a) straight lines parallel to the wire
(b) concentric circles centred on the wire
(c) radial lines starting from the wire
(d) ellipses

2. A vertical wire carries current upwards. Seen from above, the field lines are: ★★★

(a) clockwise
(b) anticlockwise
(c) straight up
(d) straight down

3. If the distance from a straight wire is doubled, the magnetic field becomes: ★★☆

(a) double
(b) half
(c) four times
(d) unchanged

4. The rule used to find the direction of the field around a wire is: ★★★

(a) Fleming's left-hand rule
(b) right-hand thumb rule
(c) Ohm's law
(d) Snell's law

5. At the centre of a current-carrying circular loop, the field lines are: ★★★

(a) concentric circles
(b) almost straight, perpendicular to the loop
(c) parallel to the plane of the loop
(d) absent

6. The resultant field at a point midway between two long parallel wires carrying equal currents in the same direction is: CBSE 2023 ★★☆

(a) double of one wire
(b) along the wires
(c) zero
(d) vertically upward

7. Which change does NOT increase the field at the centre of a coil? ★★☆

(a) more turns
(b) more current
(c) a larger radius
(d) a smaller radius

8. A current flows east to west in a horizontal wire. The field directly below it points: ★★★

(a) north to south
(b) south to north
(c) east to west
(d) vertically up

9. **Assertion (A):** The field at the centre of a current-carrying circular coil increases when its number of turns is increased.
Reason (R): The current in each turn has the same direction, so the fields of all turns add up. CBSE 2023 ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The magnetic field of a straight current-carrying wire is the same at all distances from it.
Reason (R): Field lines of a straight wire are concentric circles. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Concentric circles in a perpendicular plane.
2	(b)	Thumb up: fingers curl anticlockwise from above.
3	(b)	B ∝ 1/r, so double r gives half B.
4	(b)	Maxwell's right-hand thumb rule.
5	(b)	Large arcs look straight at the centre.
6	(c)	The two fields are equal and opposite.
7	(c)	Larger radius means weaker field at the centre.
8	(a)	Thumb west, fingers curl N to S below.
9	(a)	Fields of all turns add; R explains A.
10	(d)	A false: B falls with distance. R true.

Short and long answer questions

1. Draw a labelled diagram to show the pattern of magnetic field lines produced by a current-carrying straight conductor. Mark the direction of the field lines. CBSE 2024 **(2 marks)**
2. Mona noticed that the deflection of a compass needle fell as she moved it away from a current-carrying wire. (a) Why? (b) Name one change in the circuit that would increase the deflection. SQP 2024 **(2 marks)**
3. State the right-hand thumb rule. Draw the field lines of a current-carrying circular loop and list three factors on which the field at its centre depends. **(5 marks)**

Model answers

1 Straight conductor

- Concentric circles centred on the wire, closer near it; for current upward the arrows run anticlockwise seen from above (1 mark circles, 1 mark arrows)

2 Mona's compass

- (a) B ∝ 1/r: the field weakens with distance
- (b) Increase the current, e.g. add a cell or lower the rheostat resistance

3 Thumb rule and loop

- Thumb along the current, curled fingers give the field direction
- Circles near the wire, almost straight lines at the centre, perpendicular to the loop
- Factors: current, number of turns, radius (smaller radius, stronger field)

Case-based question (4 marks)

A vertical wire passes through a horizontal card. With current flowing upwards, a student places compasses at points P (near the wire) and Q (farther away).

1. What is the shape of the field lines on the card? **(1)**
2. Seen from above, which way do the lines run? **(1)**
3. At which point is the field stronger, and why? What happens to the directions if the current is reversed? **(2)**

Day 37 checklist: tick before you sleep

- [ ] I can describe the field pattern around a straight wire
- [ ] I can state and apply the right-hand thumb rule
- [ ] I know B ∝ I and B ∝ 1/r for a straight wire
- [ ] I can draw the field of a circular loop and list its three factors
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 37 COMPLETE +100 XP earned. Next: Level 38



BUJHO *Samajh gaya! Thumb current ki taraf, ungliyan field ki taraf, aur loop ke centre pe lines seedhi.*

PAHELI *Shabaash! Kal Level 38: solenoid aur electromagnet.*

Day 37 hit-list: Eduspark PYQ analysis

Topic	Rating
Right-hand thumb rule and its use	★★★
Field pattern of a straight wire (diagram)	★★★
Factors: current and distance	★★★
Field of a circular loop (diagram)	★★☆
Effect of number of turns	★★☆
Field midway between two wires	★☆☆



NITESH SIR SAYS

*Haath sahi chuno toh direction kabhi galat nahi hogi: **right hand**, thumb current ki taraf. **Kal milte hai Day 38 pe!***

Case-based answers

- (i) Concentric circles centred on the wire
- (ii) Anticlockwise (right-hand thumb rule)
- (iii) At P, since B ∝ 1/r; reversing the current reverses every direction

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 38 LEVEL 38

PHYSICS | CHAPTER 8 | PART 3 ASTRA SERIES

SOLENOID & ELECTROMAGNET

Goal: make a magnet you can switch on and off.



by Nitesh Prakash

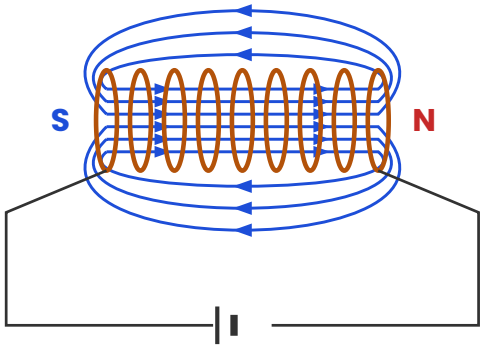
01 Solenoid | 02 Uniform Field | 03 Strength | 04 Electromagnet | 05 Comparison | +100 XP today | Level 38 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 THE SOLENOID

DEFINITION

- **Solenoid:** a coil of many circular turns of **insulated copper wire** wrapped closely in the shape of a **cylinder**
- When current flows, it produces a field just like a **bar magnet**: one end behaves as a **north pole**, the other as a **south pole**



Inside: parallel lines S → N (uniform) | Outside: N → S

Field pattern of a solenoid ★★★

1. **Outside:** field lines run from the **N end to the S end**, like a bar magnet.
2. **Inside:** lines run from **S to N** and are **parallel straight lines**.
3. Parallel, equally spaced lines mean the field inside is the **same at all points**: a **uniform field**.
4. Inside and outside lines together form **closed loops**.

Activity: make your own solenoid ★★☆

1. Wind insulated copper wire closely on a pencil or a cardboard tube and connect it to a cell and a key.
2. Close the key and bring a compass near each end.
3. One end **attracts** the north pole of the needle and the other **repels** it: the coil is now a magnet; reversing the cell swaps the ends.



★★★ *Solenoid ke andar field uniform: parallel, barabar doori wali lines. Board mein pakka, Nitesh Bhaiya ki guarantee!*

Solenoid vs bar magnet

	Solenoid	Bar magnet
Field pattern	same: N to S outside, S to N inside	
Freely suspended	both rest in the north–south direction	
Switch off	field disappears	stays a magnet
Reverse polarity	reverse the current	not possible



MEME BREAK

*Solenoid ne bar magnet ka pura field pattern **copy** kar liya, Mr Bean style! Bas switch off karte hi magnetism gayab.*

Which end is north?

- Curl the fingers of your right hand along the current in the turns: the **thumb points to the N end**
- Reverse the current and the N and S ends swap

2 STRENGTH OF THE SOLENOID'S FIELD ★★★

It depends on (class notes)

1. **Number of turns**: more turns, stronger field
2. **Current** through the coil: more current, stronger field
3. **Material of the core**: a soft iron core makes it much stronger
4. **Radius of the coil**: a narrower coil gives a stronger field at its centre



BUJHO *Solenoid ke andar field uniform hai, toh beech aur end pe same strength hogi?*

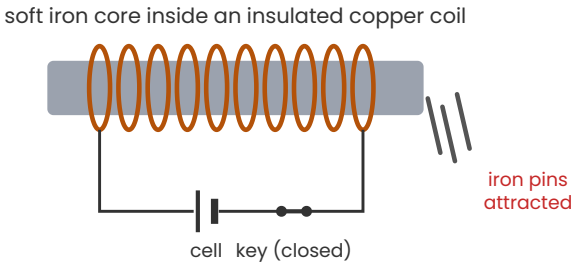
PAHELI *Andar ke lambe hisse mein haan, **uniform**. Sirf bilkul ends ke paas lines phailne lagti hain, wahan field thoda kam hota hai.*



3 ELECTROMAGNET ★★★ MOST ASKED

DEFINITION

- **Electromagnet:** a **temporary magnet** made by placing a **soft iron core** inside a current-carrying solenoid
- It is a magnet **only while current flows**; it is demagnetised when the current is switched off



Four properties (class notes)

1. **Temporary:** magnetism lasts only while the current flows.
2. **Polarity can be reversed** by reversing the direction of the current.
3. **Strength can be changed** by changing the current (or the number of turns).
4. It is much stronger than the solenoid alone because the soft iron core gets magnetised.

Note on the class notes: polarity is **reversed** (N and S swap) by reversing the current; it is the **strength** that is increased or decreased.



MEME BREAK

*Switch off karte hi magnet gayab! Yahi hai electromagnet ka **temporary** nature: current gaya, magnetism gaya.*

Activity: making an electromagnet

- Wind about 50 turns of insulated wire on a large iron nail; connect a cell and a key
- Close the key: the nail picks up pins. Open it: the pins drop
- Add more turns or one more cell: it picks up **more** pins

Why soft iron, not steel? ★★☆

- Soft iron is **magnetised strongly** by the solenoid and **loses its magnetism at once** when the current stops
- Steel keeps its magnetism after the current stops, so a steel core gives a **permanent magnet**
- A copper or wooden core does not get magnetised at all

Uses of electromagnets

- ✔ **Cranes** in scrap yards lift and drop heavy iron loads by switching the current on and off
- ✔ **Electric bells**, relays and electric door locks
- ✔ **Loudspeakers** and headphones
- ✔ Separating iron pieces from waste and from ores

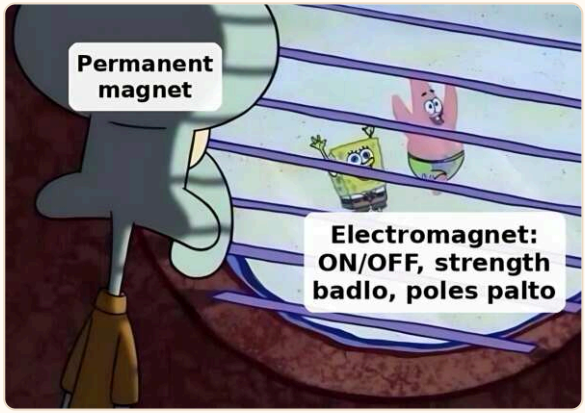
CLASS TRICK

"Permanent magnet ka properties theek ulta hain electromagnet se :)"

Electromagnet: temporary, strength badlo, polarity palto. Permanent magnet: sab ulta, sab fixed.

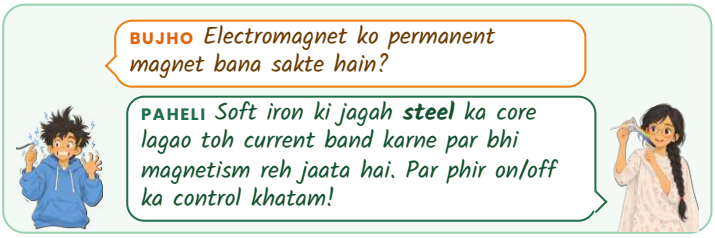
4 ELECTROMAGNET VS PERMANENT MAGNET ★★★

Property	Electromagnet	Permanent magnet
Nature	temporary	permanent
Strength	can be changed	fixed
Polarity	can be reversed	cannot be reversed
Core	soft iron	steel, alloys such as alnico
Needs current?	yes	no
Strength possible	very strong	comparatively weak



MEME BREAK

*Electromagnet: on/off, strength aur poles sab control mein. Permanent magnet window se dekhta reh gaya: "main toh bas **fixed** hoon bhai."*



Increase the strength (material fixed) ★★★

- **Increase the current** through the coil
- **Increase the number of turns** in the coil

Asked as: "List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed." **CBSE 2020**

Day 38 mind map: revise in 2 minutes

SOLENOID
Many close turns of insulated copper wire in a cylinder

FIELD ★★★
Like a bar magnet; uniform, parallel lines inside

SOLENOID & ELECTROMAGNET

STRENGTH
Turns, current, core material, coil radius

ELECTROMAGNET
Soft iron core in a solenoid; temporary magnet

VS PERMANENT
Electromagnet: variable, reversible; permanent: fixed

Conceptual Q&A: think, then read ★★★

1 Can a freely suspended current-carrying solenoid stay in any direction?
> No. It behaves like a bar magnet and rests in the **north–south** direction; reversing the current turns it round

2 Why is the field inside a long solenoid uniform?
> The fields of all the turns point the same way inside and add up evenly, giving parallel, equally spaced lines

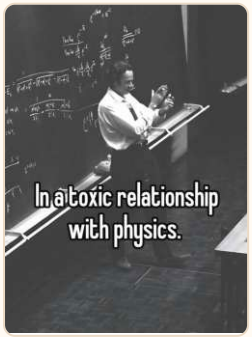
3 A copper rod is put inside the solenoid. Does it become a magnet?
> No. Copper is not a magnetic material; only a magnetic material such as soft iron gets magnetised

4 Why are field lines inside a solenoid drawn S to N?
> Field lines are closed loops: they leave the N end outside and must return through the inside

× WRONG	✓ RIGHT
Field inside a solenoid is zero	Strong and uniform: parallel lines S to N
Electromagnets use a steel core	Soft iron core: loses magnetism quickly
Reversing current increases strength	Reversing current reverses the polarity
Any metal core will do	Only magnetic materials (soft iron) work

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

For a long solenoid, $B = \mu_0 nI$, where n is the number of turns per metre; inside a long solenoid the radius hardly matters.
Electric motor, electromagnetic induction and the generator are no longer in the CBSE Class 10 syllabus.



MEME BREAK
*Physics ke saath toxic relationship? Solenoid, electromagnet, sab ek hi kahani hai: **current se magnet**. Pyaar ho jayega!*

5 PRACTICE MCQS

1. A solenoid is a coil of many turns of insulated copper wire wound in the shape of a: ★★☆
(a) sphere (b) cylinder
(c) cone (d) flat spiral

2. The magnetic field inside a long straight current-carrying solenoid is: CBSE 2025 ★★★
(a) zero (b) maximum at the centre only
(c) uniform throughout its length (d) maximum at the ends

3. The field pattern of a current-carrying solenoid resembles that of a: ★★★
(a) straight wire (b) horseshoe magnet
(c) bar magnet (d) circular loop only

4. Which core makes the best electromagnet? ★★★
(a) steel (b) copper
(c) soft iron (d) wood

5. The polarity of an electromagnet can be reversed by: ★★☆
(a) increasing the current (b) reversing the current
(c) adding more turns (d) removing the core

6. Which of these is NOT a factor for the strength of a solenoid's field? ★☆☆
(a) number of turns (b) current
(c) nature of the core (d) colour of the wire

7. Steel is used to make permanent magnets because it: ★★☆
(a) loses magnetism quickly (b) retains magnetism
(c) is a poor conductor (d) is non-magnetic

8. Electromagnets are used in all of these EXCEPT: ★★☆
(a) electric bell (b) scrap-yard crane
(c) loudspeaker (d) dry cell

9. **Assertion (A):** A soft iron core is used in an electromagnet.
Reason (R): Soft iron is magnetised strongly and loses its magnetism as soon as the current is switched off. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The magnetic field lines inside a solenoid are parallel straight lines.
Reason (R): The field inside a long solenoid is uniform. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Case-based question (4 marks)

A scrap-yard crane lifts old cars with a large electromagnet and drops them into a crusher.

1. Which core material should the electromagnet have, and why? **(1)**
2. How does the operator drop the load? **(1)**
3. Give two ways to lift heavier loads with this core. **(1)**
4. How can the poles of the electromagnet be swapped? **(1)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Turns wound closely as a cylinder.
2	(c)	Parallel, equally spaced lines inside.
3	(c)	One end N, other end S.
4	(c)	Strong, temporary magnetisation.
5	(b)	N and S swap with the current.
6	(d)	Colour has no effect.
7	(b)	Steel keeps its magnetism.
8	(d)	A dry cell is a source, not a magnet.
9	(a)	R explains why soft iron is used.
10	(a)	Uniform field is shown by parallel lines.

Short and long answer questions

1. Define the term solenoid. Draw the pattern of field lines around a current-carrying solenoid and state how it can be used to magnetise soft iron. SQP 2025 **(3 marks)**
2. Why can two field lines never cross? Draw the field lines of a long current-carrying solenoid, state the conclusion from the lines inside, and name two factors on which its field depends. CBSE 2025 **(5 marks)**
3. A current-carrying solenoid is suspended freely. Will it stay in any direction? What happens when the current is reversed? CBSE 2020 **(2 marks)**

Model answers

1 Solenoid and soft iron

- A coil of many close turns of insulated copper wire in the shape of a cylinder
- Diagram: N to S outside, parallel lines S to N inside
- Soft iron placed inside is magnetised and becomes an electromagnet

2 Field lines and factors

- If they crossed, the field there would have two directions: impossible
- Parallel lines inside: the field is uniform there
- Factors: number of turns, current (also the core material)

3 Suspended solenoid

- No: it settles north–south like a bar magnet
- Reversing the current swaps its poles, so it turns through 180°

Case-based answers

- (i) Soft iron: magnetised strongly, loses magnetism at once when switched off
- (ii) Switch off the current: the core is demagnetised
- (iii) Increase the current or the number of turns
- (iv) Reverse the current



BUJHO *Solenoid ke bahar bhi field uniform hoga na?*

PAHELI *Nahi! Bahar lines mudti hain. Uniform sirf **andar** hai.*

Day 38 checklist: tick before you sleep

- [ ] I can define a solenoid and draw its field lines
- [ ] I know why the field inside a solenoid is uniform
- [ ] I can list four factors for the strength of a solenoid's field
- [ ] I can compare an electromagnet with a permanent magnet
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 38 COMPLETE +100 XP earned. Next: Level 39



BUJHO *Samajh gaya! Solenoid = bar magnet ki copy. Soft iron daalo toh electromagnet.*

PAHELI *Shabaash! Kal Level 39: current wale taar par force aur ghar ki wiring.*

Day 38 hit-list: Eduspark PYQ analysis

Topic	Rating
Field lines of a solenoid (diagram)	★★★
Uniform field inside a solenoid	★★★
Electromagnet: definition and soft iron core	★★★
Electromagnet vs permanent magnet	★★☆
Factors for the strength of the field	★★☆
Uses of electromagnets	★☆☆



NITESH SIR SAYS

*Solenoid samajh liya toh electromagnet free mein mil gaya: coil + current + soft iron. **Kal milte hai Day 39 pe!***

Coming up: Day 39

Force on a current-carrying conductor, Fleming's left-hand rule, AC and DC, and domestic electric circuits.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 39 · LEVEL 39

PHYSICS | CHAPTER 8 | PART 4 · ASTRA SERIES

FLEMING'S RULE & HOME CIRCUITS

Goal: find the force direction, then wire a safe home.



by Nitesh Prakash

01 Force on Wire | 02 Fleming's LHR | 03 AC vs DC | 04 Home Wiring | 05 Fuse & Safety | +100 XP today | Level 39 / 60

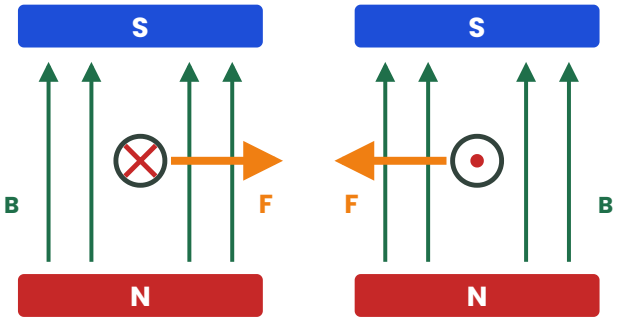
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 FORCE ON A CURRENT-CARRYING WIRE ★★★

A current-carrying wire makes its own magnetic field (Day 37). So when it is placed in the field of a magnet, the two fields push on each other: the **magnet exerts a force on the conductor**, and (as Ampère suggested) the conductor exerts an equal and opposite force on the magnet.

Activity: the jumping aluminium rod

A light aluminium rod AB hangs horizontally from two wires between the poles of a horseshoe magnet (N below, S above, so the field points **upward**). It is connected in series with a battery, a key and a rheostat.



- Key closed: the rod is **pushed sideways** (displaced)
- Current reversed: the rod moves the **opposite way**
- Magnet flipped (field reversed): direction again reverses
- Rheostat lowered (more current): bigger displacement

KEY RESULTS

- The force is **maximum** when the current is **perpendicular** to the field
- The force is **zero** when the current is **parallel** to the field
- Its direction depends on **both** the current and the field
- Its size grows with current, field strength and length of wire in the field



BUJHO *Current bhi ulta, magnet bhi ulta: toh rod bhi ulti jayegi na?*

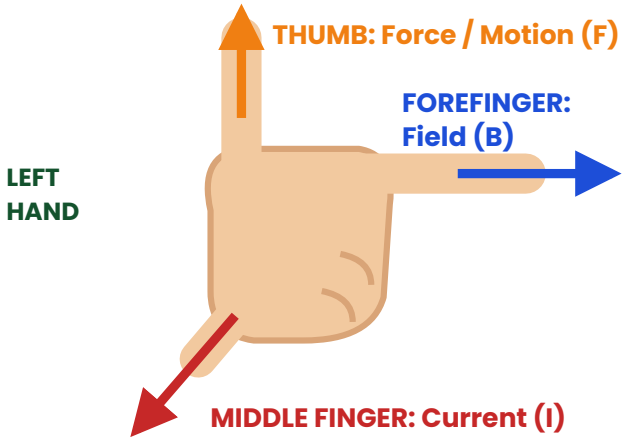
PAHELI *Nahi! Do baar ulta matlab seedha. Force ki direction* ***wahi rahegi.*** *Sirf ek ko reverse karo tab direction badlegi.*



2 FLEMING'S LEFT-HAND RULE ★★★ MOST ASKED

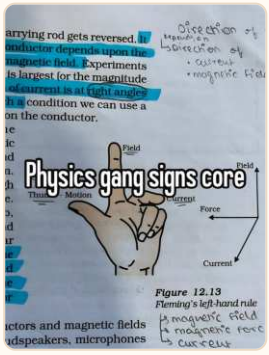
STATEMENT

Stretch the **thumb, forefinger and middle finger of the left hand** so that they are mutually perpendicular. If the **forefinger** points along the **magnetic field** and the **middle finger** along the **current**, then the **thumb** points along the **motion or force** on the conductor.



MEMORY TRICK: FBI ON THE LEFT HAND

Force = thumb, **B** (field) = forefinger, **I** (current) = middle finger. Thumb upar, F-B-I order mein ungliyaan kholo.



MEME BREAK

Exam hall mein sabke haath hawa mein! ***Left hand*** *hi use karna, right hand wala thumb rule Day 37 ka hai.*

Solved Example 1: find the field NCERT

A positively charged alpha particle moving **towards west** is deflected **towards north**. Find the direction of the magnetic field.

- Current = direction of positive charge = **west** (middle finger)
- Force = **north** (thumb)
- Forefinger then points **upward**: the field is **vertically upward**

Solved Example 2: an electron beam

An electron beam moves to the **right** through a field pointing **into the page**. Which way is the force?

- Electrons move right, so conventional current is to the **left**
- Field into the page, current left: thumb points **downward**
- Force on the beam is **towards the bottom of the page**

Devices using this force: electric motor, electric fan, loudspeakers and moving-coil meters (galvanometer, ammeter). Their working is not in the current syllabus.

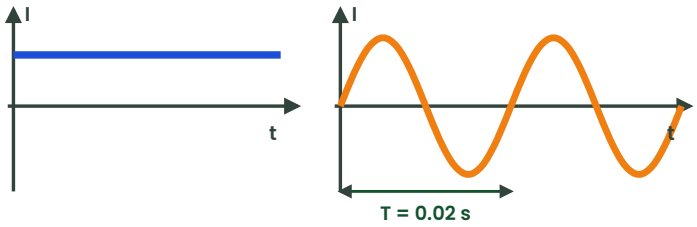
FROM CLASS NOTES: MAGNETISM INSIDE YOU

Nerve impulses are tiny currents, so the **heart and the brain** produce weak but significant magnetic fields. **MRI** (magnetic resonance imaging) uses magnetic fields to form images of the inside of the body.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Size of the force: F = B I L sin θ, where θ is the angle between the wire and the field. At θ = 90°, F = BIL (maximum); at θ = 0°, F = 0. That is why the rod moves most when it is perpendicular to the field.

3 DC VS AC: KINDS OF CURRENT ★★☆

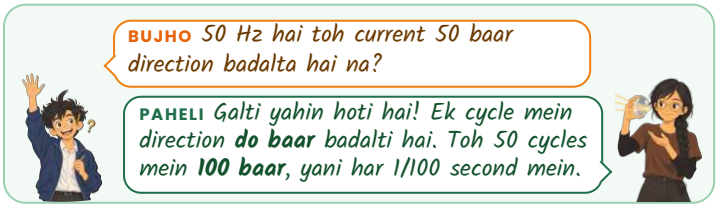


DC: one direction — AC: reverses direction

Point	DC (direct current)	AC (alternating current)
Direction	always one direction	reverses periodically
Sources	cells, batteries	power stations, AC generators
Storage	can be stored in cells	cannot be stored
Transmission	big losses over long distances	long distances with **little energy loss**

IN INDIA

- Frequency of AC = **50 Hz**: 50 complete cycles every second
- Time period T = 1/f = 1/50 = **0.02 s**
- Direction reverses **twice per cycle**, i.e. after every **1/100 s**, or **100 times a second**



QUICK CHECK

Direction changes of Indian AC in one minute: 2 per cycle × 50 cycles per second × 60 s = **6000**.

4 DOMESTIC ELECTRIC CIRCUITS ★★★ MOST ASKED

STEP 1 Power reaches the house through overhead poles or underground cables

↓ ↓

STEP 2 Main fuse, then the electricity meter (kWh)

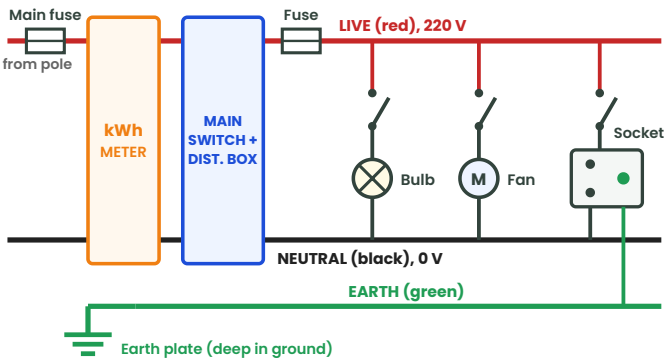
↓ ↓

STEP 3 Main switch and distribution box: separate circuits, each with its own fuse

The three wires

Wire	Insulation	Job
Live (positive)	red	brings current in; at 220 V
Neutral (negative)	black	carries current back; at 0 V
Earth	green	safety wire to a metal plate buried deep in the earth near the house

Potential difference between live and neutral in India = **220 V**. Newer wiring may use brown (live), blue (neutral) and green-yellow (earth); for the exam, write the NCERT colours.



Domestic circuit: every appliance is in **parallel**, with its switch and fuse in the **live** wire.

Why parallel, and why two circuits?

- ✔ **Parallel:** every appliance gets the full 220 V and can be switched on or off independently
- ✔ **15 A circuit** for high-power appliances: geyser, air conditioner, refrigerator, electric iron, room heater
- ✔ **5 A circuit** for low-power ones: bulbs, fans, TV, mobile chargers

Earthing: why the third wire? CBSE 2025

Appliances with a **metal body** (electric iron, toaster, refrigerator, table fan) have their body joined to the earth wire. If live wire touches the body, the leaked current flows to the earth through this **low-resistance path**. The body stays at the potential of the earth, so the user does not get a severe shock.



★★★ *Live red, neutral black, earth green, 220 V, 50 Hz: ye five facts har saal aate hain, Nitesh Bhaiya ki guarantee!*

5 SHORT CIRCUIT & OVERLOADING

★★★ MOST ASKED

SHORT CIRCUIT	OVERLOADING
Live and neutral wires come in **direct contact**, because the insulation is damaged or the appliance is faulty. Resistance becomes almost zero, so the current **rises suddenly** to a very large value.	Current drawn exceeds the safe limit because **too many appliances** run on one socket, or the **supply voltage rises** accidentally. Wires overheat slowly.
Result: sparks, heating, fire	**Result:** overheating, fire risk



MEME BREAK

*Insulation ghisi, live ne neutral ko chhua: **short circuit!** Current achanak itna ki sparks hi sparks.*

Electric fuse: the most important safety device

- A short piece of wire of **low melting point** and relatively high resistance, made of a suitable alloy
- Connected in **series**, in the **live wire**, at the start of the circuit
- Works on the **heating effect of current** ($H = I^2Rt$)
- If the current exceeds its rating, the wire **melts** and breaks the circuit, saving wiring and appliances
- Common ratings: 1 A, 2 A, 3 A, 5 A, 10 A, 15 A
- **MCB** (miniature circuit breaker): a switch that **trips automatically** on excess current; reset it instead of replacing a wire

Precautions against overloading NCERT

- ✔ Do not connect too many appliances to a single socket
- ✔ Use a fuse or MCB of the correct rating in every circuit
- ✔ Run high-power appliances on the 15 A circuit only
- ✔ Replace damaged or old wiring and loose plugs

Solved Example 3: will the fuse blow? NCERT

An electric oven of **2 kW** runs on a 220 V domestic circuit of current rating **5 A**. What happens?

1. Given: P = 2 kW = 2000 W, V = 220 V
2. Formula: I = P / V
3. I = 2000 / 220 = **9.09 A**
4. 9.09 A > 5 A, so the circuit is overloaded: the **fuse melts** and cuts off the supply. Run the oven on a 15 A circuit.



MEME BREAK

*220 V AC ki power: **unlimited power!** Isliye fuse, MCB aur earthing ke bina kabhi kaam nahi.*



MEME BREAK

*Metal body + earthing = chill. Metal body, no earthing, live wire chhua = **jhatka**. Third wire mat bhoolo!*

Solved Example 4: choose the fuse

Which fuse suits a 1 kW electric iron on 220 V: 2 A, 5 A or 15 A?

1. I = P / V = 1000 / 220 = **4.55 A**
2. Pick the rating just above the normal current: **5 A**. A 2 A fuse would melt at once; a 15 A fuse would not protect the iron.

× WRONG	✔ RIGHT
Fuse in the neutral wire	Fuse and switch always in the live wire
Fuse wire: high melting point	Low melting point, so it melts first
Earth wire carries current all the time	Only when current leaks to the body
Appliances in series	In parallel: same 220 V each



BUJHO *Fuse ko neutral wire mein laga doon toh bhi toh circuit tootega na?*

PAHELI *Circuit tootega, par appliance phir bhi **220 V live** se juda rahega. Chhuoge toh shock! Isliye fuse hamesha **live wire** mein.*



Short and long answer questions

1. State Fleming's left-hand rule. A wire carrying current from south to north is placed in a magnetic field acting vertically upward. Find the direction of the force on it. **(3 marks)**
2. Explain the function of an electric fuse in a domestic circuit. Why is it always connected in the live wire? CBSE 2024 **(3 marks)**
3. What is short-circuiting? How is it different from overloading? Name two devices that protect a circuit from both. **(3 marks)**

Exam tips

- ✔ Draw the three fingers at right angles and label F, B, I: diagrams fetch marks
- ✔ Direction answers need a word: up, down, north, south, into or out of the page
- ✔ For electrons, first flip the direction to get conventional current
- ✔ Fuse questions: always write "in series, in the live wire"

6 PRACTICE MCQS

Phone side mein, pen haath mein! 10 MCQs khud solve karo, phir key dekho.



1. The direction of force on a current-carrying conductor in a magnetic field is given by: ★★★
(a) right-hand thumb rule (b) Fleming's left-hand rule
(c) Ohm's law (d) Joule's law

2. The force on a current-carrying conductor in a magnetic field is maximum when the angle between current and field is: ★★★
(a) 0° (b) 45°
(c) 90° (d) 180°

3. In Fleming's left-hand rule, the middle finger shows the direction of: ★★☆
(a) magnetic field (b) force
(c) motion (d) current

4. A proton enters vertically downwards into a uniform magnetic field acting horizontally from west to east. The force on it acts towards the: SQP 2025 ★★★
(a) north (b) south
(c) east (d) west

5. The frequency of AC supplied in India is: ★★★
(a) 50 Hz (b) 60 Hz
(c) 100 Hz (d) 220 Hz

6. The insulation colour of the live wire as per NCERT is: ★★☆
(a) black (b) green
(c) red (d) blue

7. An electric fuse works on the: ★★★
(a) magnetic effect of current (b) heating effect of current
(c) chemical effect of current (d) induction of current

8. Domestic appliances are connected in parallel so that each: ★★☆
(a) draws the same current (b) gets the same voltage, 220 V
(c) has the same resistance (d) uses less power

9. **Assertion (A):** In domestic circuits, the earth wire is connected to a metal plate buried deep inside the earth.
Reason (R): The earth wire keeps the metallic body of an appliance at the potential of the earth, so the user does not get a severe shock. CBSE 2025 ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The current in an AC supply of 50 Hz changes its direction 50 times in one second.
Reason (R): The current in AC reverses its direction twice in every cycle. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Fleming's LHR gives the force direction.
2	(c)	F is maximum when I is perpendicular to B.
3	(d)	Middle finger = current.
4	(b)	v down, B east: thumb points south.
5	(a)	India: 50 Hz.
6	(c)	Live red, neutral black, earth green.
7	(b)	Fuse melts by heating (I^2Rt).
8	(b)	Parallel: same 220 V for each.
9	(a)	R correctly explains A.
10	(d)	It changes 100 times a second; R is true.

Model answers: short and long answer

1 Fleming's rule and the force
Statement as in Section 2. Middle finger north (current), forefinger up (field): the thumb points **east**. Force: **towards east**.

2 Fuse
- A low-melting-point wire in series; it **melts** when current exceeds its rating and cuts the supply
- In the live wire, so the appliance is **cut off from the 220 V side** and is safe to touch

3 Short circuit vs overloading
- Short circuit: live touches neutral, so current shoots up suddenly
- Overloading: too many appliances or a voltage surge; current exceeds the rating
- Protection: **fuse** and **MCB**

Day 39 checklist: tick before you sleep

- ☐ I can explain the aluminium rod activity
- ☐ I can apply Fleming's left-hand rule
- ☐ I can compare AC and DC and explain 50 Hz
- ☐ I can draw a home circuit with all 3 wires
- ☐ I solved all 10 MCQs and checked the key



My score: ____ / 5
5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 39 COMPLETE +100 XP earned. Next: Level 40



BUJHO *Samajh gaya! Left hand se force, live wire mein fuse, metal body ke liye earthing: done!*

PAHELI *Shabaash! Kal Level 40: Boss Level aur poore Physics ka mock test.*



NITESH SIR SAYS
Left hand force ke liye, right hand field ke liye. Haath mat ulto, marks mat gawao. ***Kal milte hai Day 40 pe!***

DAY 40
LEVEL 40

PHYSICS | CHAPTER 8 | BOSS LEVEL + MOCK TEST

BOSS LEVEL + PHYSICS MOCK TEST

Goal: clear Chapter 8, then test all four physics chapters.



by Nitesh Prakash

01 Boss Level | 02 Mock Test | 03 Answer Key | 04 Marking Points | 05 Checklist | +200 XP today | Level 40 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 8 BOSS LEVEL

PAPER PATTERN

Time: 40 min | Max. marks: 24 | Questions: 14
Q1–10: 1 mark each; Q11–13: 3 marks each; Q14: 5 marks
Plan: Q1–10 in 10 min, Q11–13 in 15, Q14 in 10, re-check 5



BUJHO *Boss Level aa gaya... main thoda so jaaun? Kal se pakka padhunga.*

PAHELI *Uth jao Bujho! 40 minute, bina book dekhe. Phir answer key se khud check karna.*



SECTION A: OBJECTIVE — 10 × 1 = 10 marks

1. Inside a long current-carrying solenoid, the magnetic field lines are: ★★★

(a) circles (b) parallel straight lines
(c) absent (d) spreading outwards

2. Two magnetic field lines never intersect because, at the crossing point: ★★☆

(a) compass points two ways (b) field is zero
(c) lines cancel (d) poles merge

3. The direction of the magnetic field around a straight current-carrying wire is given by: ★★★

(a) Fleming's left-hand rule (b) the right-hand thumb rule
(c) Ohm's law (d) Snell's law

4. As we move away from a straight current-carrying wire, the magnetic field: ★★☆

(a) increases (b) decreases
(c) stays the same (d) becomes zero at once

5. The field at the centre of a circular coil grows if we increase: ★★☆

(a) the radius (b) the resistance
(c) the number of turns (d) the length of leads

6. Outside a bar magnet, field lines go from: ★★★

(a) S pole to N pole (b) N pole to S pole
(c) the middle to the ends (d) nowhere: they are closed

7. A soft iron core is placed inside a solenoid to: ★★★

(a) reduce the current (b) make it a weak magnet
(c) make a strong electromagnet (d) stop heating

8. An electron beam moving east to west enters a field acting vertically downward. The force on it is towards the: ★★☆

(a) north (b) south
(c) east (d) west

ASSERTION–REASON OPTIONS

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** A current-carrying solenoid behaves like a bar magnet.
Reason (R): One end of the solenoid acts as a north pole and the other end as a south pole. ★★★

10. **Assertion (A):** The fuse is always connected in the live wire.
Reason (R): A fuse wire has a very high melting point. ★★☆

SECTION B: SHORT ANSWER — 3 × 3 = 9 marks

11. Draw the pattern of magnetic field lines produced by a current-carrying straight conductor and mark their direction. Which rule gives this direction? **(3)** CBSE 2024 ★★★

12. Name two factors on which the field of a current-carrying solenoid depends. List two ways to strengthen an electromagnet whose core material is fixed. **(3)** CBSE 2025 ★★★

13. A 1.1 kW heater and a 220 W TV run together on a 220 V line with a 5 A fuse. Will the fuse blow? Show the calculation. **(3)** ★★★

SECTION C: LONG ANSWER — 1 × 5 = 5 marks

14. (a) Describe an activity to show the field pattern of a current-carrying solenoid and compare it with a bar magnet. (b) Give two differences between an electromagnet and a permanent magnet. (c) State Fleming's left-hand rule. **(5)** ★★★

Diagram scale aur arrows ke saath banao



EXAM RULE

*Shortcut mat lo! Ray diagram aur circuit diagram dono pencil aur scale se, **arrows ke saath**.*

2 PHYSICS MOCK TEST: DAYS 21-39

PAPER PATTERN

Time: 75 min | Max. marks: 40 | Questions: 22, all compulsory
Marks: A (MCQ) 1, B (very short) 2, C (short) 3, D (long) 5, E (case-based) 4
Time plan: A 15 min, B 12, C 15, D 10, E 8; last 15 min for checking



REALITY CHECK
*Formula ratta, par numerical mein unit aur sign bhool gaye? Result: **2/20**. Har step likho, marks pakke!*

SECTION A: MULTIPLE CHOICE — 12 × 1 = 12 marks

1. The focal length of a spherical mirror of radius of curvature 20 cm is: ★★★
(a) 20 cm (b) 10 cm
(c) 40 cm (d) 5 cm

2. An object at the centre of curvature of a concave mirror forms an image that is: ★★★
(a) virtual and erect
(b) at F, highly diminished
(c) at C, real, inverted, same size
(d) at infinity

3. The mirror used as a rear-view mirror in vehicles is: ★★★
(a) plane (b) convex
(c) concave (d) any curved mirror

4. Glass has refractive index 1.5. Speed of light in glass is: ★★☆
(a) 1.5×10^8 m/s (b) 2×10^8 m/s
(c) 3×10^8 m/s (d) 4.5×10^8 m/s

5. The power of a lens of focal length −25 cm is: ★★★
(a) −4 D (b) +4 D
(c) −0.25 D (d) +0.04 D

6. The near point of a normal human eye is at: ★★☆
(a) infinity (b) 2.5 cm
(c) 25 cm (d) 1 m

7. Myopia is corrected by using a: ★★★
(a) concave lens (b) convex lens
(c) bifocal lens (d) prism

8. The sky looks blue because of: ★★☆
(a) dispersion (b) scattering of light
(c) reflection (d) total internal reflection

9. The magnetic field lines around a straight current-carrying wire are: ★★☆
(a) parallel lines (b) ellipses
(c) radial lines (d) concentric circles

10. Three resistors of 6 Ω each are connected in parallel. The equivalent resistance is: ★★★
(a) 2 Ω (b) 3 Ω
(c) 6 Ω (d) 18 Ω

Q11–12: use the Assertion–Reason options on page 1.

11. **Assertion (A):** Danger signal lights are red in colour.
Reason (R): Red light is scattered the least by fog and smoke. ★★★

12. **Assertion (A):** The resistivity of a wire depends on its length.
Reason (R): The resistance of a wire is given by R = ρl/A. ★★☆

SECTION B: VERY SHORT ANSWER — 5 × 2 = 10 marks

13. Draw a ray diagram for an object placed between F and P of a concave mirror. State the nature of the image. **(2)** ★★★

14. A concave lens (f = 15 cm) forms an image 10 cm from it. Find the object distance and the magnification. **(2)** NCERT ★★★

15. What is dispersion of white light? Which colour bends the most through a prism? **(2)** ★★☆

16. Why are the coils of electric toasters and irons made of an alloy rather than a pure metal? **(2)** NCERT ★★☆

17. What is the power of accommodation of the eye? Which muscles make it possible? **(2)** ★★☆

SECTION C: SHORT ANSWER — 3 × 3 = 9 marks

18. A 5 cm tall object is placed 25 cm from a convex lens of focal length 10 cm. Find the position, size and nature of the image. **(3)** NCERT ★★★

19. Resistors of 5 Ω, 10 Ω and 30 Ω are in parallel across 12 V. Find each current, total current and total resistance. **(3)** NCERT ★★★

20. An electric iron of resistance 20 Ω draws 5 A. Find (a) the heat produced in 30 s and (b) its power. **(3)** NCERT ★★★

SECTION D: LONG ANSWER — 1 × 5 = 5 marks

21. What is myopia? Give two causes and draw ray diagrams of the defect and its correction. A person cannot see beyond 1.2 m: find the lens type and power. **(5)** ★★★

SECTION E: CASE-BASED — 1 × 4 = 4 marks

22. A student records these readings for a nichrome wire: **(4)** ★★★

V (volt)	1.0	2.0	3.0	4.0
I (ampere)	0.1	0.2	0.3	0.4

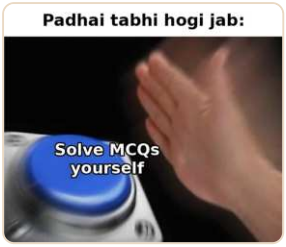
(i) Which law does the data verify? **1**
(ii) Find the resistance of the wire. **1**
(iii) A wire of the same material and thickness but twice as long replaces it. Find its resistance, with reason. **2**

EXAM TIP

Numericals: Given, Formula, Substitution, Answer with unit. **End of paper:** check with the key on pages 3–4.

3 ANSWER KEY AND MARKING POINTS

Pehle khud solve kiya? Tabhi answer key kholo, warna button dabao phir se!



Boss Level: MCQ answers

Q	Ans	Why
1	(b)	Uniform field: parallel straight lines
2	(a)	A compass cannot point two ways at once
3	(b)	Right-hand thumb rule
4	(b)	Field weakens with distance from the wire
5	(c)	More turns, stronger field
6	(b)	Outside: N to S; inside: S to N
7	(c)	Soft iron is strongly magnetised
8	(a)	Current east, B down: force north
9	(a)	R explains A
10	(c)	R is false: fuse wire has a low melting point

Boss Level: model answers, Q11–13

11 Field around a straight wire — 1 + 1 + 1

- Concentric circles centred on the wire, in a plane perpendicular to it; closer near the wire
- Arrows: current upward gives anticlockwise circles seen from above
- Rule: **right-hand thumb rule**

12 Solenoid and electromagnet — 1½ + 1½

- Field depends on the **number of turns** and the **current** (and the core material)
- To strengthen: **increase the number of turns** and **increase the current**

13 Will the fuse blow? — 1 + 1 + 1

1	Total power P = 1100 W + 220 W = 1320 W
2	I = P / V = 1320 / 220 = **6 A**
3	6 A > 5 A: the circuit is overloaded and the **fuse melts**

Boss Level: marking points, Q14

14 Solenoid, magnets, Fleming — 2 + 2 + 1

- (a) Pass a solenoid through a cardboard sheet, sprinkle iron filings, switch on and tap: the pattern matches a bar magnet; inside, lines are parallel (uniform field); one end acts as N, the other as S
- (b) Electromagnet: temporary, strength can be changed, polarity can be reversed. Permanent magnet: always magnetic, fixed strength and polarity
- (c) Thumb, forefinger, middle finger of the left hand at right angles: forefinger field, middle finger current, thumb force

Mock Test: Section A answers

Q	Ans	Why
1	(b)	f = R/2 = 10 cm
2	(c)	At C: real, inverted, same size
3	(b)	Convex: erect image, wide view
4	(b)	$v = c/n = 3\times10^8/1.5$
5	(a)	P = 1/f = 1/(−0.25 m) = −4 D
6	(c)	25 cm for a normal eye
7	(a)	Concave (diverging) lens
8	(b)	Blue is scattered more than red
9	(d)	Concentric circles around the wire
10	(a)	6/3 = 2 Ω
11	(a)	R explains A
12	(d)	A is false: resistivity is a material property; R is true

Mock Test: Section B, Q13–14

13 Concave mirror, object between F and P — 1 + 1

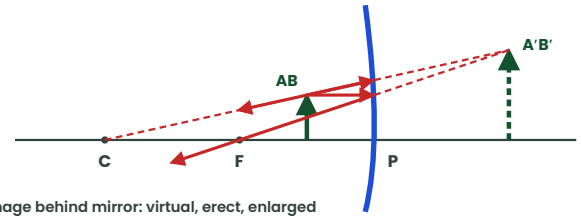


Image behind mirror: virtual, erect, enlarged

14 Concave lens — 1 + 1

1	Given: f = −15 cm, v = −10 cm (virtual image, same side)
2	1/u = 1/v − 1/f = −1/10 + 1/15 = −1/30, so **u = −30 cm**
3	m = v/u = (−10)/(−30) = **+1/3**: erect, diminished

Mock Test: Section B, Q15–17

15 Dispersion — 1 + 1

Splitting of white light into its seven colours (VIBGYOR) by a prism. **Violet** bends the most, red the least.

16 Alloy coils — 1 + 1

An alloy has a **higher resistivity** than a pure metal, and it **does not oxidise (burn) easily** at high temperature.

17 Power of accommodation — 1 + 1

The ability of the eye lens to **change its focal length** to see near and far objects clearly; done by the **ciliary muscles**.

Physics hit-list: Eduspark PYQ analysis

Topic	Rating
Ray diagrams for mirrors and lenses (all cases)	★★★
Mirror and lens formula numericals with signs	★★★
Myopia, hypermetropia and their correction	★★★
Series and parallel circuit numericals	★★★
Field lines: bar magnet, wire, loop, solenoid	★★★

Mock Test: Section C, Q18–20

18 Convex lens numerical 1 + 1 + 1

1. Given: h = +5 cm, u = −25 cm, f = +10 cm
2. 1/v = 1/f + 1/u = 1/10 − 1/25 = 3/50, so **v = +16.7 cm**
3. m = v/u = 16.7/(−25) = −0.67; h′ = m × h = **−3.3 cm**. Real, inverted, diminished, on the other side of the lens

19 Parallel resistors 1 + 1 + 1

1. Same V = 12 V across each: I_1 = 12/5 = **2.4 A**, I_2 = 12/10 = **1.2 A**, I_3 = 12/30 = **0.4 A**
2. Total I = 2.4 + 1.2 + 0.4 = **4 A**
3. $1/R_p$ = 1/5 + 1/10 + 1/30 = 10/30, so $\mathbf{R_p = 3\ \Omega}$ (check: 12/3 = 4 A)

20 Heating effect 2 + 1

1. $H = I^2Rt = 5^2 \times 20 \times 30 = \mathbf{1.5 \times 10^4\ J}$
2. $P = I^2R = 25 \times 20 = \mathbf{500\ W}$

Mock Test: Section D and E

21 Myopia 1 + 1 + 2 + 1

- Near objects clear, far objects blurred; the image forms **in front of the retina**
- Causes: **excessive curvature** of the eye lens; **elongation** of the eyeball
- Diagrams: rays from a far object meet in front of the retina; a concave lens makes them meet on the retina
- Far point 1.2 m, so f = −1.2 m; P = 1/f = **−0.83 D**, a **concave** lens

22 Case: Ohm's law 1 + 1 + 2

- (i) **Ohm's law**: V ∝ I at constant temperature
- (ii) R = V/I = 1.0/0.1 = **10 Ω**
- (iii) R ∝ l (same material and area), so double length gives **20 Ω**

Physics formula sheet: Days 21–39

Chapter	Must-know formulae and values
Light	f = R/2; mirror 1/v + 1/u = 1/f, m = −v/u; lens 1/v − 1/u = 1/f, m = v/u; n = c/v; P = 1/f (f in m), D
Human eye	near point 25 cm, far point infinity; myopia: concave lens, P = −1/far point (m); hypermetropia: convex lens
Electricity	I = Q/t; V = W/Q; V = IR; R = ρl/A; series $R = R_1 + R_2 + \ldots$; parallel $1/R = 1/R_1 + 1/R_2 + \ldots$; $H = I^2Rt$; $P = VI = I^2R = V^2/R$; 1 kWh = 3.6×10^6 J
Magnetism	right-hand thumb rule (field), Fleming's left-hand rule (force); India: 220 V, 50 Hz AC

Sign convention (New Cartesian): distances from the pole or optical centre; along the incident light +ve; real object u is -ve; f of concave mirror and concave lens -ve.



FORMULA SHEET

*Physics ke formula dekh ke darr lagta hai? Bas yaad rakho: **1/v − 1/u = 1/f**, V = IR, H = I²Rt. Simple hai boss!*

× WRONG	✔ RIGHT
u = +25 cm for a real object	u = −25 cm (against incident light)
P = 1/f with f in cm	f in metres: P = 1/(−0.25) = −4 D
Parallel $R = R_1 + R_2$	$1/R = 1/R_1 + 1/R_2$; R is smaller than each
Right hand for force on a wire	Left hand (Fleming) for force; right hand for field

Score tracker

Paper	Max.	My marks
Chapter 8 Boss Level	24	____
Physics Mock Test	40	____
Total	**64**	____

55+ Topper zone | 45–54 Strong, polish the numericals | 32–44 Revise the weak chapter | below 32 Redo Days 21–39 summaries



RESULT DAY

*Mock mein 55+ aaye? Ab toh **Jethalal wali smile** banti hai!*

Day 40 checklist: tick before you sleep

- ☐ I finished the Boss Level in 40 minutes
- ☐ I wrote the mock test in one sitting
- ☐ I checked every answer with the key
- ☐ I reworked every numerical I got wrong
- ☐ I listed my weakest chapter to revise



My score: ____ / 5

5/5 = Jalwa hai! Weak chapter ka summary phir se padho.

LEVEL 40 COMPLETE

+200 XP earned. Physics complete. Book total: 5000 XP



BUJHO *Physics khatam! Light, eye, electricity, magnet: sab boss levels clear!*

PAHELI *Shabaash! Ab Day 41 se Biology shuru. Naya chapter, naya level!*



NITESH SIR SAYS

*Physics mein formula kam, concept zyada. Diagram saaf, unit sahi, sign convention pakka. **Kal milte hai Day 41 pe!***

PHYSICS | CHAPTER 8 COMPLETE | PHYSICS COMPLETE

CHAPTER 8 SWAHAA!

Magnetic Effects of Electric Current | Days 36–40



BUJHO *Field lines, thumb rule, solenoid, Fleming ka left hand aur mock test bhi! Chapter 8 ke saath poori Physics Swahaa!*

PAHELI *Trophy phir se tumhari, Bujho! 40 levels clear. Ab Biology ki duniya mein chalte hain.*

CHAPTER XP EARNED
+600 XP

LEVELS CLEARED
5 levels (Days 36–40)

BOOK PROGRESS
Level 40 / 60 5000 XP total

WHAT YOU HAVE MASTERED

- ✔ Magnetic field and field lines
- ✔ Field of a circular loop and a solenoid
- ✔ Force on a conductor, Fleming's left-hand rule
- ✔ Right-hand thumb rule, straight wire
- ✔ Electromagnets and their uses
- ✔ AC, DC and domestic electric circuits

NEXT CHAPTER Day 41: Biology begins — *Physics cleared!*